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1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.

2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.

3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

CHINA
AFIX

CK APPD

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0006532879

ENGINEERING RELEASED

2016-07-06

D11 MLB - PVT

LAST_MODIFICATION=Wed Jul 6 08:55:26 2016

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Schematic & PCB Callouts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-00482	1	SCH_MLB,D11-12	SCH	CRITICAL	?
820-00229	1	PCBF,MLB,D11-12	PCB	CRITICAL	?

SCH 051-00482
BRD 820-00229
MCO 056-01585

System Block Diagram:

<rdar://problem/16684269>

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NAND BOM Options

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00169	1	NAND, H, 32GB, 16nm, MLC	U1701	CRITICAL	NAND_32G
335S00182	1	NAND, H, 128GB, 16nm, TLC	U1701	CRITICAL	NAND_128G
335S00183	1	NAND, T, 256GB, 3Dv3, TLC	U1701	CRITICAL	NAND_256G
138S00867	5	CAP_XSR, 180F, 20%, 6.3V, 0.65MM, MRZTL, 0402	C1748, C1713, C1716, C1721, C1733	CRITICAL	NAND_32G
138S00003	5	CAP_XSR, 150F, 20%, 6.3V, 0.65MM, MRZTL, 0402	C1748, C1713, C1716, C1721, C1733	CRITICAL	NAND_128G
138S00003	5	CAP_XSR, 150F, 20%, 6.3V, 0.65MM, MRZTL, 0402	C1748, C1713, C1716, C1721, C1733	CRITICAL	NAND_256G

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00201	335S00169	ALTERNATE	U1701	T, 15nm, MLC, 32GB
335S00209	335S00169	ALTERNATE	U1701	S, 16nm, MLC, 32GB
335S00195	335S00182	ALTERNATE	U1701	SS, 1Ynm, TLC, 128GB
335S00180	335S00182	ALTERNATE	U1701	T, 15nm, TLC, 128GB
335S00179	335S00182	ALTERNATE	U1701	SD, 15nm, TLC, 128GB
335S00148	335S00183	ALTERNATE	U1701	SD, 3Dv2, TLC, 256GB
335S00190	335S00183	ALTERNATE	U1701	SS, 3Dv3, TLC, 256GB

#22686038: See Radar

Magnesium Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
338S00173	338S00203	ALTERNATE	U2402	Larger MAFer (-20 FLOW) Magnesium

UT LDO Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S00089	353S00015	ALTERNATE	U2501	ST, LDO REG, 2.920V, CSP 0.65x0.45

Mamba LDO Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S00932	353S00576	ALTERNATE	U3001	ST, LDO REG, 2.75V

I2C5 Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00234	335S00233	ALTERNATE	U1101	I2C5 ALTERNATE

Active Diode Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S00106	376S00047	ALTERNATE	Q2101	DIODES INC. ACT DIODE

DDR PLL Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
155S00095	155S00068	ALTERNATE	FL1501	FERR BD, 1600MHZ, 160MH, 20MH, 0305

Power Inductor Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00118	152S00075	ALTERNATE	ALL	IND, PMU, SML3, 1.2 OH, 3.5A, 0.600 OHM, 2018
152S00077	152S00397	ALTERNATE	ALL	IND, PMU, SML3, 1.0 OH, 3.25A, 0.150 OHM, 2018
152S00121	152S00081	ALTERNATE	ALL	IND, PMU, SML3, 47 OH, 2.5A, 0.440 OHM, 2012
152S00123	152S1936	ALTERNATE	ALL	IND, PMU, SML3, 15 OH, 0.75A, 0.600 OHM, 2015
152S00402	152S00366	ALTERNATE	ALL	IND, MLT, 20H, 1.2A, 0.220 OHM, 0003
152S00297	152S1843	ALTERNATE	ALL	CINTEC 2012 30H
152S00365	152S00297	ALTERNATE	ALL	CINTEC 2012 30H
152S00398	152S00204	ALTERNATE	ALL	IND, PMU, 0.220H, 20H, 0.75, 220MH, 2012
152S00120	152S00077	ALTERNATE	ALL	For CHESTNUT 1400000P only
152S00117	152S00074	ALTERNATE	ALL	IND, PMU, SML3, 1.0 OH, 3.5A, 0.600 OHM, 2018

updated 11/12

updated 11/12

reverted 11/13

For Chestnut inductor; so it doesn't interfere with PMU inductor Buck 7 alts
Except BUCKS LX (BUCKS LX is Taiyo only)

Global R/C Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
118S0764	118S0717	ALTERNATE	ALL	RES, 3.62K, 0.1%, 0201
138S0702	138S0657	ALTERNATE	ALL	CAP_XSR, 4.3UF, 4V, 0010
138S00006	138S0835	ALTERNATE	ALL	CAP, 3-TERM, 4.3UF, 4V, 0402
138S0648	138S0652	ALTERNATE	ALL	CAP_XSR, 4.7UF, 6.3V, 0.65MM, 0402, 74310
132S0400	132S0436	ALTERNATE	ALL	CAP_XSR, 0.220F, 6.3V, 0305, TDK
138S00024	138S0906	ALTERNATE	ALL	CAP, 10H, 1.00H, 1.50, 10H, 1H, 0402, 74310/TDK
138S0706	138S0739	ALTERNATE	ALL	CAP, CER, 10F, 20V, 10H, 10H, 0201, MURATA
138S0945	138S0739	ALTERNATE	ALL	CAP, CER, 10F, 20V, 10H, 10H, 0201, KYOCERA
132S0436	132S0400	ALTERNATE	ALL	CAP, CER, XSR, 0.220F, 20V, 0.3V, 20H

Global Ferrite Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
155S00067	155S0581	ALTERNATE	ALL	FERR, 2400MH, 0.380MH DCR, 0201
155S0581	155S00067	ALTERNATE	ALL	FERR, 2400MH, 0.380MH DCR, 0201
155S00012	155S00168	ALTERNATE	ALL	FLTR, 65 OHMS, 0605
155S00194	155S0610	ALTERNATE	ALL	FERR BD, 1500MH, TDK
155S00200	155S0610	ALTERNATE	ALL	FERR BD, 1500MH, TY
152S00489	152S00456	ALTERNATE	ALL	FERR BD, 0.470H, TY

Global Varistor Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S0168	377S0140	ALTERNATE	ALL	VARIATOR, 6.0V, 100PF, 0305

Acc Buck Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
371S00087	371S00064	ALTERNATE	D2700	D300E, SHOTTKY, 30V, 200MA, 0201
152S00558	152S00557	ALTERNATE	L2700	IND, ML2, 0.47UH, 2.5A, 50MHZ, 1608
376S00166	376S00164	ALTERNATE	Q2700, Q2701	PFET, 12V, CSP4

Load Switch OMIT TABLE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S01007	1	ONSEMI, IC, LOAD, SWITCH, MLCSM4	U2710, NFCSM_RF	CRITICAL	

N&V 15uF Cap Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00005	138S00003	ALTERNATE	(C0101, C1001, C1001)	CAP_XSR, 150F, 6.3V, 0.65MM, 0402, TY
138S00005	138S00003	ALTERNATE	(C0077, C1001, C1001)	CAP_XSR, 150F, 6.3V, 0.65MM, 0402, TY
138S00005	138S00003	ALTERNATE	(C0010, C1001, C1001)	CAP_XSR, 150F, 6.3V, 0.65MM, 0402, TY
138S00005	138S00003	ALTERNATE	(C0010, C1001, C1001)	CAP_XSR, 150F, 6.3V, 0.65MM, 0402, TY
138S00005	138S00003	ALTERNATE	(C1001, C1001, C1001)	CAP_XSR, 150F, 6.3V, 0.65MM, 0402, TY
138S00005	138S00003	ALTERNATE	(C0010, C1001, C1001)	CAP_XSR, 150F, 6.3V, 0.65MM, 0402, TY
138S00005	138S00003	ALTERNATE	(C0010, C1001, C1001)	CAP_XSR, 150F, 6.3V, 0.65MM, 0402, TY
138S00005	138S00003	ALTERNATE	(C1001, C1001, C1001)	CAP_XSR, 150F, 6.3V, 0.65MM, 0402, TY
138S00005	138S00003	ALTERNATE	(C0010, C1001, C1001)	CAP_XSR, 150F, 6.3V, 0.65MM, 0402, TY
138S00005	138S00003	ALTERNATE	(C0010, C1001, C1001)	CAP_XSR, 150F, 6.3V, 0.65MM, 0402, TY
138S00005	138S00003	ALTERNATE	(C0010, C1001, C1001)	CAP_XSR, 150F, 6.3V, 0.65MM, 0402, TY
138S00005	138S00003	ALTERNATE	(C0010, C1001, C1001)	CAP_XSR, 150F, 6.3V, 0.65MM, 0402, TY
138S00048	138S00003	ALTERNATE	ALL	CAP_XSR, 150F, 6.3V, 0.65MM, 0402, KYOCERA

MUR-KYTO 150F

CPU

GPU + GPU_SRAM



D11 EEEE CALLOUTS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-6838	1	EEEE CODE FOR 639-01812	EEEE_GY2T	CRITICAL	EEEE_D11_BEST_JP
825-6838	1	EEEE CODE FOR 639-01813	EEEE_GY2V	CRITICAL	EEEE_D11_SUPREME_JP
825-6838	1	EEEE CODE FOR 639-01814	EEEE_GY2W	CRITICAL	EEEE_D11_EXTREME_JP
825-6838	1	EEEE CODE FOR 639-02138	EEEE_H3RN	CRITICAL	EEEE_D11_BEST_ROW
825-6838	1	EEEE CODE FOR 639-02139	EEEE_H3RP	CRITICAL	EEEE_D11_SUPREME_ROW
825-6838	1	EEEE CODE FOR 639-02148	EEEE_H3RQ	CRITICAL	EEEE_D11_EXTREME_ROW
825-6838	1	EEEE CODE FOR 639-02468	EEEE_H8C1	CRITICAL	EEEE_D11_BEST_CH
825-6838	1	EEEE CODE FOR 639-02465	EEEE_H8CK	CRITICAL	EEEE_D11_SUPREME_CH
825-6838	1	EEEE CODE FOR 639-02462	EEEE_H8C3	CRITICAL	EEEE_D11_EXTREME_CH

CAYMAN DDR Alternates

339S00258	339S00257	ALTERNATE	ALL	DDR-S, 3G, B1

CAYMAN OMIT TABLE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
339S00257	1	CAYMAN, DDR-H, 3G, B1	U0700	CRITICAL	

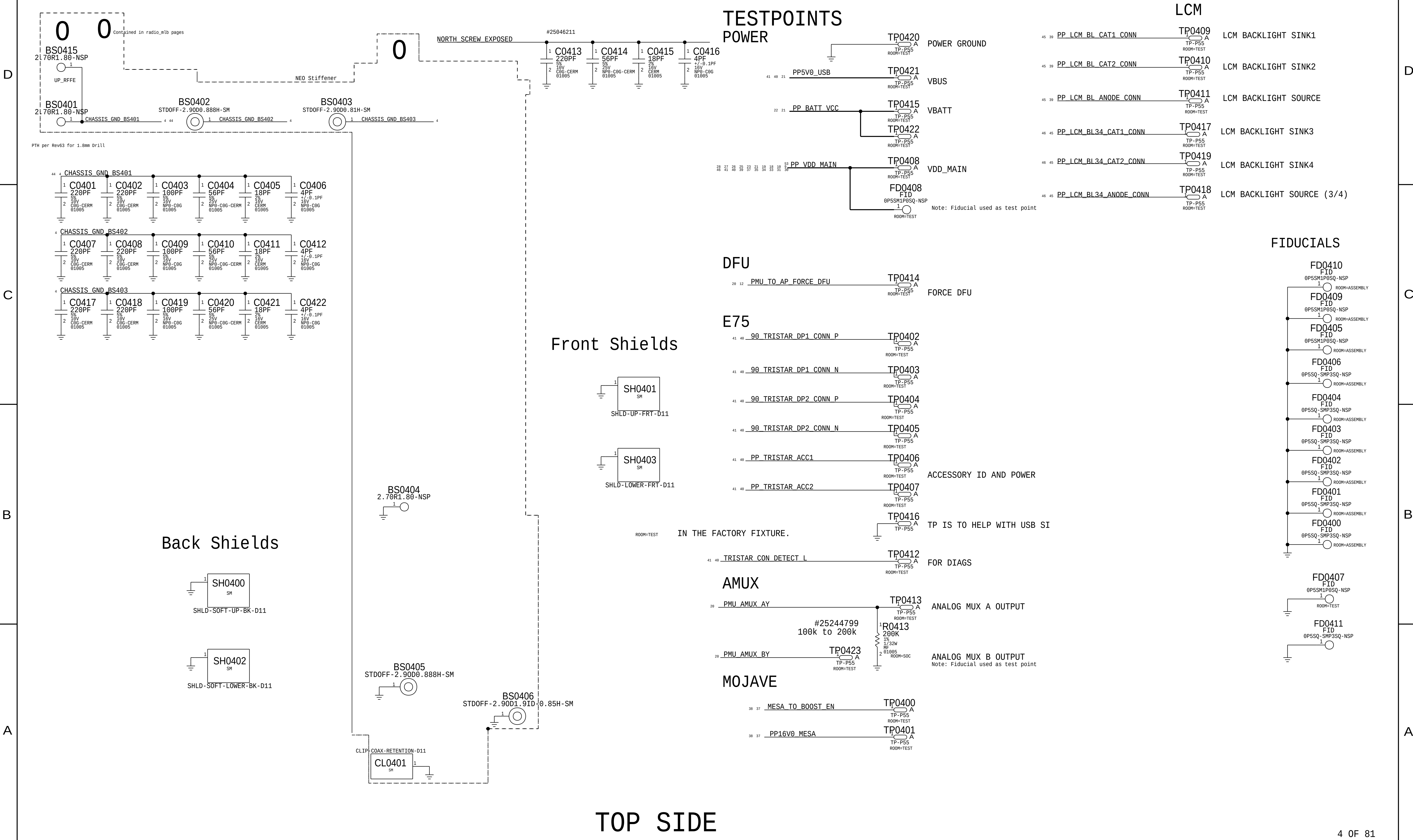
2.2uF CAP Alts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00049	138S00032	ALTERNATE	ALL	CAP, 22F, 150V, 2, 20PF, 2000, 20V, 20%, #110000A
138S00031	138S00032	ALTERNATE	ALL	CAP, 22F, 150V, 2, 20PF, 2000, 20V, 20%, #100000A

SIP Alternates

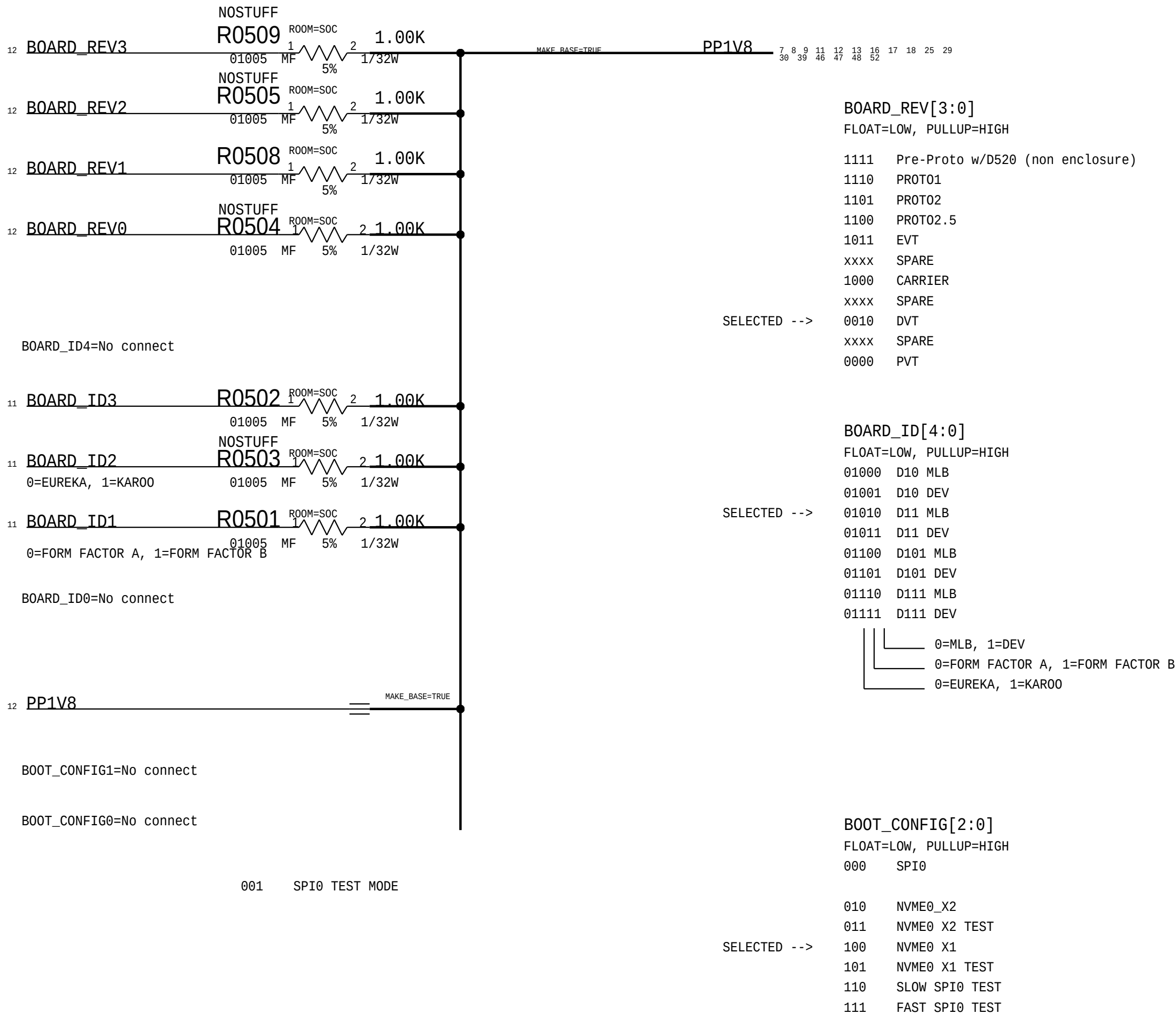
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339M00009	339M00003	ALTERNATE	ALL	TRINITY BLUE, STATS
339M00008	339M00002	ALTERNATE	ALL	NEO, STATS

Current as of D11 MC0 056-01585 rev. 63





BOOTSTRAPPING:BOARD REV BOARD ID BOOT CONFIG





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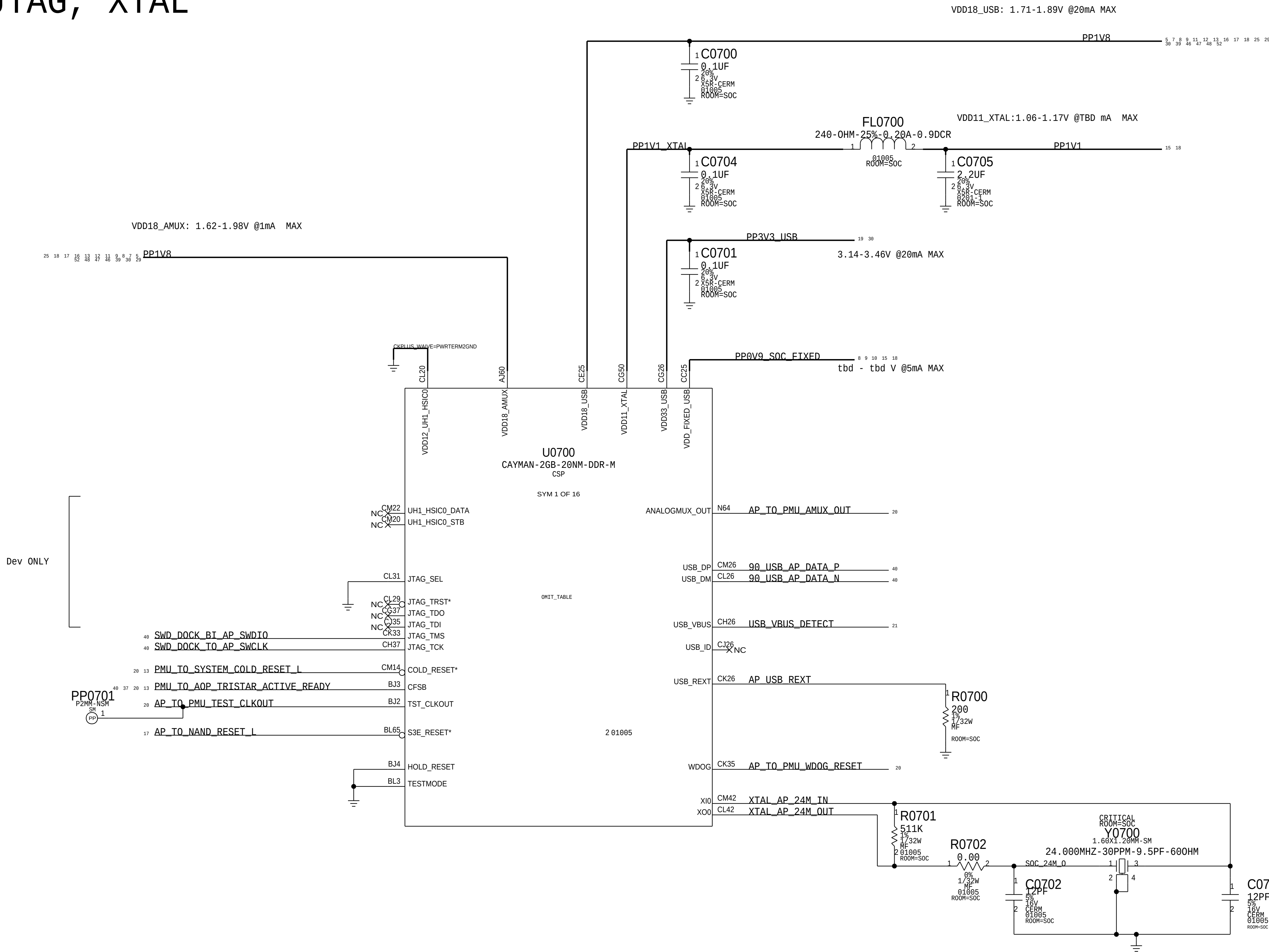
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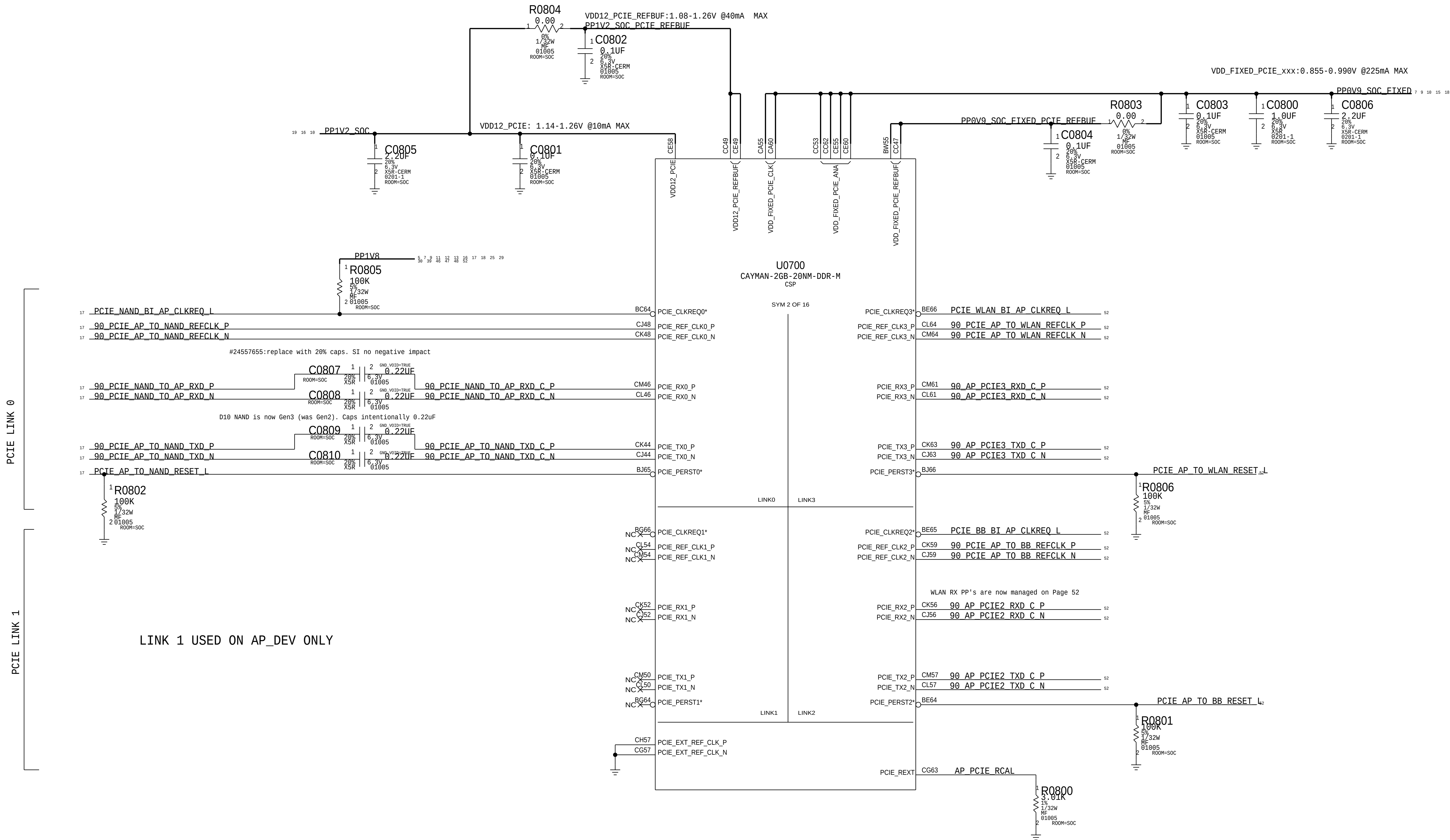
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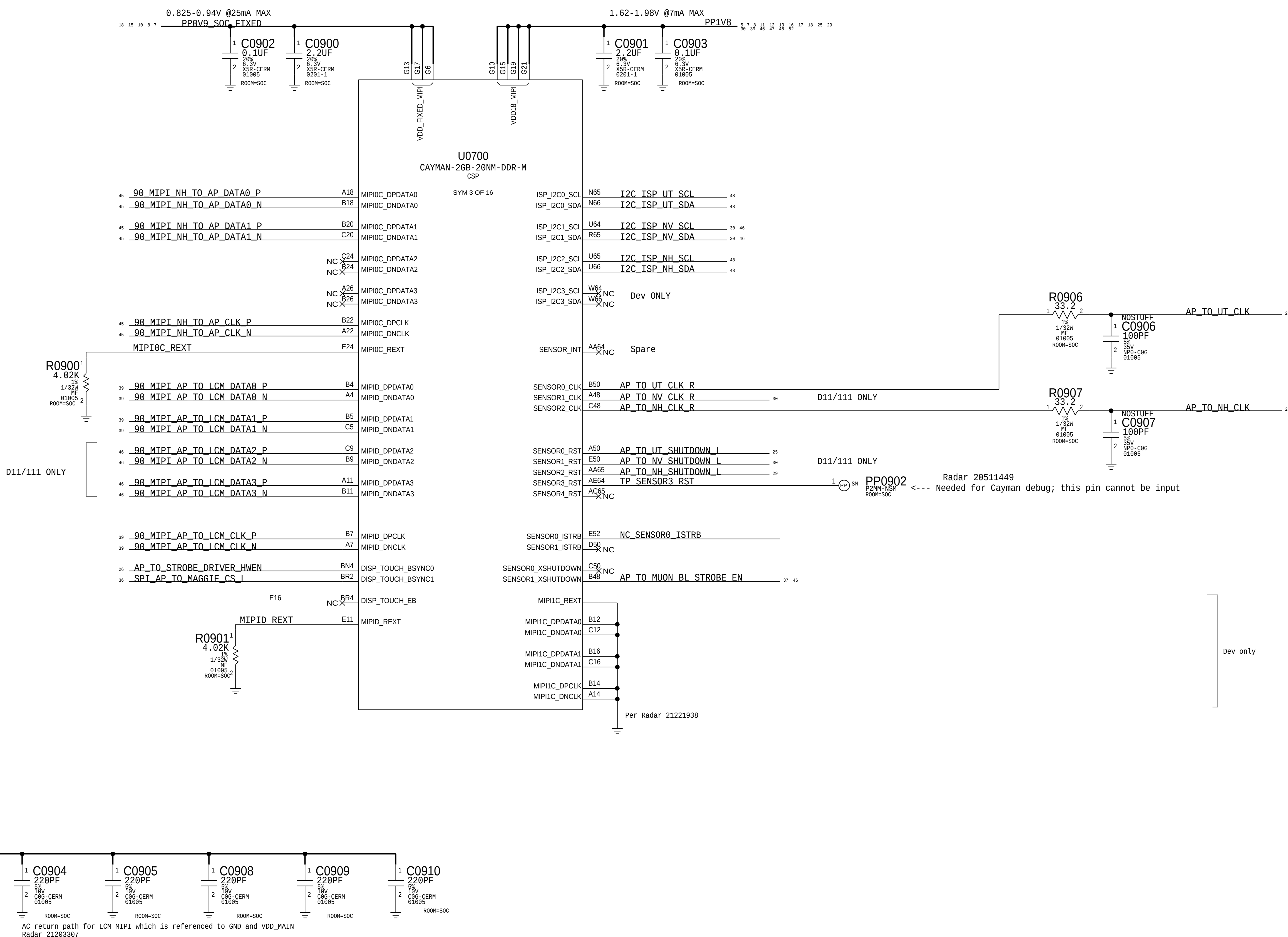
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SOC - PCIE INTERFACES



SOC - MIPI & ISP INTERFACES



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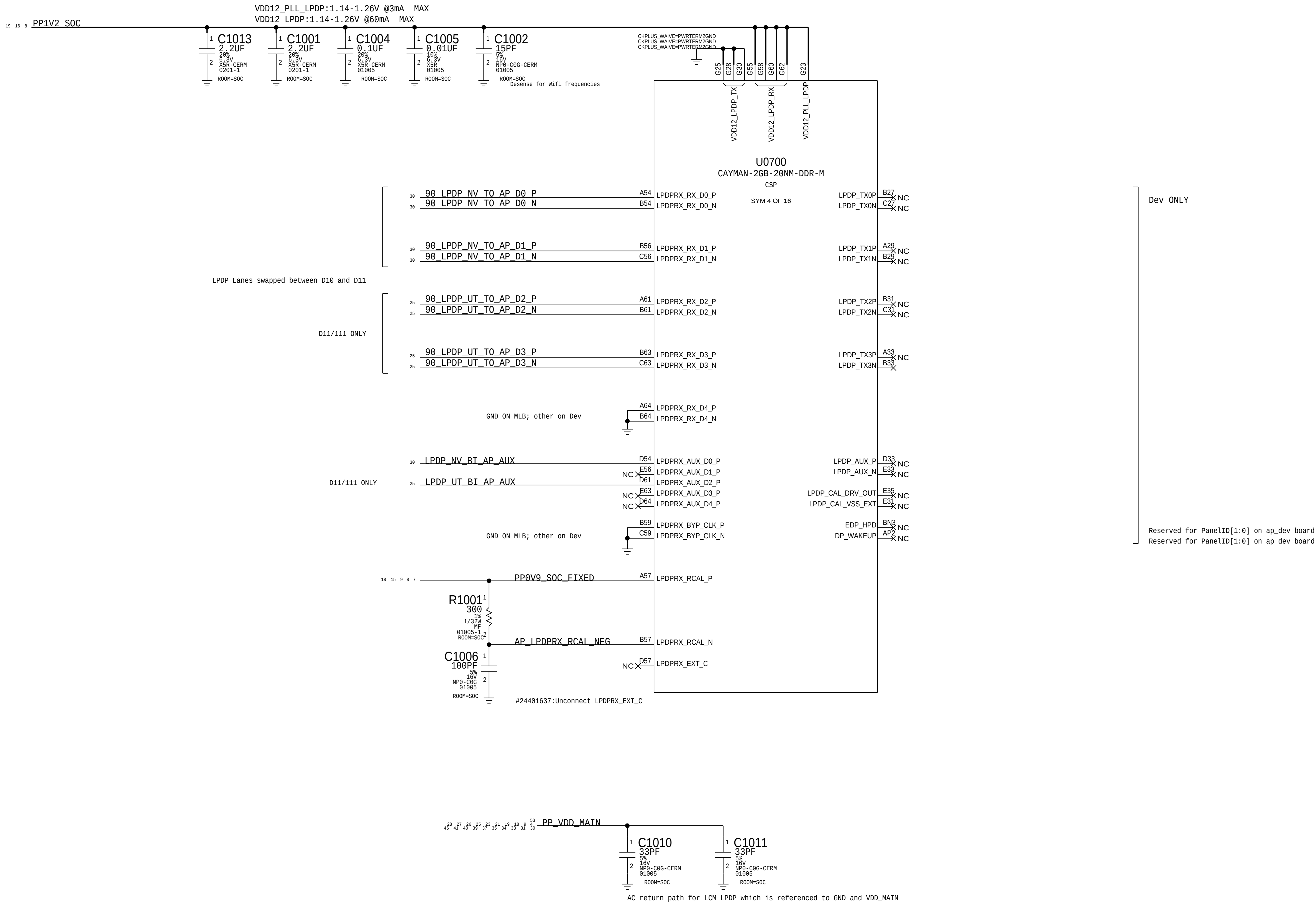
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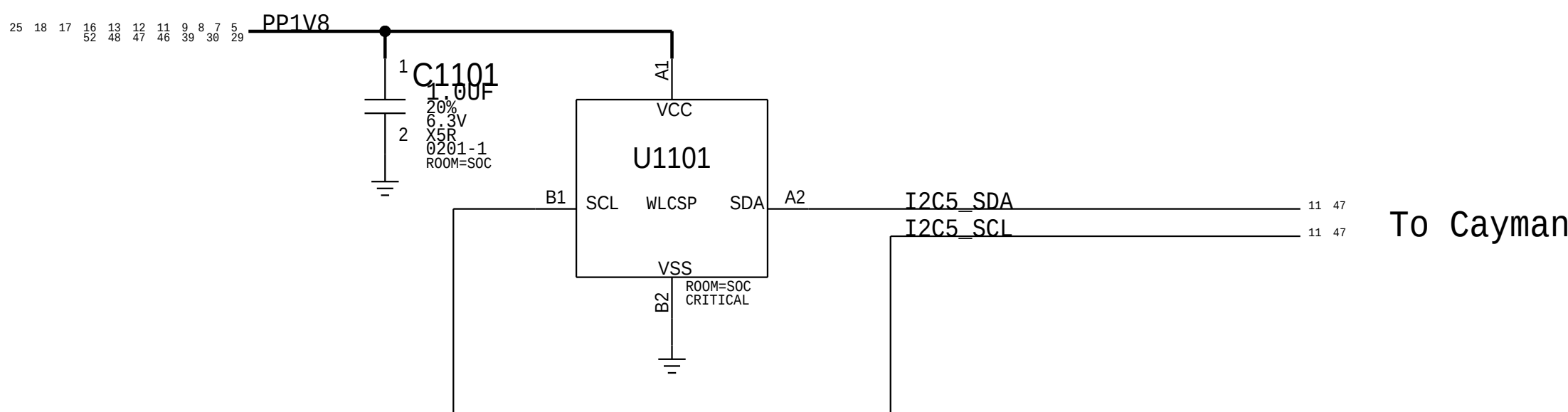
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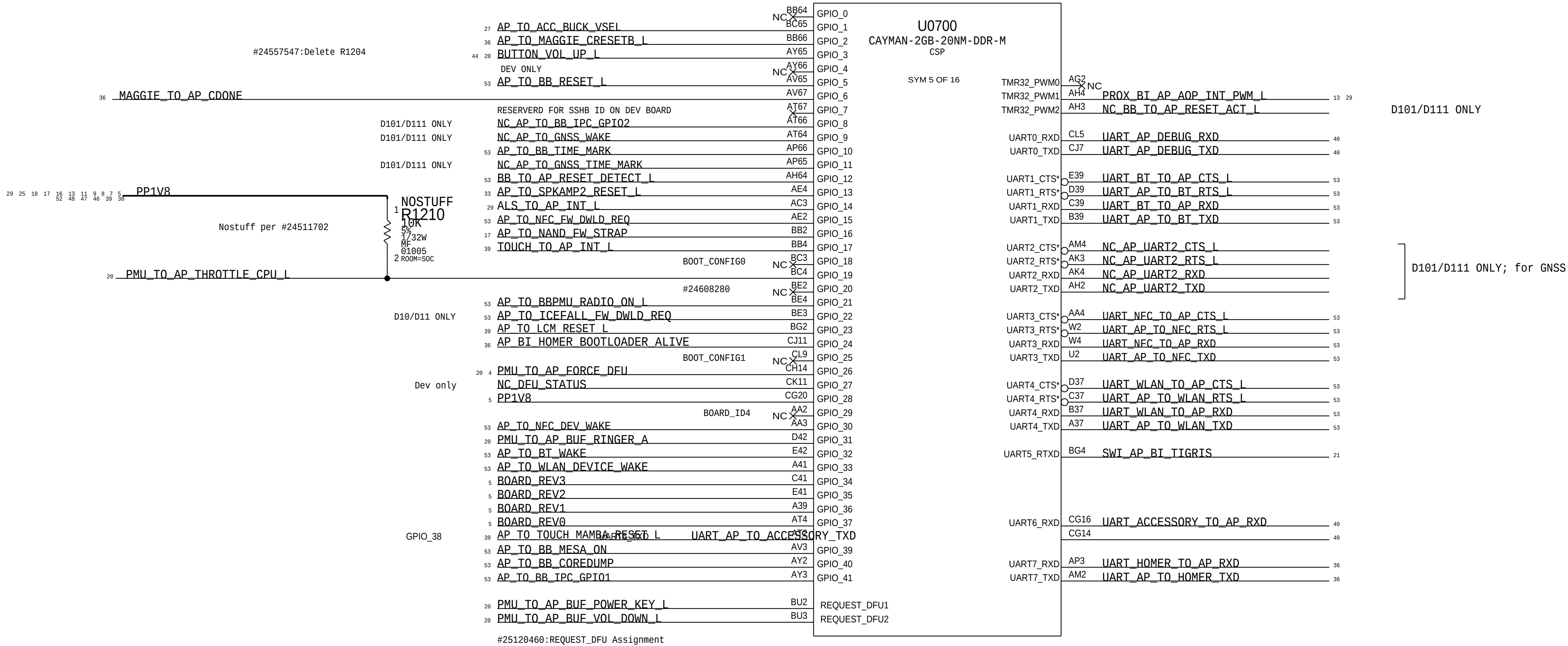
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I2C5
See Radar#25316444 for Details



SOC - GPIO INTERFACES





SOC - AOP

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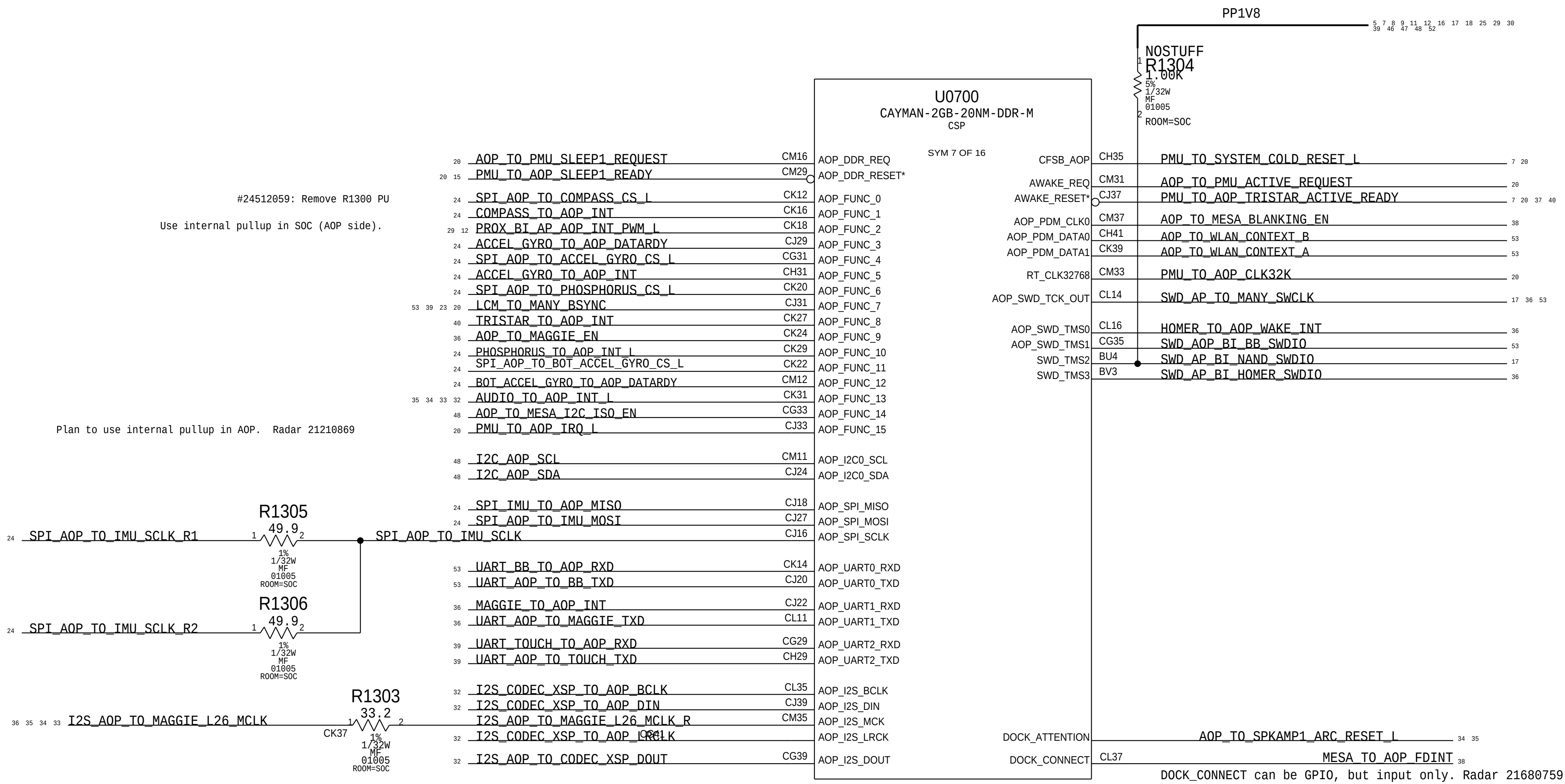
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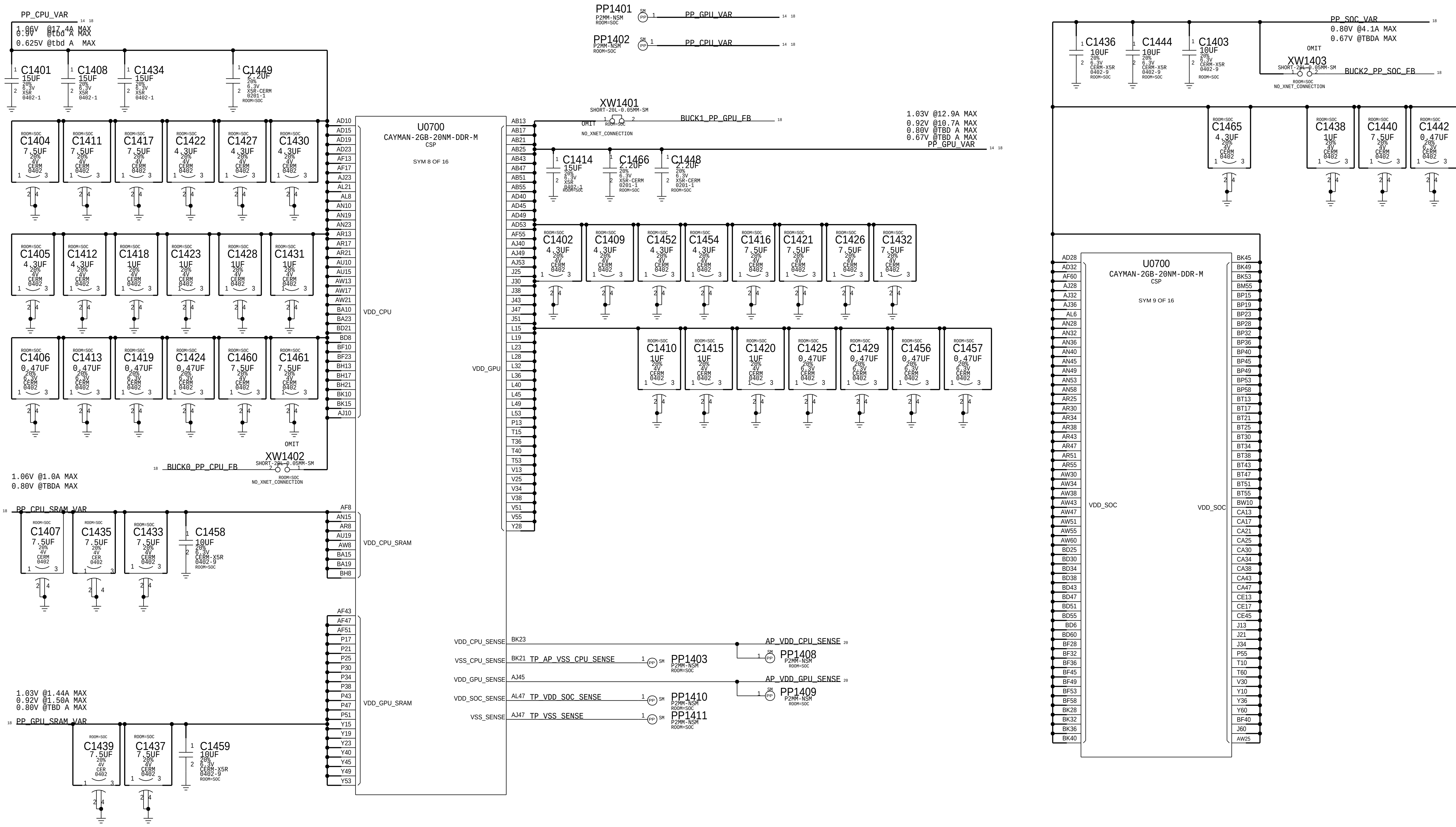
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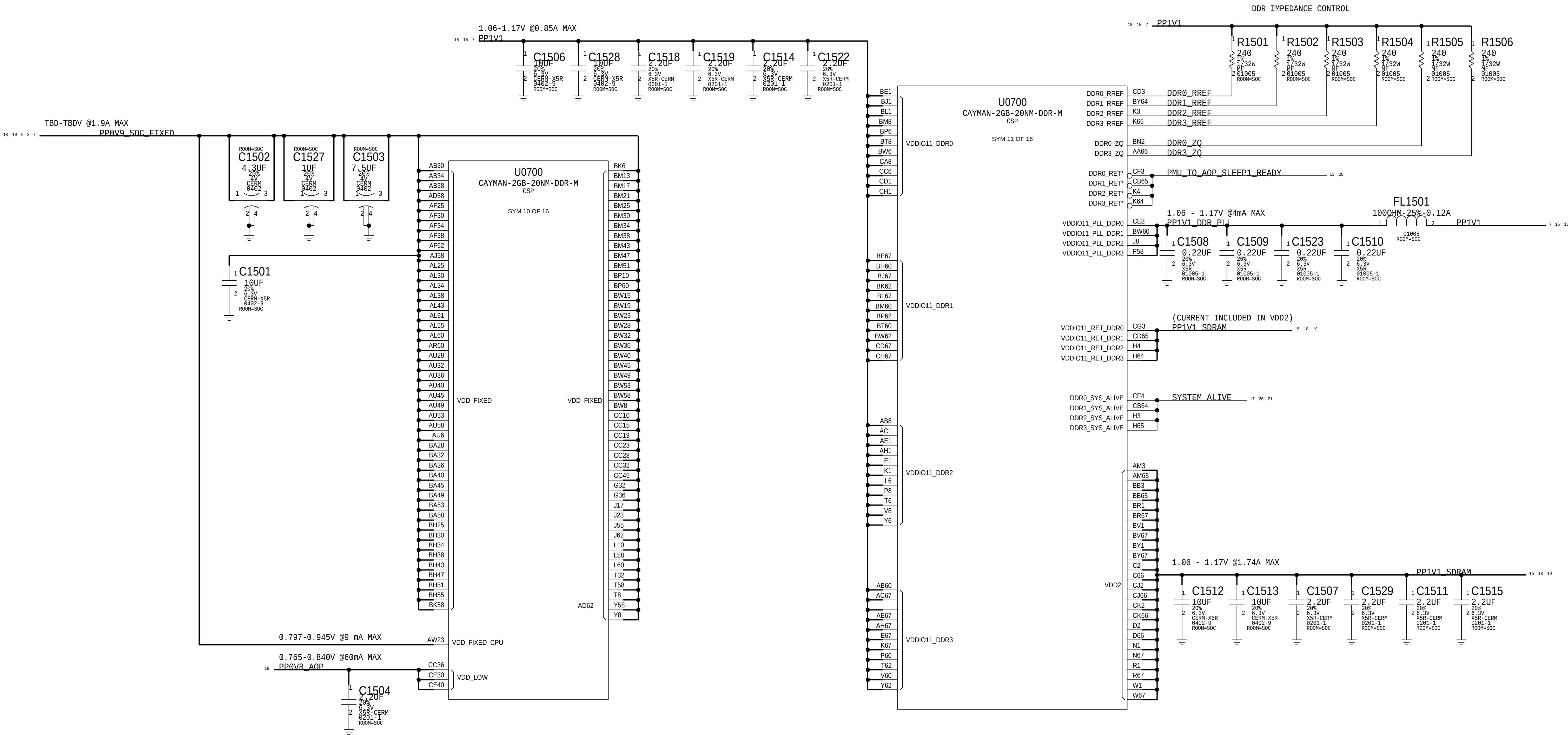
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SOC - CPU, GPU & SOC RAILS



SOC - POWER SUPPLIES

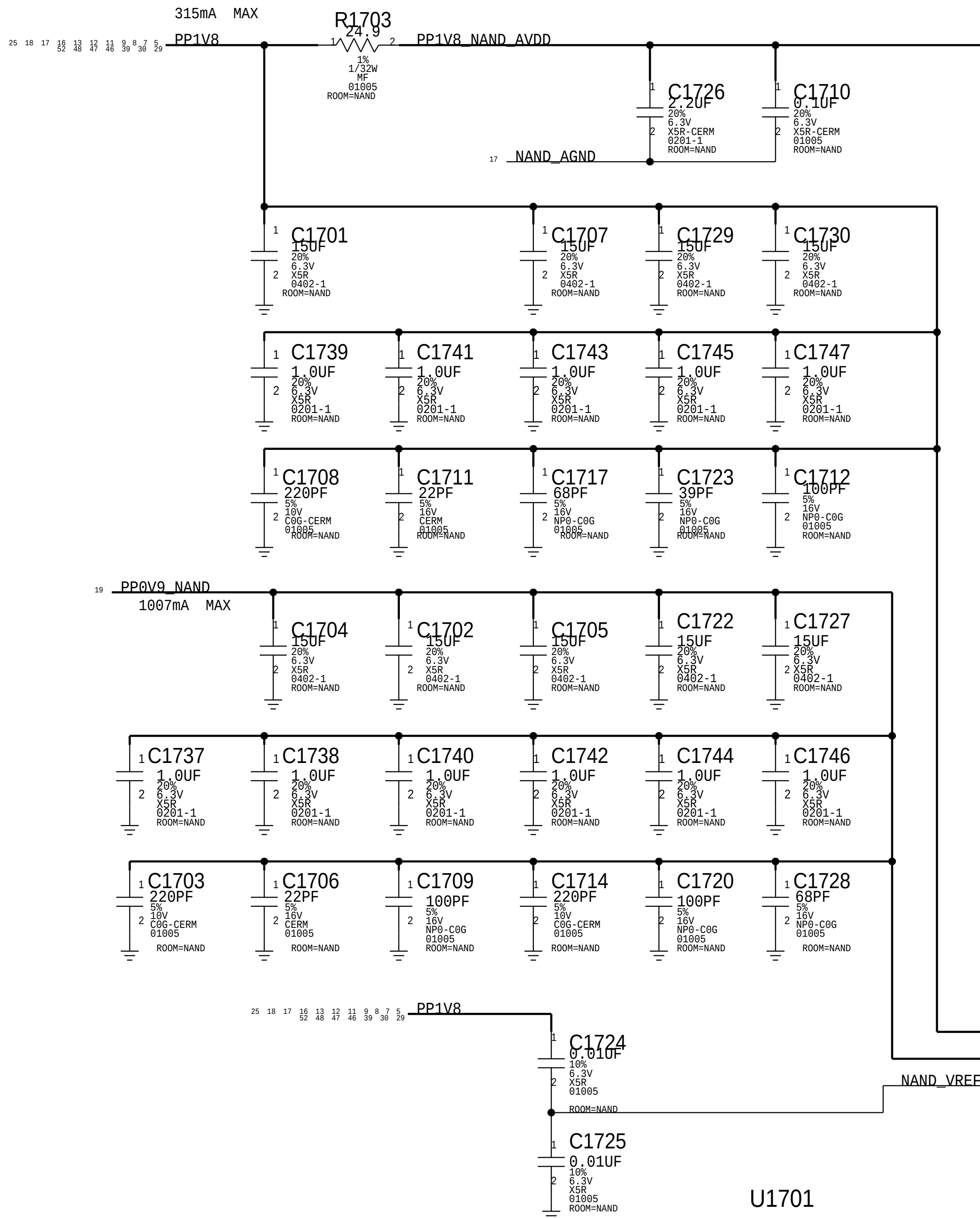
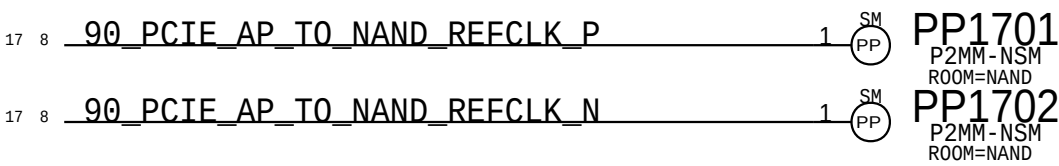


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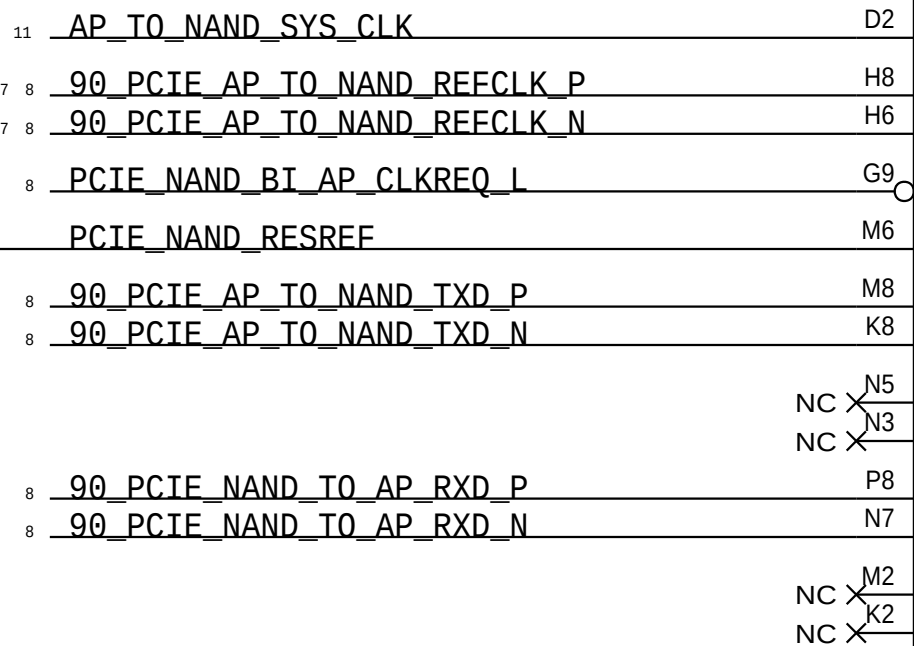
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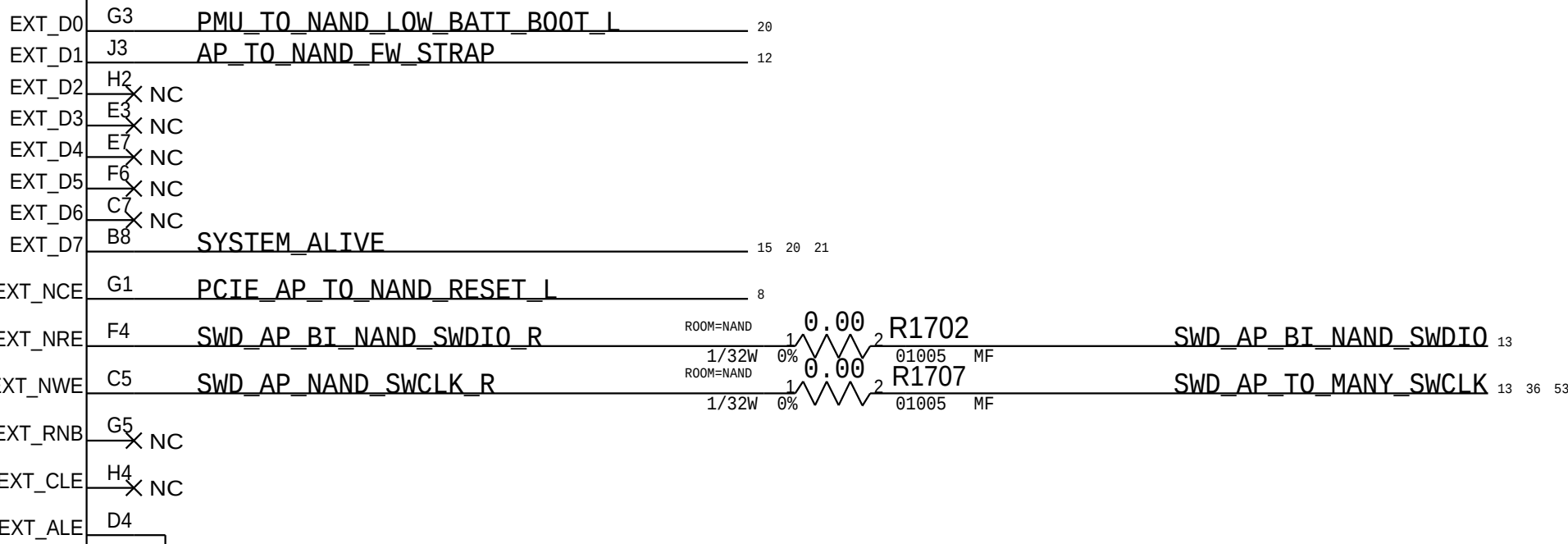
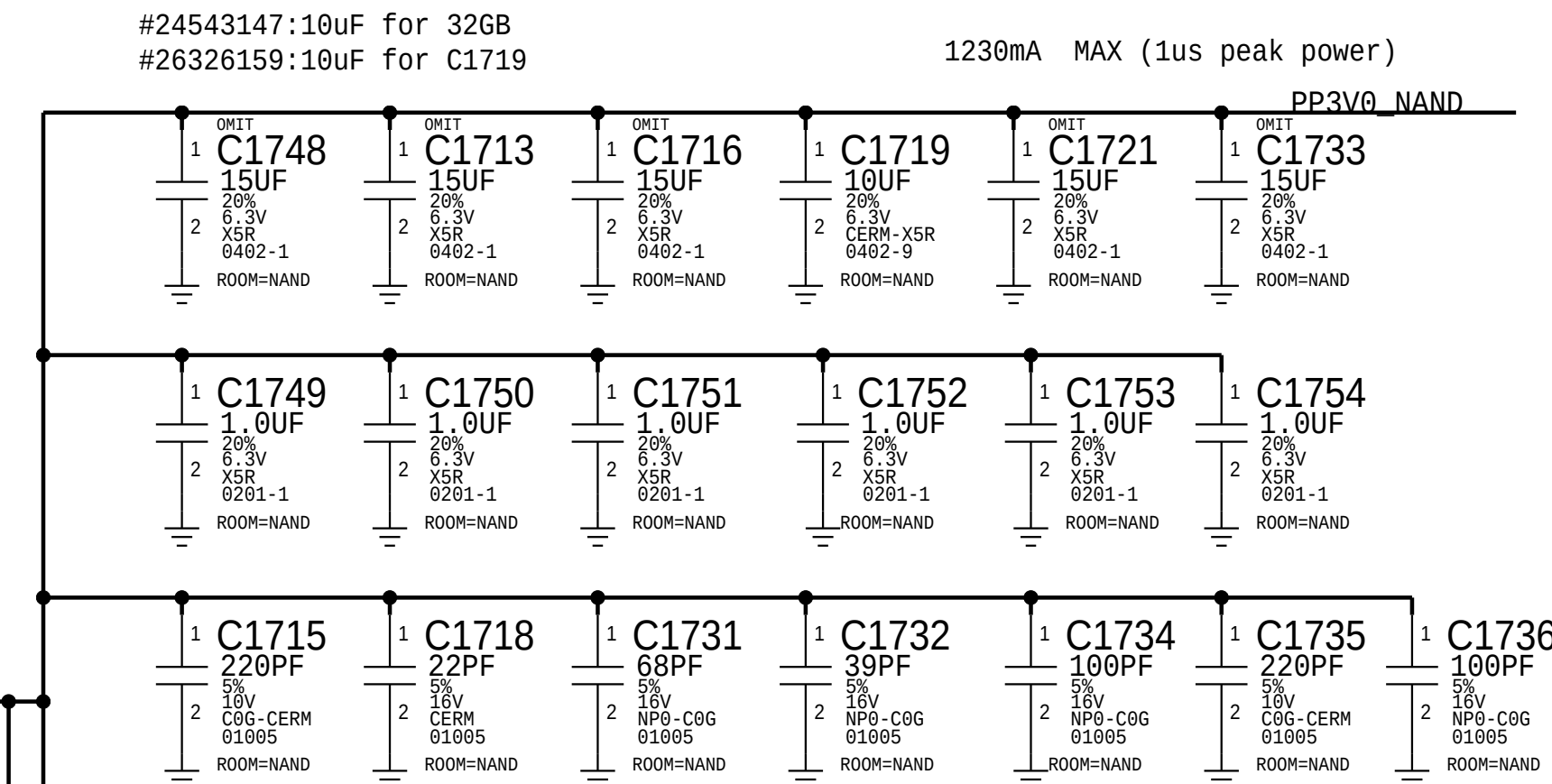
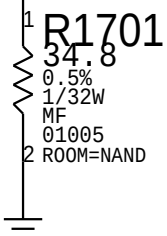
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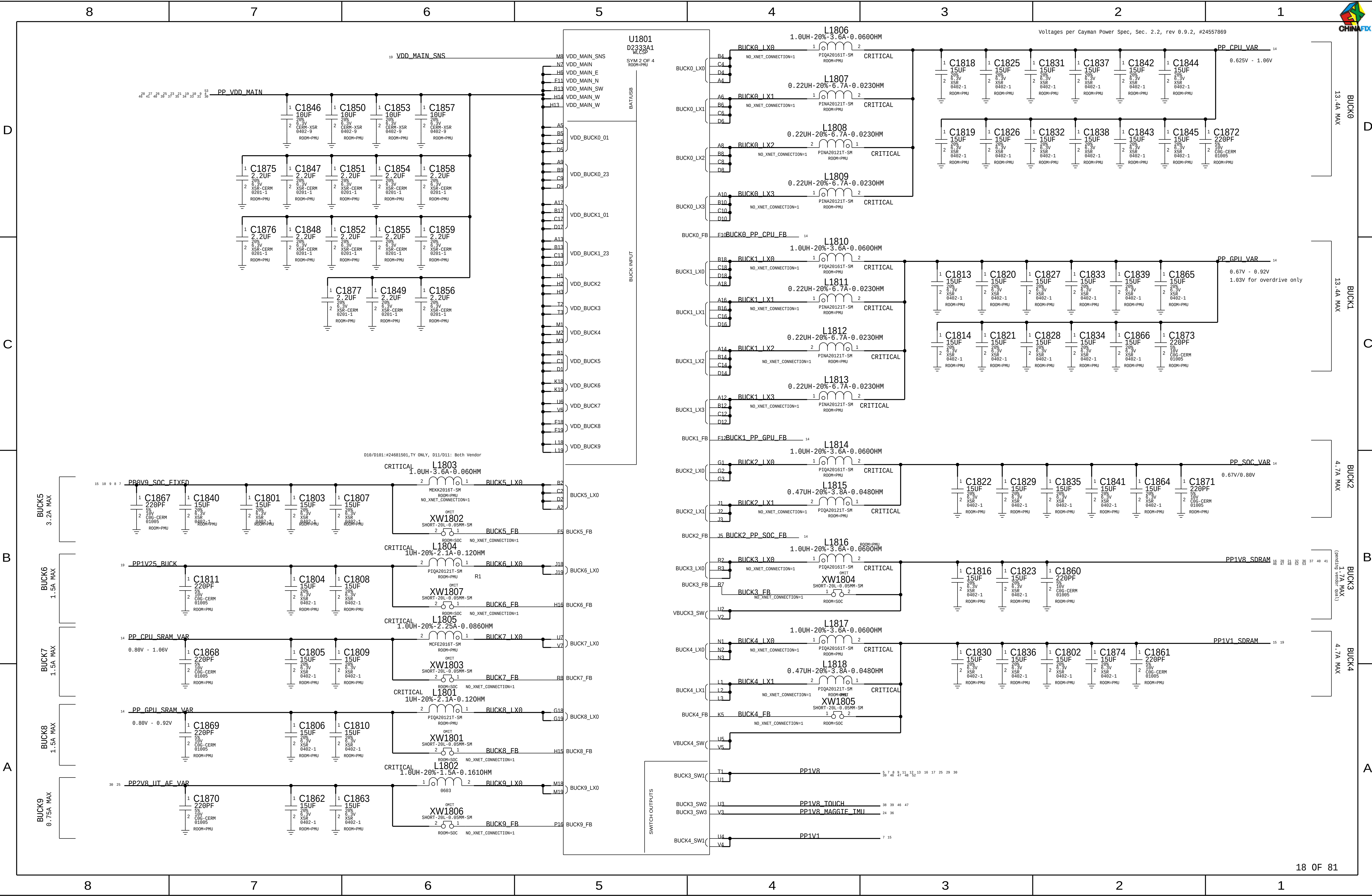
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VL6A
VER-1

ROOM=NAND
BOMOPTION=OMIT_TABLE
CRITICAL



NAND_ZQ







ADELYN LDO SPECS

LDO#	ADJ. RANGE, LOW	ADJ. RANGE, HI	ACCURACY	MAX. CURRENT
LD01 (Ca)	1.2-2.475V	2.4-3.675V	+/-1.4%	50mA
LD02 (Ca)	1.2-2.475V	2.4-3.675V	+/-2.5%	50mA
LD03 (Ca)	1.2-2.475V	2.4-3.675V	+/-2.5%	50mA
LD04 (D)	0.7-1.2V		+/-2.5%	60mA
LD05 (F)	2.5-3.6V(tbc)		+/-75mV	1000mA
LD06 (Cb)	1.2-2.475V	2.4-3.675V	+/-2.5%	250mA (500/100mA in bypass)
LD07 (Cb)	1.2-2.475V	2.4-3.675V	+/-30mV	250mA
LD08 (Cb)	1.2-2.475V	2.4-3.675V	+/-30mV	250mA
LD09 (Cb)	1.2-2.475V	2.4-3.675V	+/-25mV	250mA
LD010 (Ga)	0.7-1.2V		+/-4.5%	1150mA

LDO#	ADJ. RANGE, LOW	ADJ. RANGE, HI	ACCURACY	MAX. CURRENT
LD011 (Cb)	1.2-2.475V	2.4-3.675V	+/-30mV	250mA
LD012 (E)	1.8V		+/-5%	10mA
LD013 (Cb)	1.2-2.475V	2.4-3.675V	+/-30mV	250mA
LD014 (Gb)	0.7-1.4V		+/-3.0%	400mA
LD015 (Ca)	1.2-2.475V	2.4-3.675V	+/-2.5%	50mA
LD016 (Cb)	1.2-2.475V	2.4-3.675V	+/-30mV	250mA
LD017 (Ca)	1.2-2.475V	2.4-3.675V	+/-2.5%	50mA
LD018 (Gb)	0.7-1.4V		+/-3.0%	400mA
LD019 (Gb)	0.7-1.4V		+/-3.0%	400mA
LD0_RTC	2.5V		+/-2.0%	10mA
BUF_1V2	1.2V		+/-5.0%	10mA

D

D

C

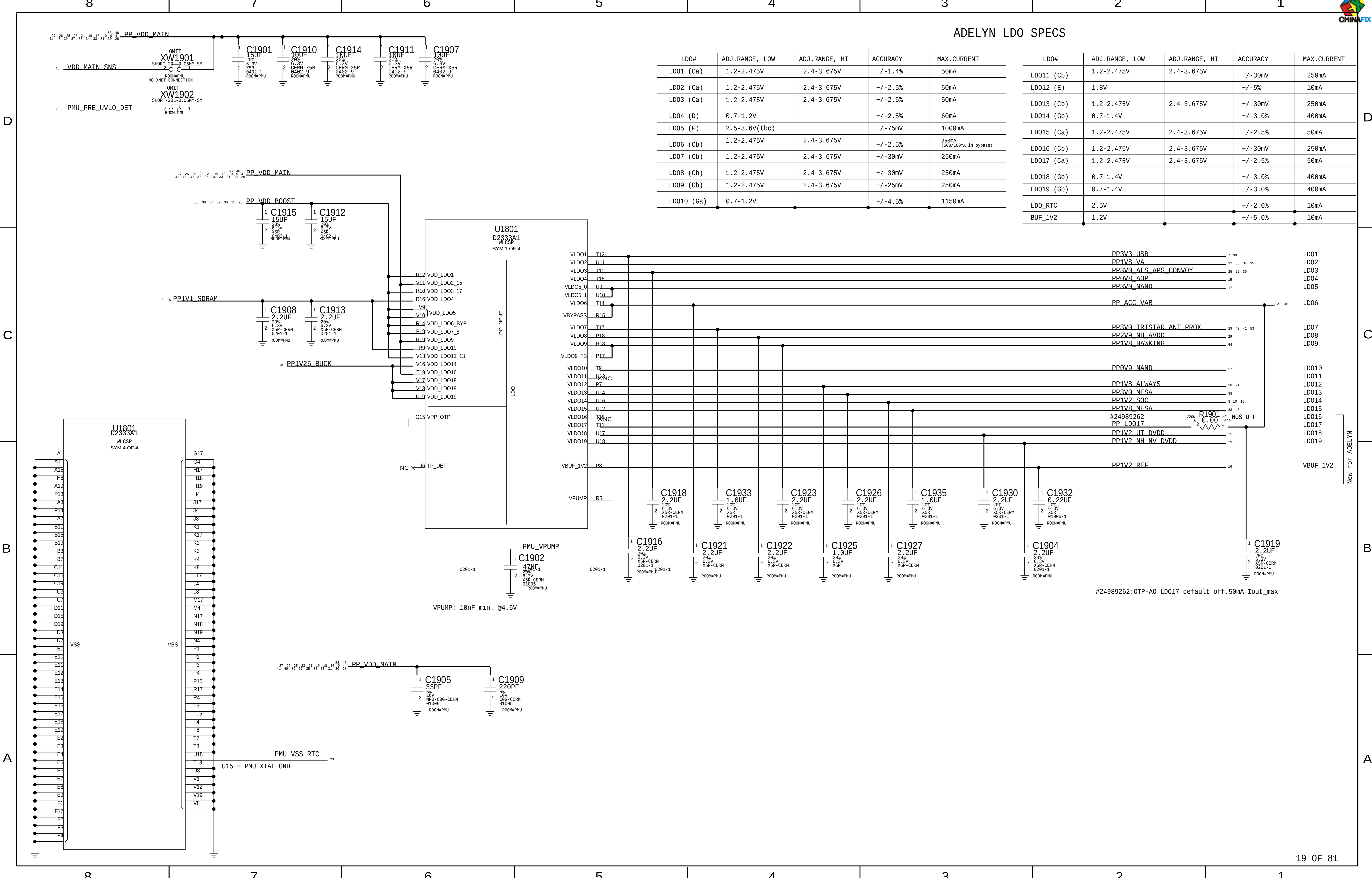
C

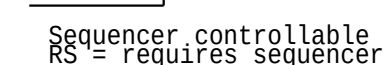
B

B

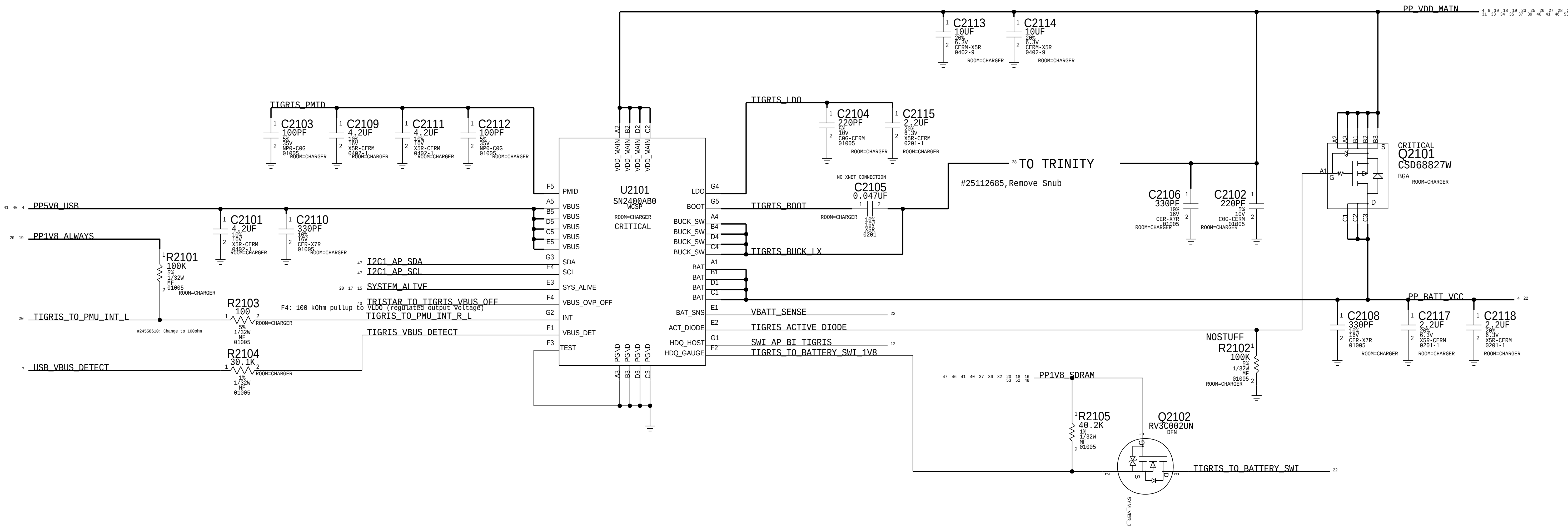
A

A



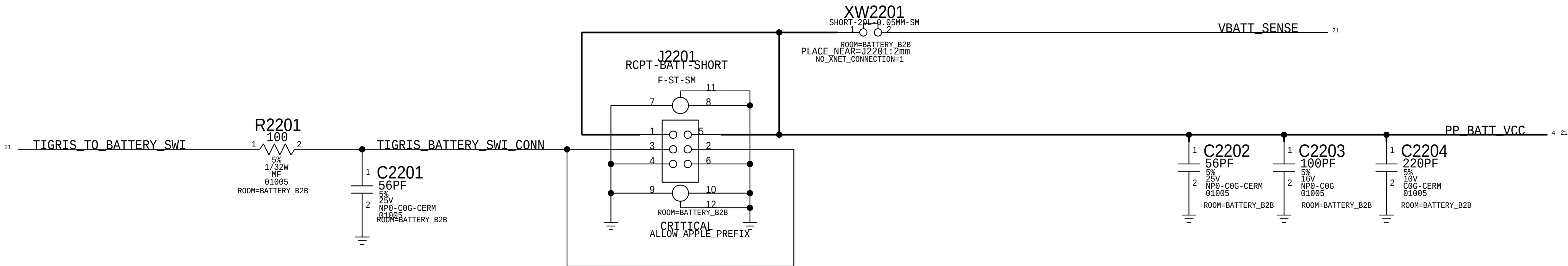


TIGRIS CHARGER

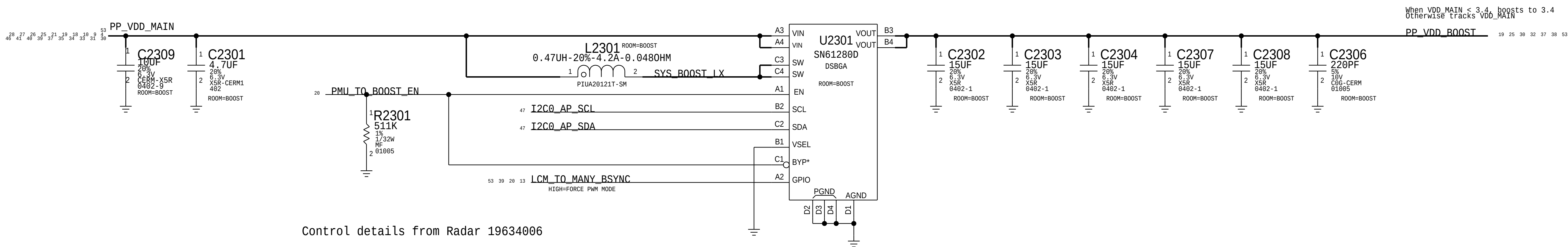


BATTERY CONNECTOR

THIS ONE ON MLB ---> 516S00172 (matches d10 mlb MCO rev 27)

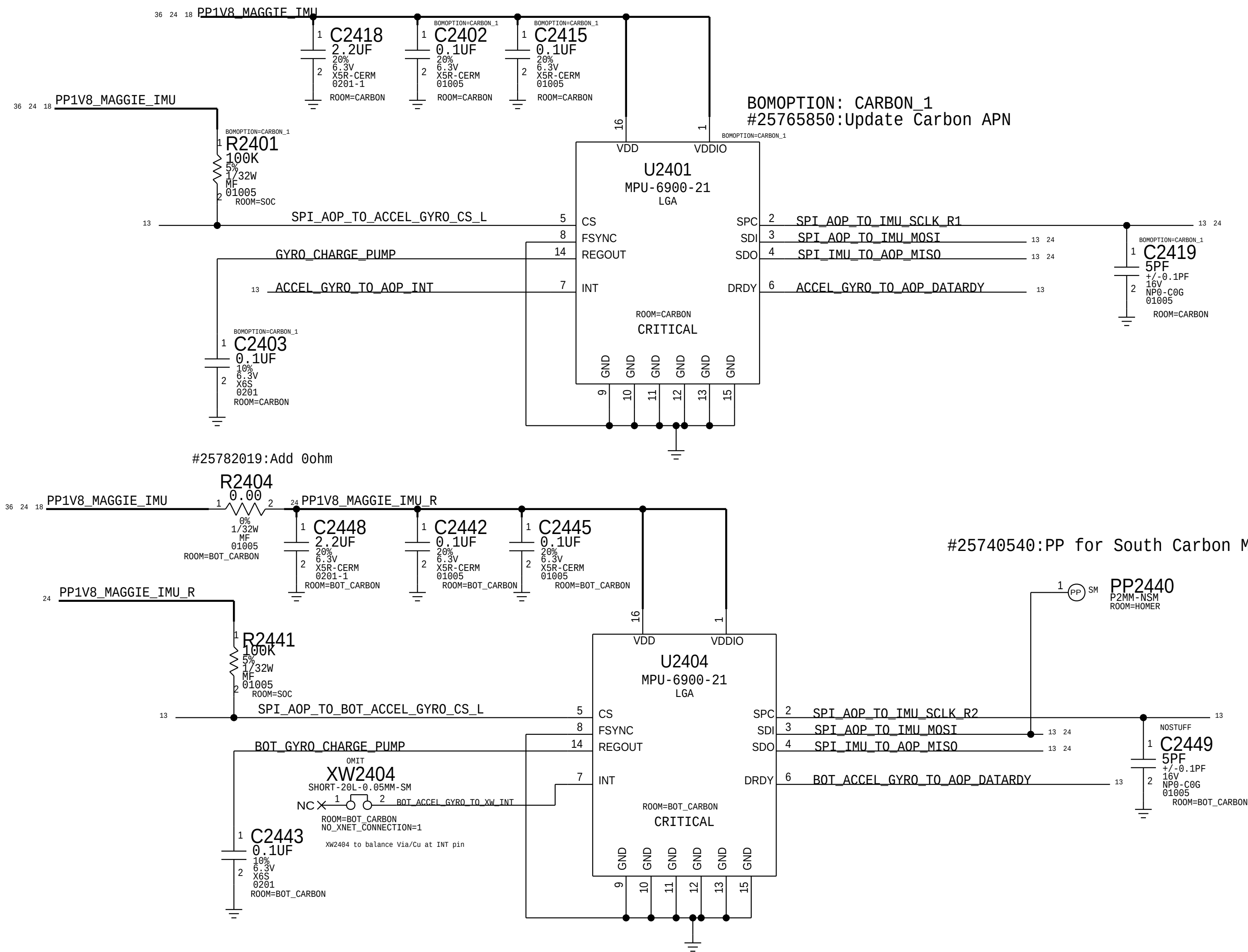


BOOST

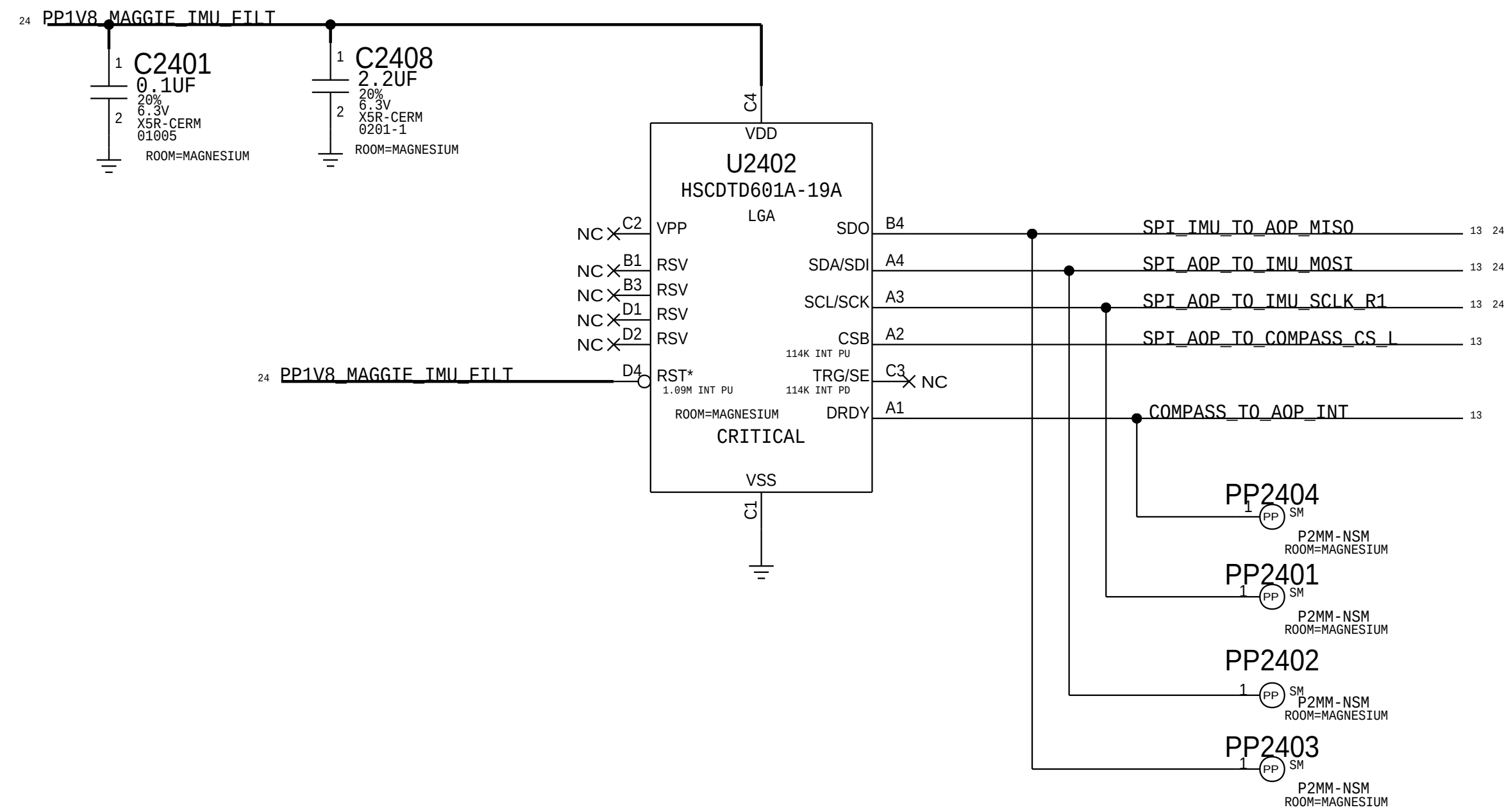


CARBON - ACCEL & GYRO

INVENSENSE, MPU-6800: C2403=0.1UF



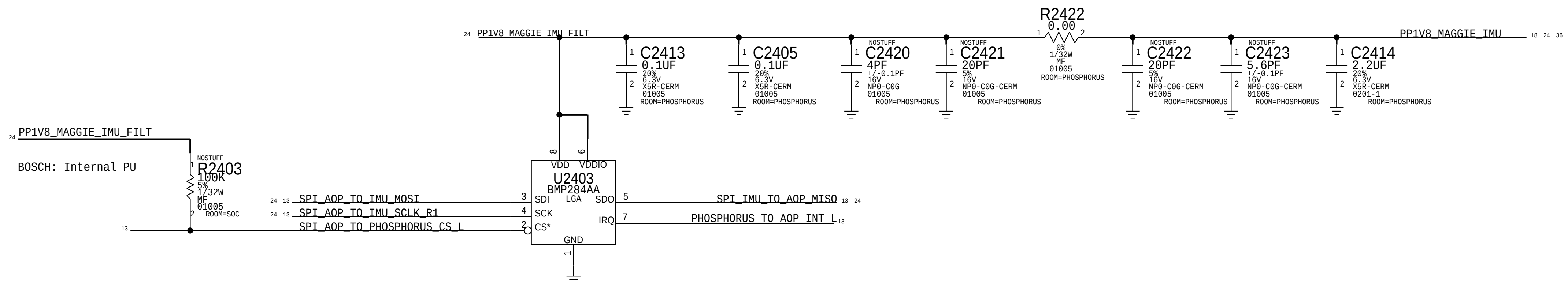
MAGNESIUM - COMPASS



PHOSPHORUS

#24593845

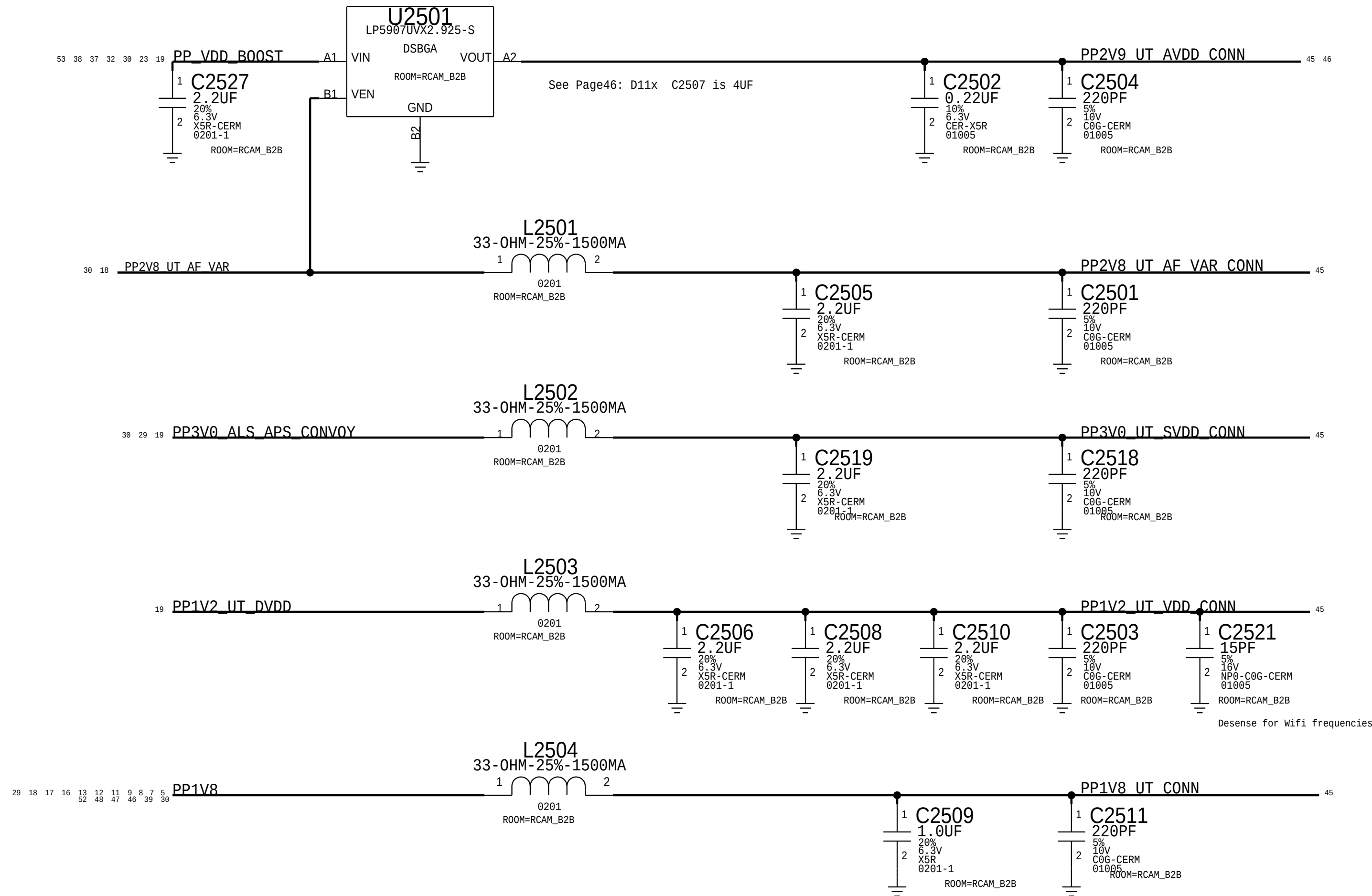
BOSCH (APN:338S00188): nostuff C2420/C2421/C2422/C2423 and R2403 PU
ST (APN:338S00230): stuff C2420-C2423, C2420R2422 with 155S00017, stuff R2403 PU
C2420=4pF(131S0253),C2405=3pF(131S0251) per #25691124



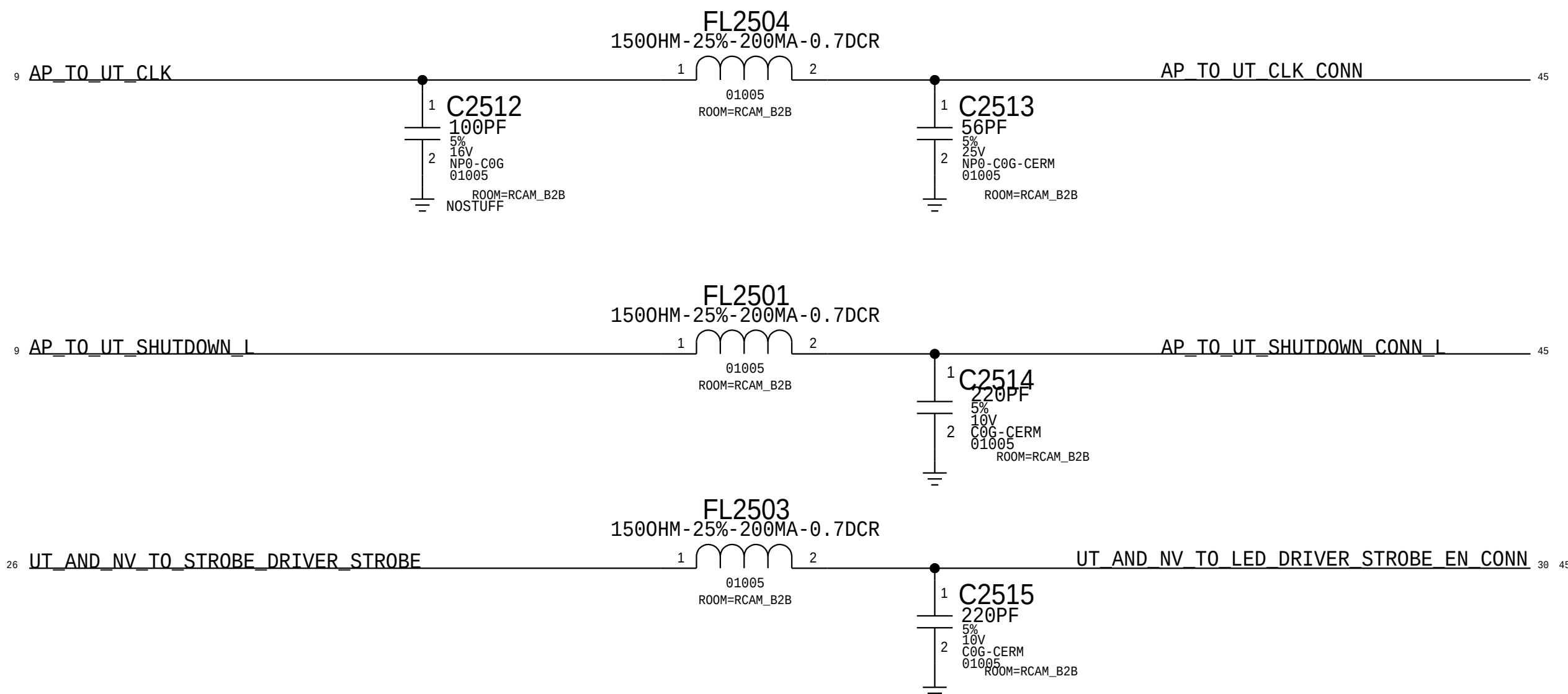
UTAH POWER

NOTE: OUTPUT IMPEDANCE MUST BE >0.005-OHM
IN ORDER TO MEET CAP ESR REQUIREMENT PER LDO SPEC.
VENDOR ALSO RECOMMENDS CIN = COUT FOR STABILITY

Scrub voltage selection

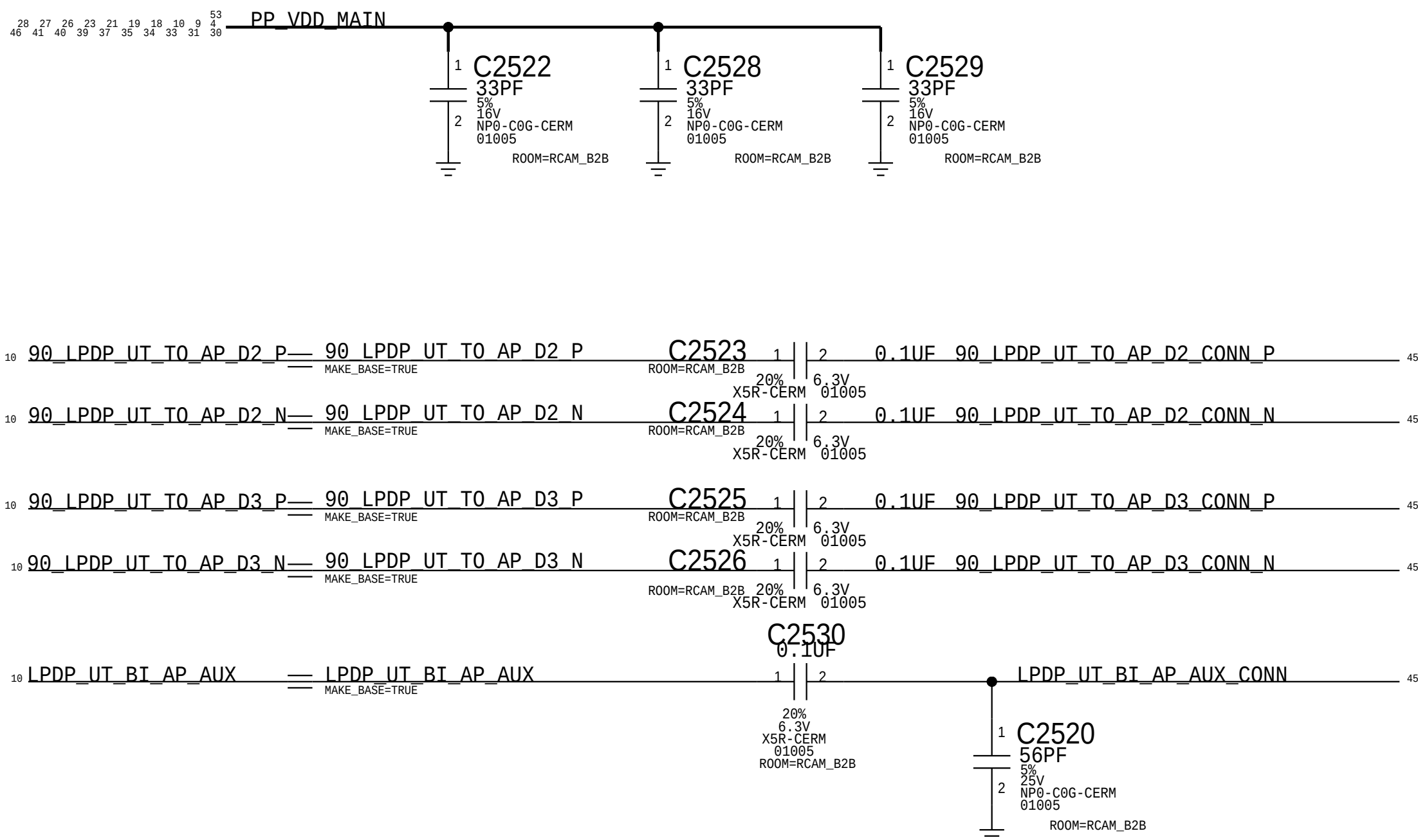


IO FILTERS



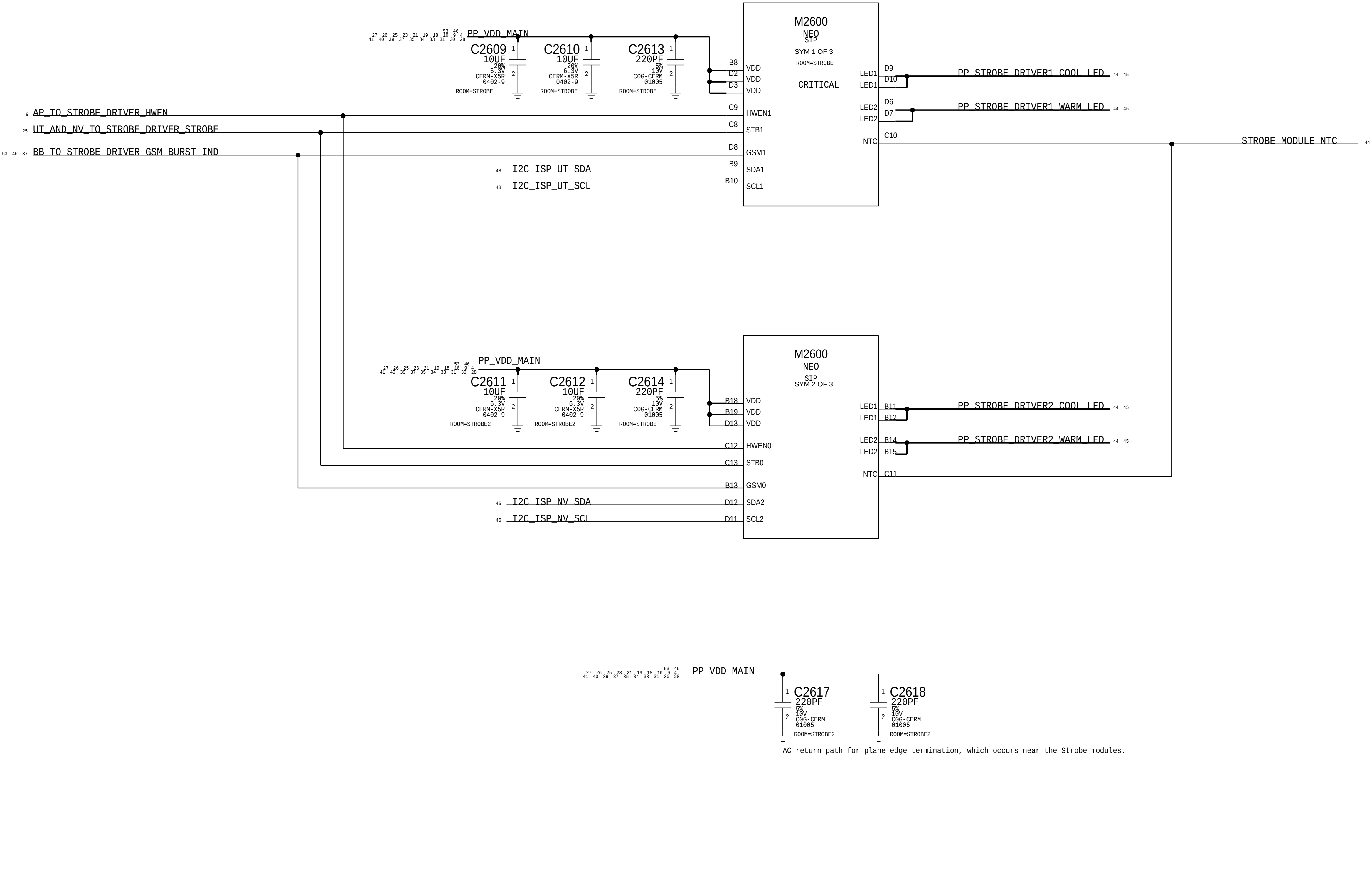
LPDP FILTERS

AC return path for LPDP which is referenced to GND and VDD_MAIN

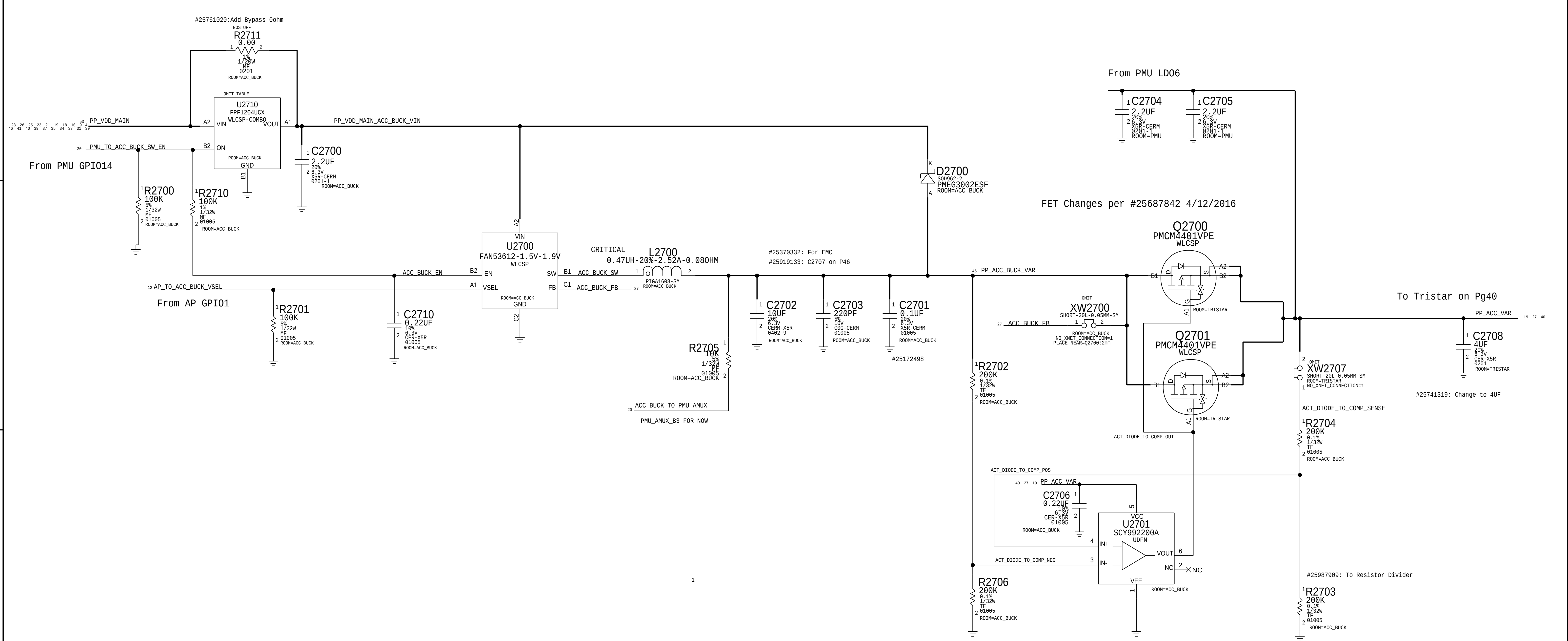


STROBE DRIVERS INSIDE NEO SIP MODULE

D10/sip_neo



ACCESSORY BUCK

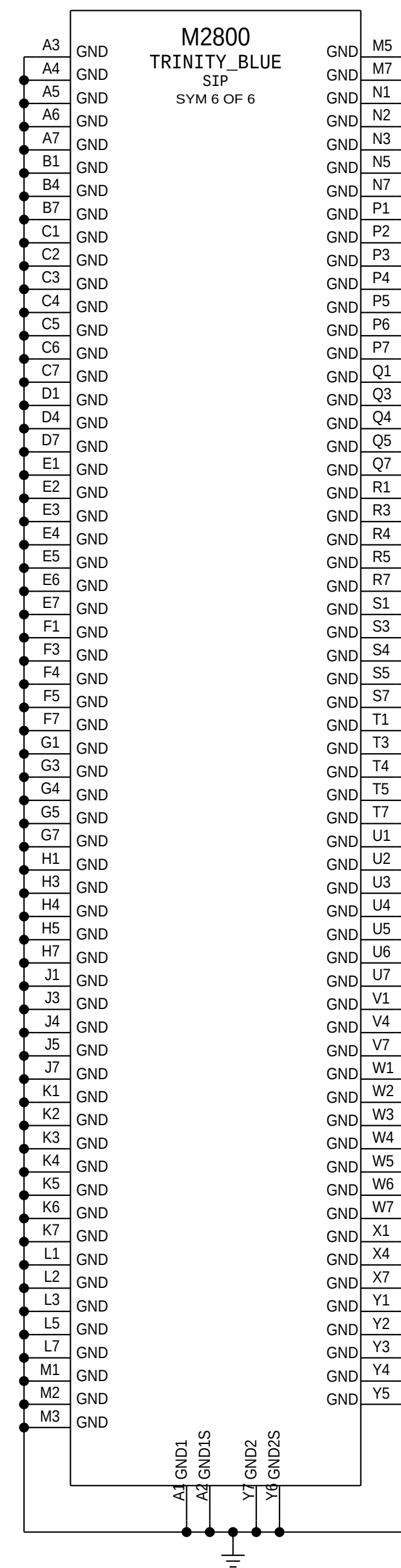
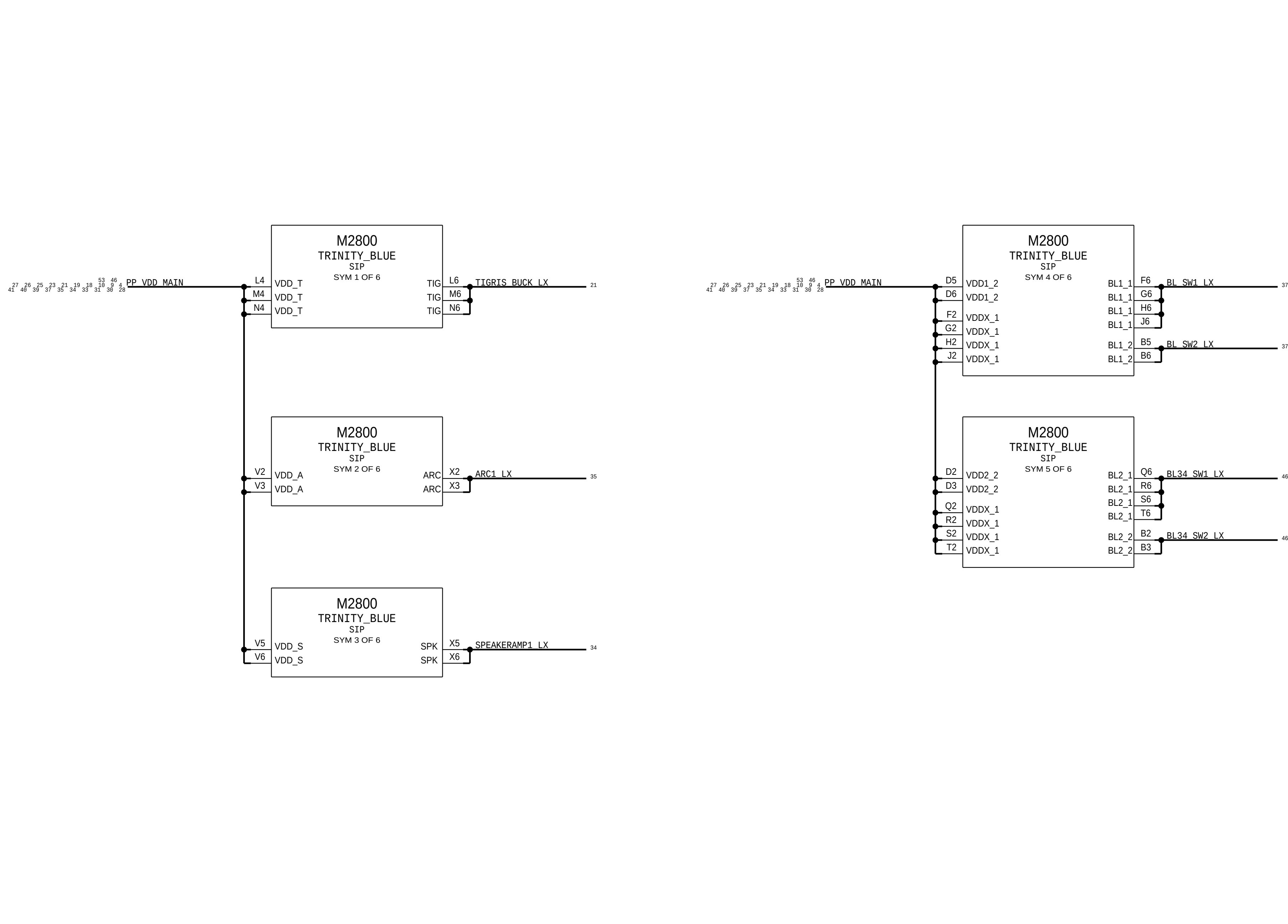


D

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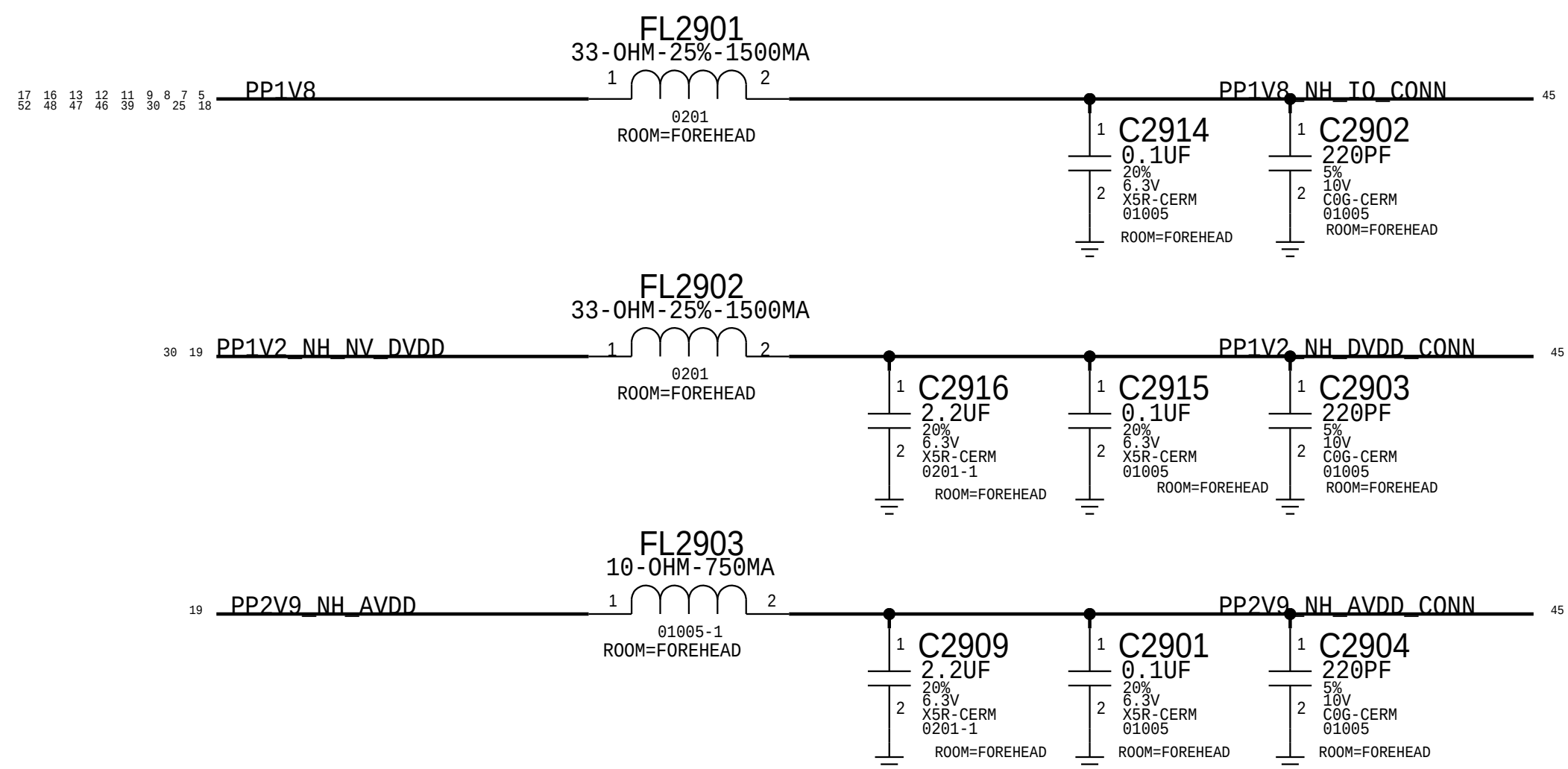
4

3

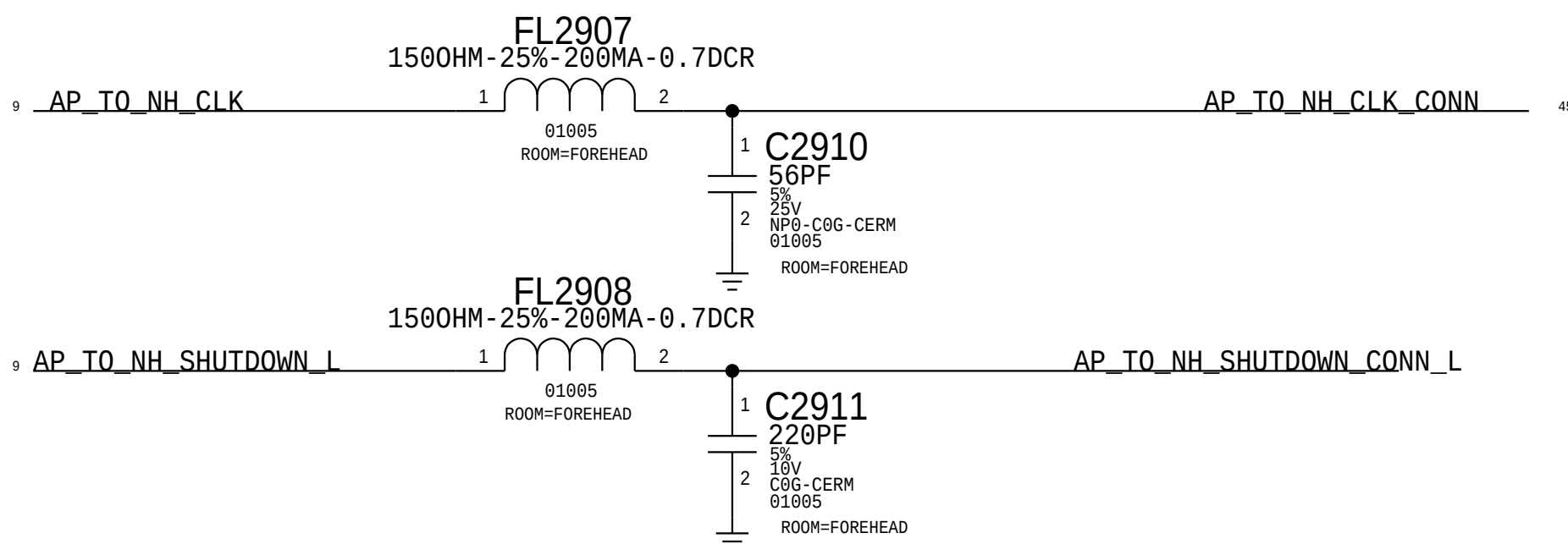
2

1

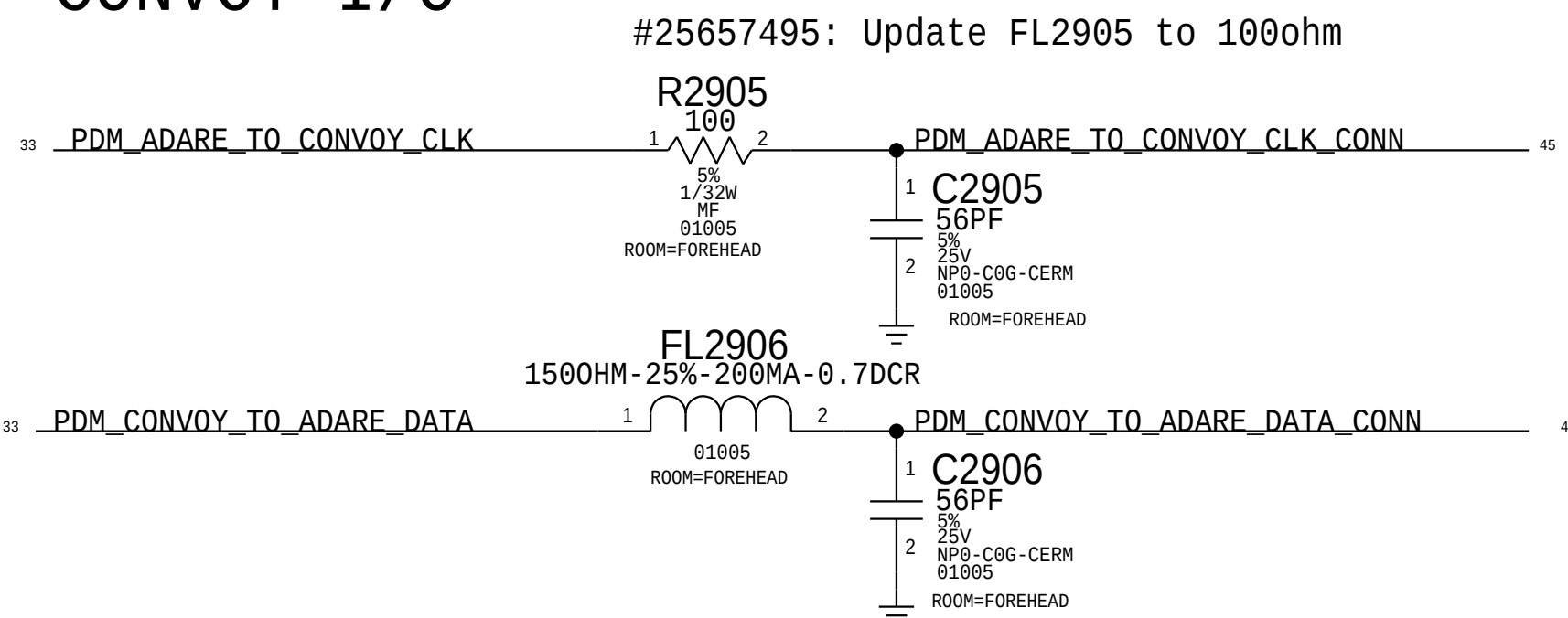
NEW HAMPSHIRE POWER



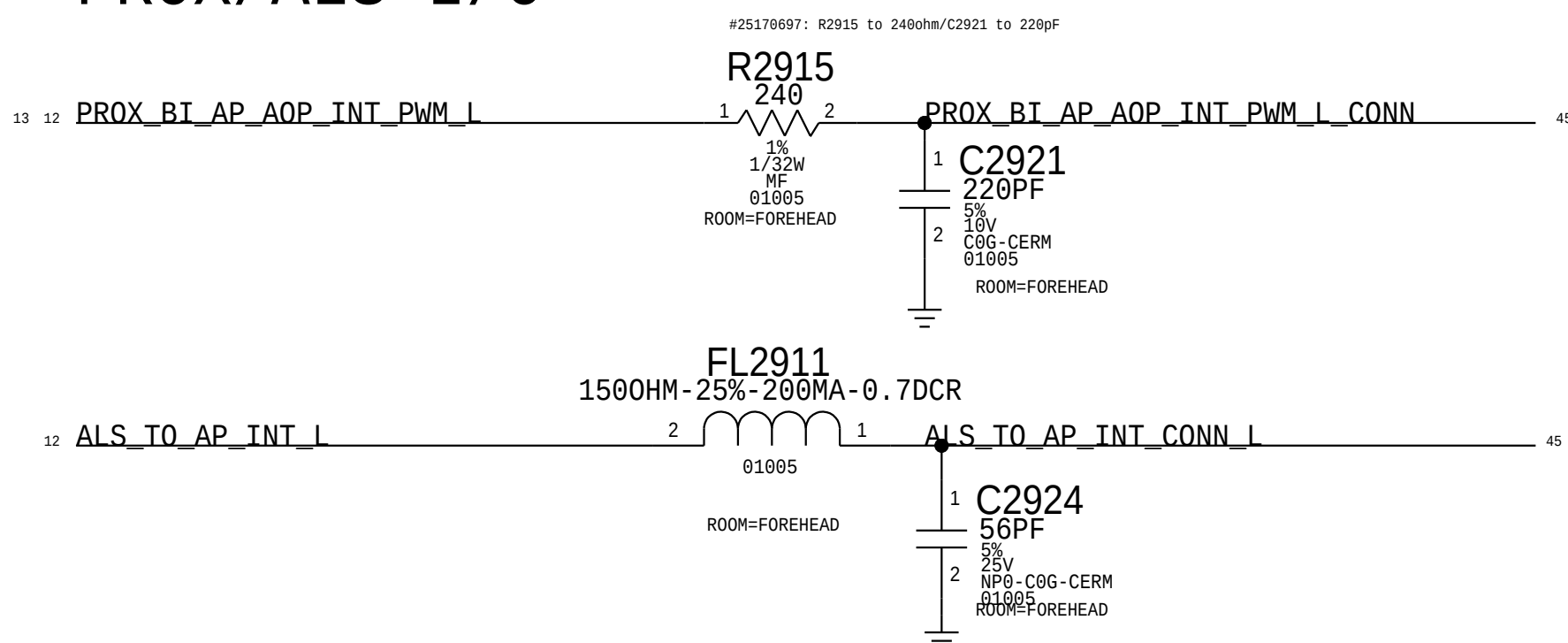
NEW HAMPSHIRE I/O



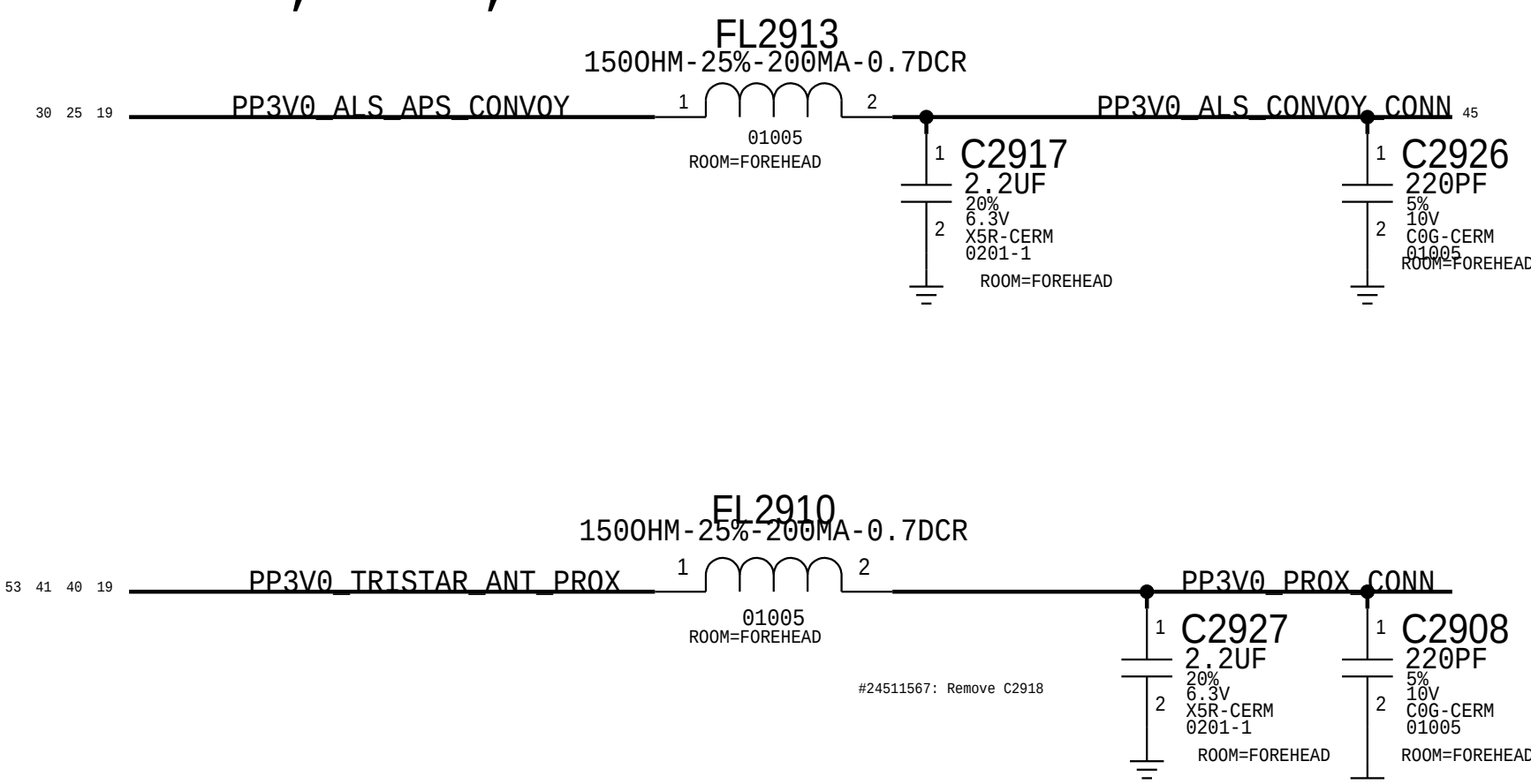
CONVOY I/O



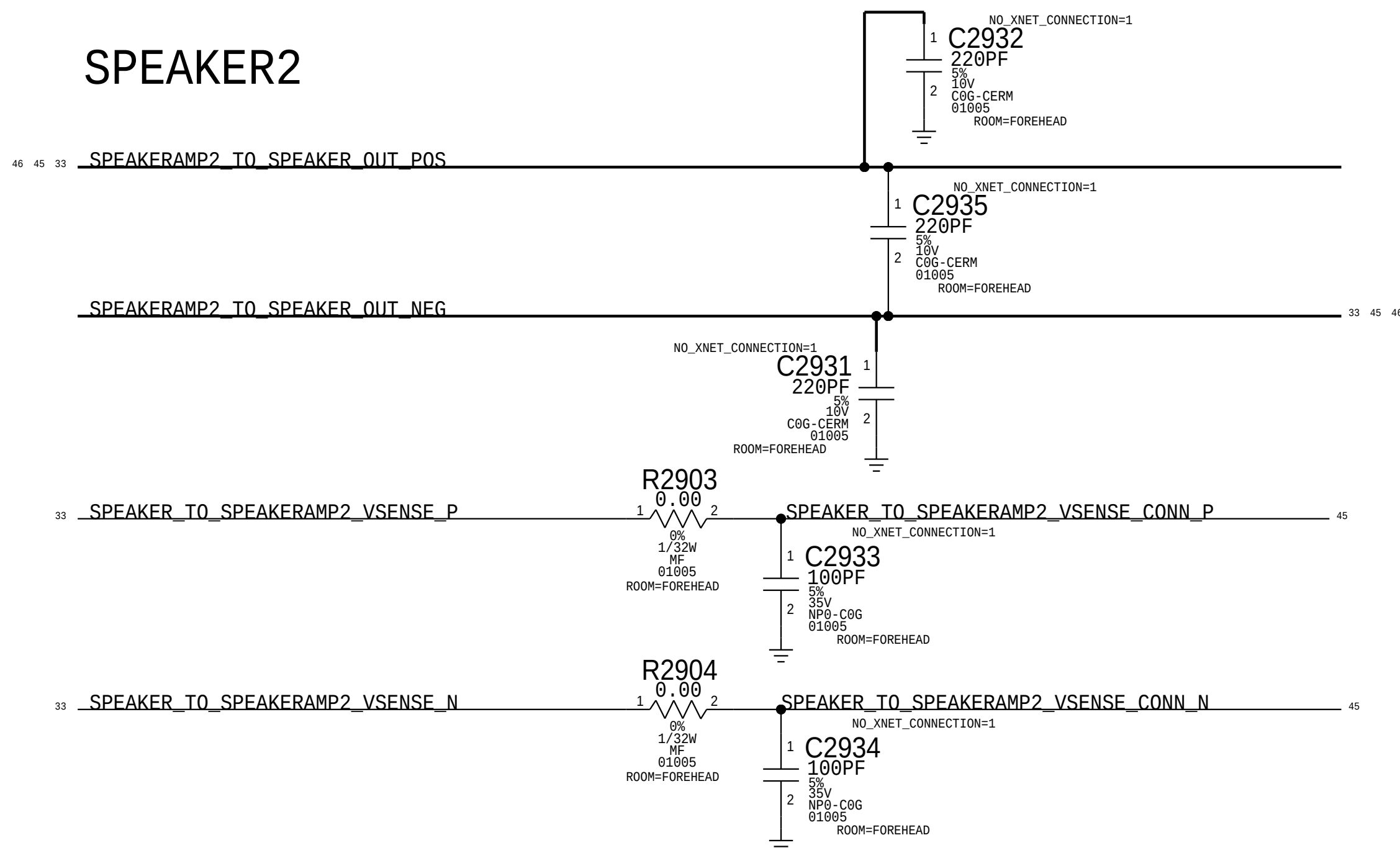
PROX/ALS I/O



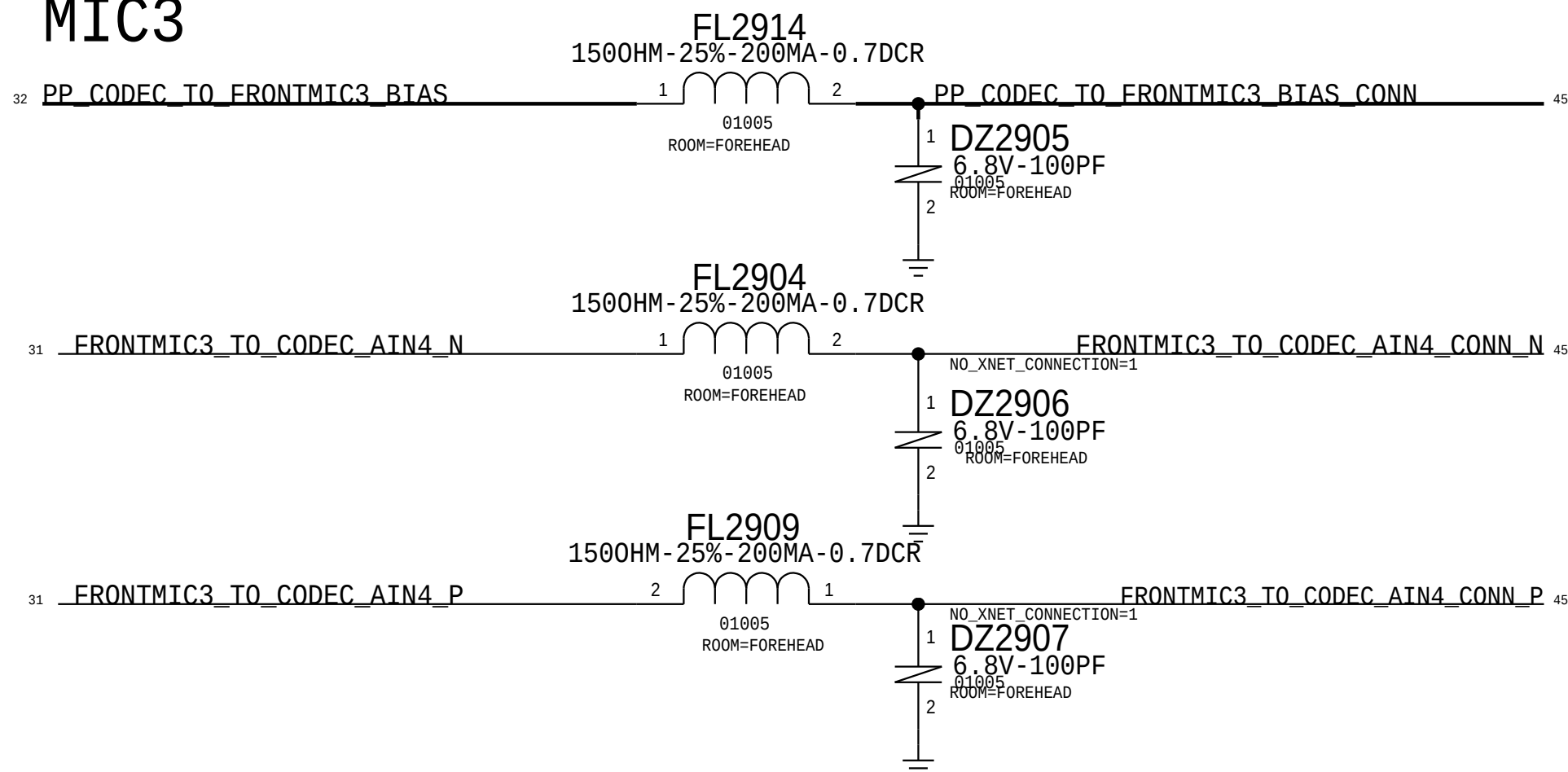
PROX, ALS, & CONVOY POWER



SPEAKER2



MIC3



8

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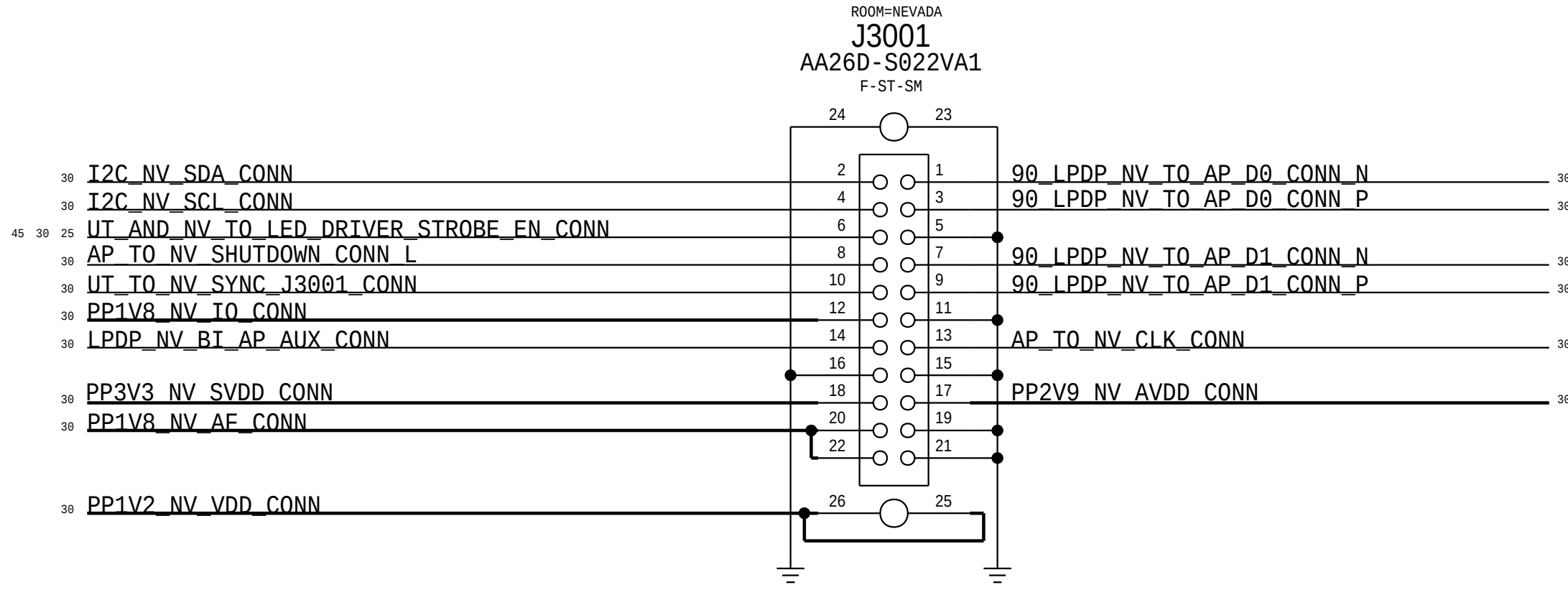
1



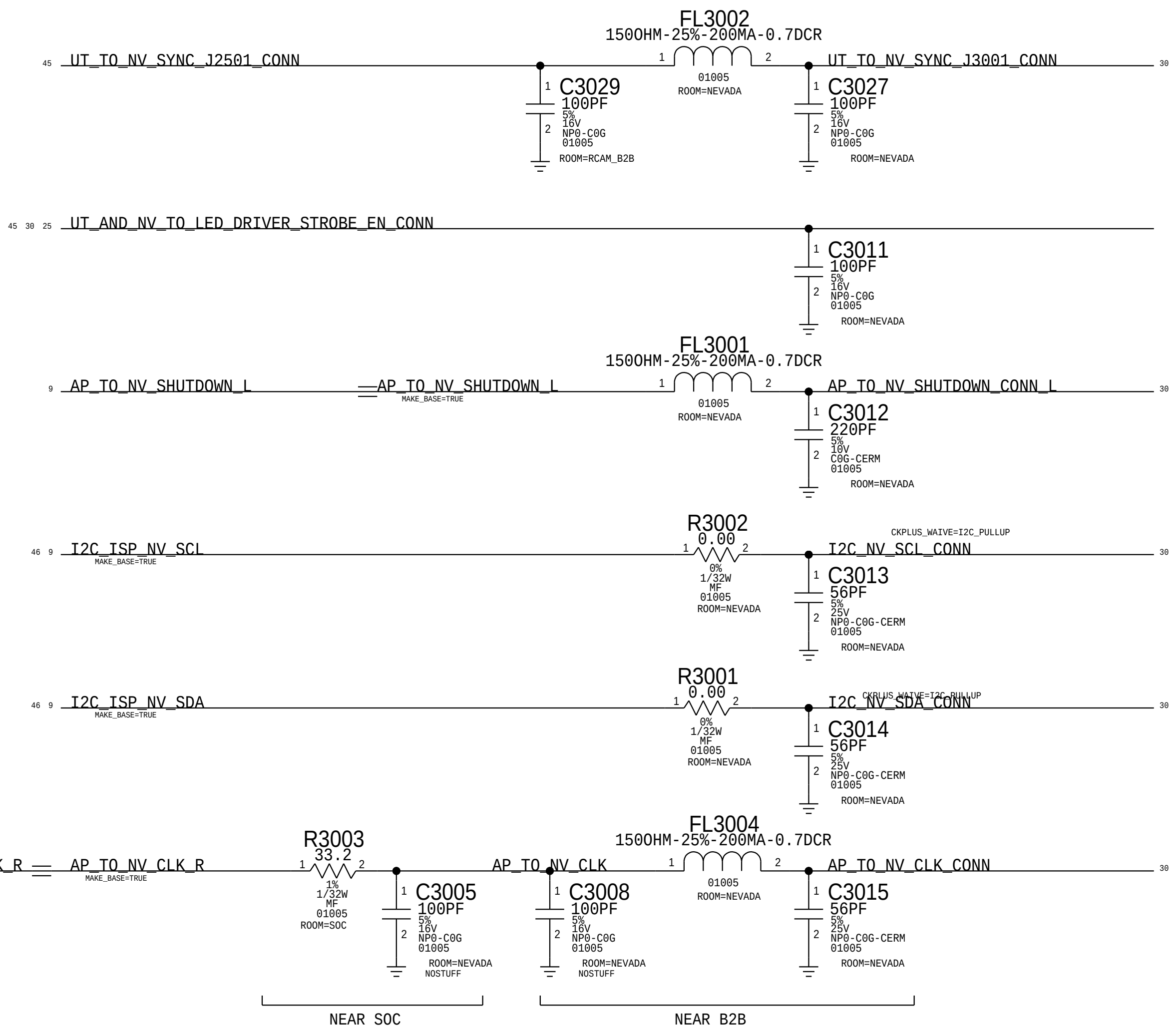
THIS PAGE UNIQUE TO LARGE FORM FACTOR
OTHER FORM FACTOR SPECIFIC PAGES:
4 - Mechanical
46 - FF Specific

NEVADA CONNECTOR

THIS ONE ---> 516S00152 RCPT (USED ON MLB)
516S00151 PLUG

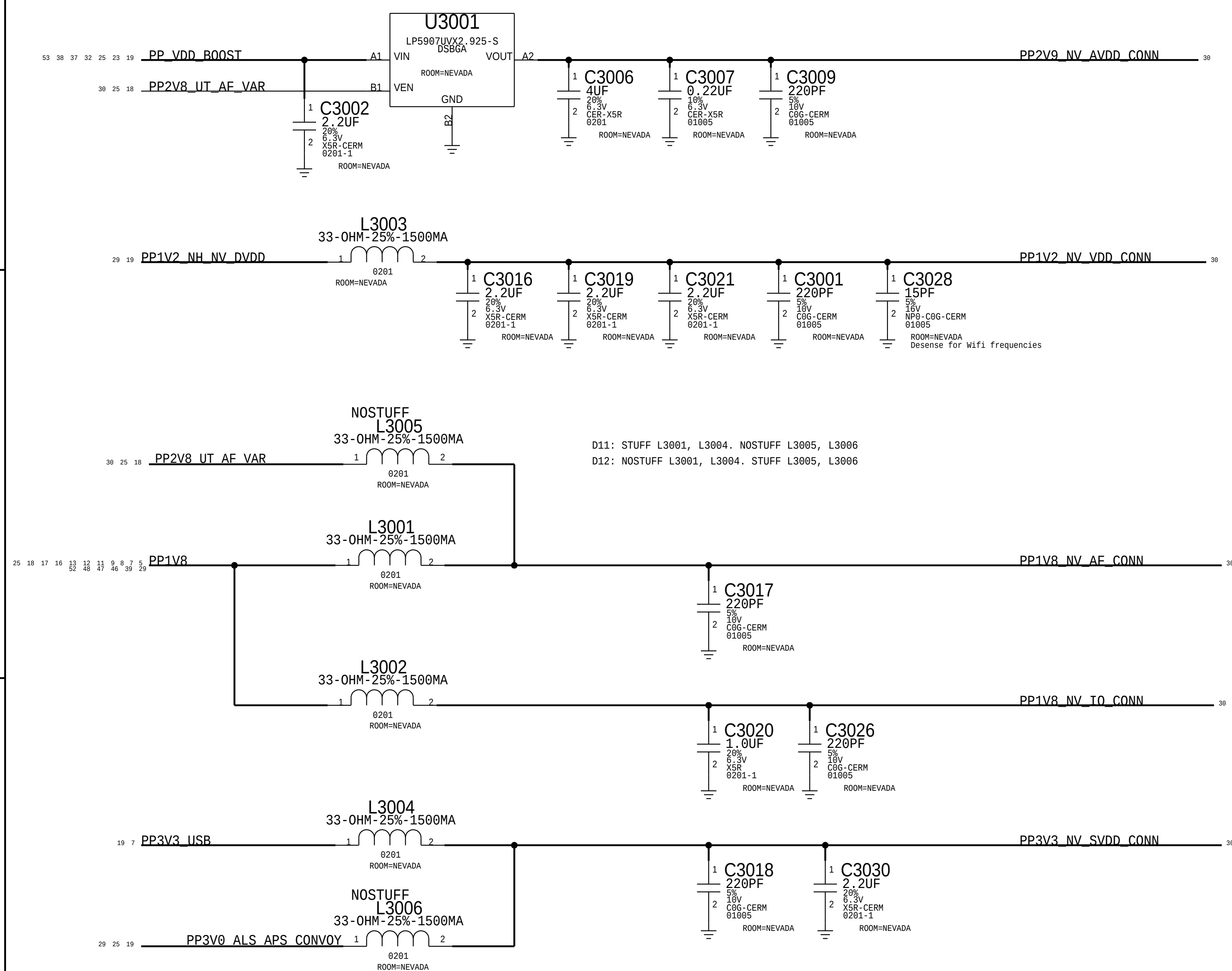


GPIO FILTERS

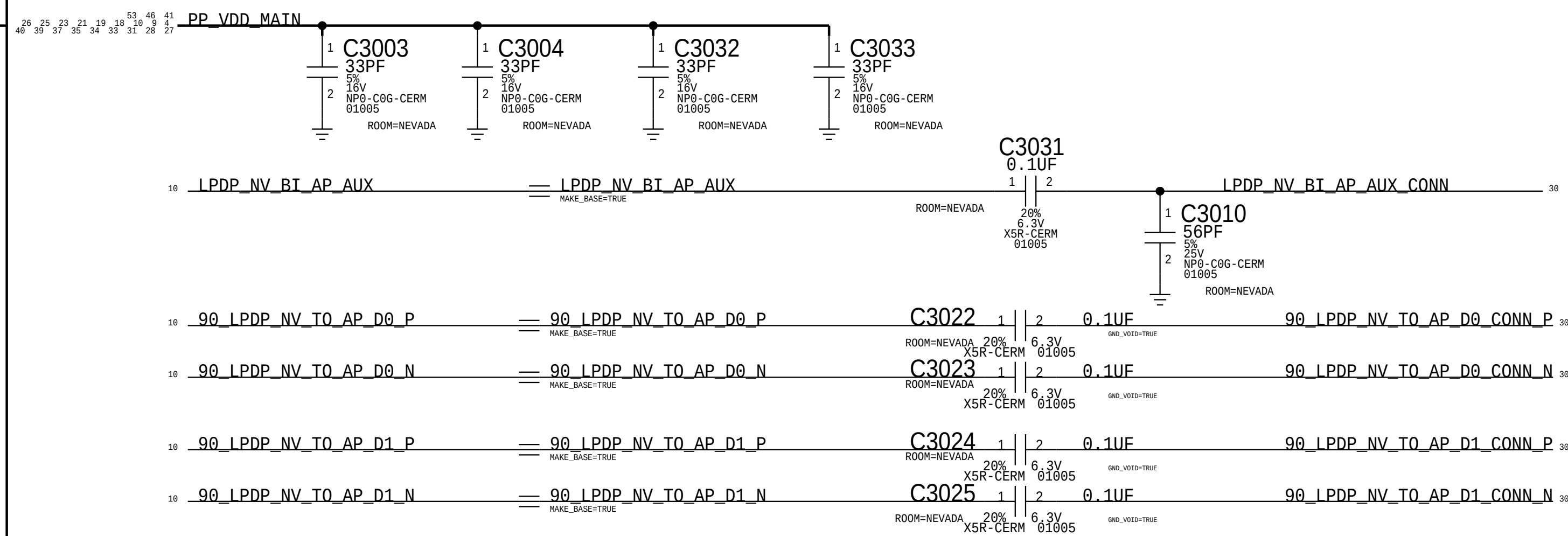


NEVADA POWER

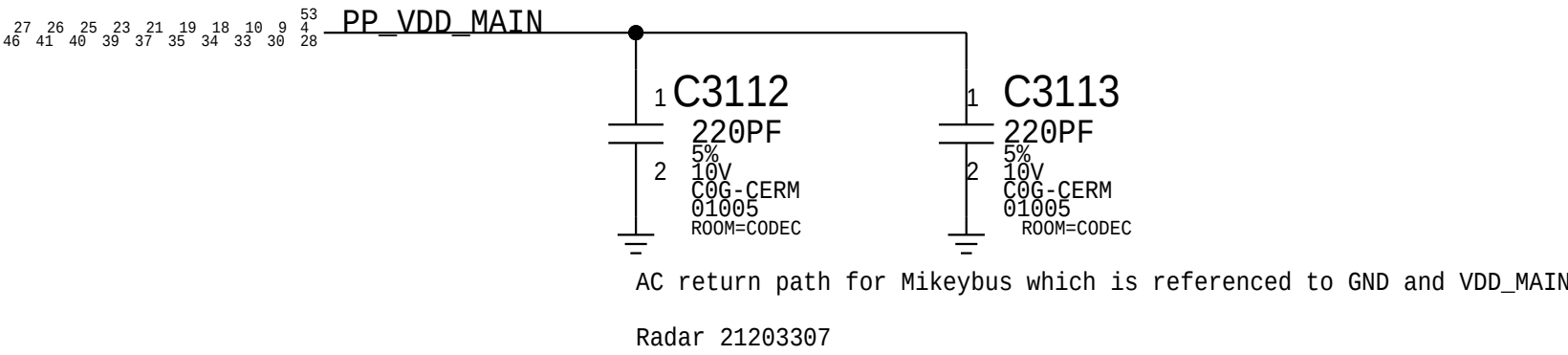
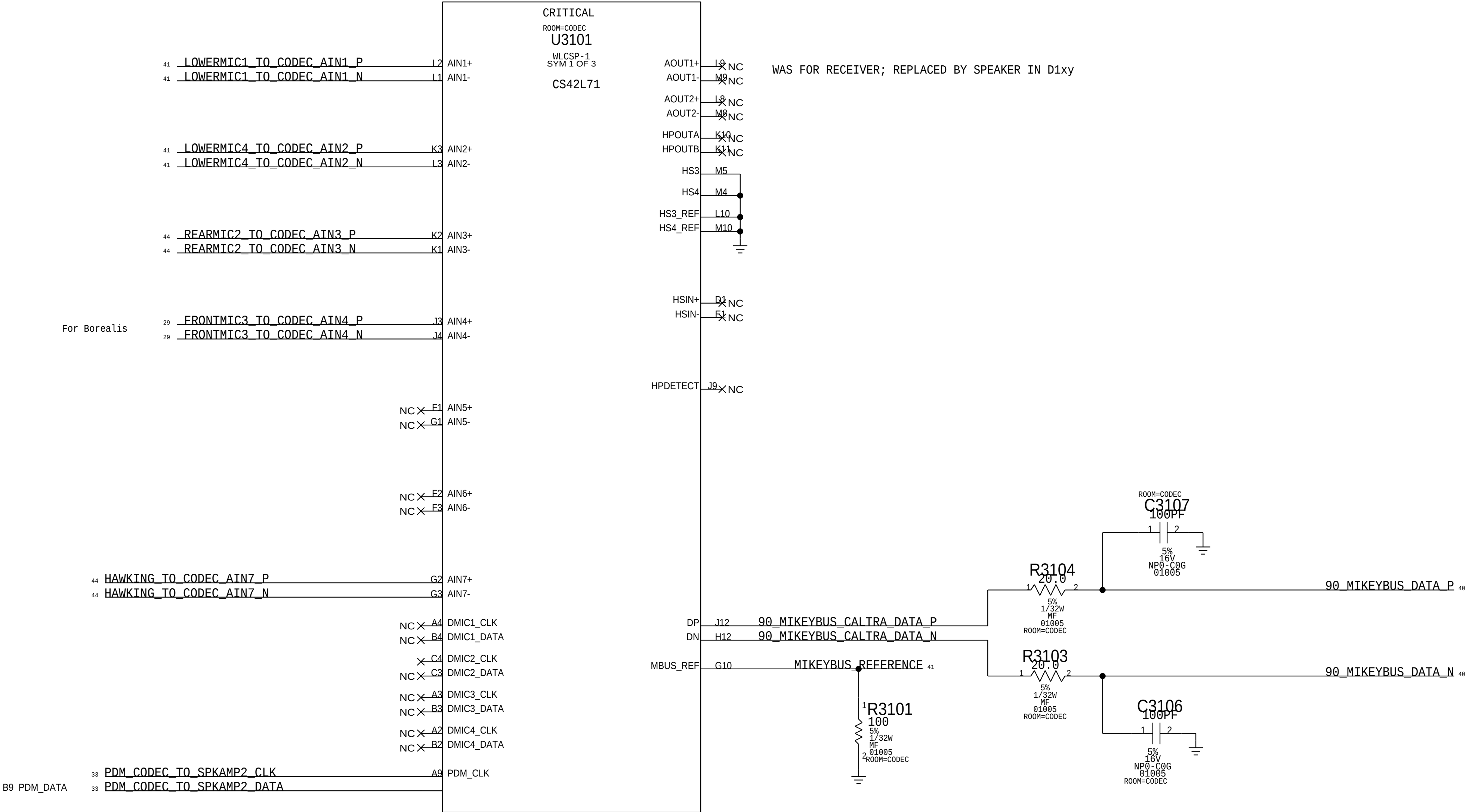
NOTE: OUTPUT IMPEDANCE MUST BE >0.005-OHM
IN ORDER TO MEET CAP ESR REQUIREMENT PER LDO SPEC.
VENDOR ALSO RECOMMENDS CIN = COUT FOR STABILITY



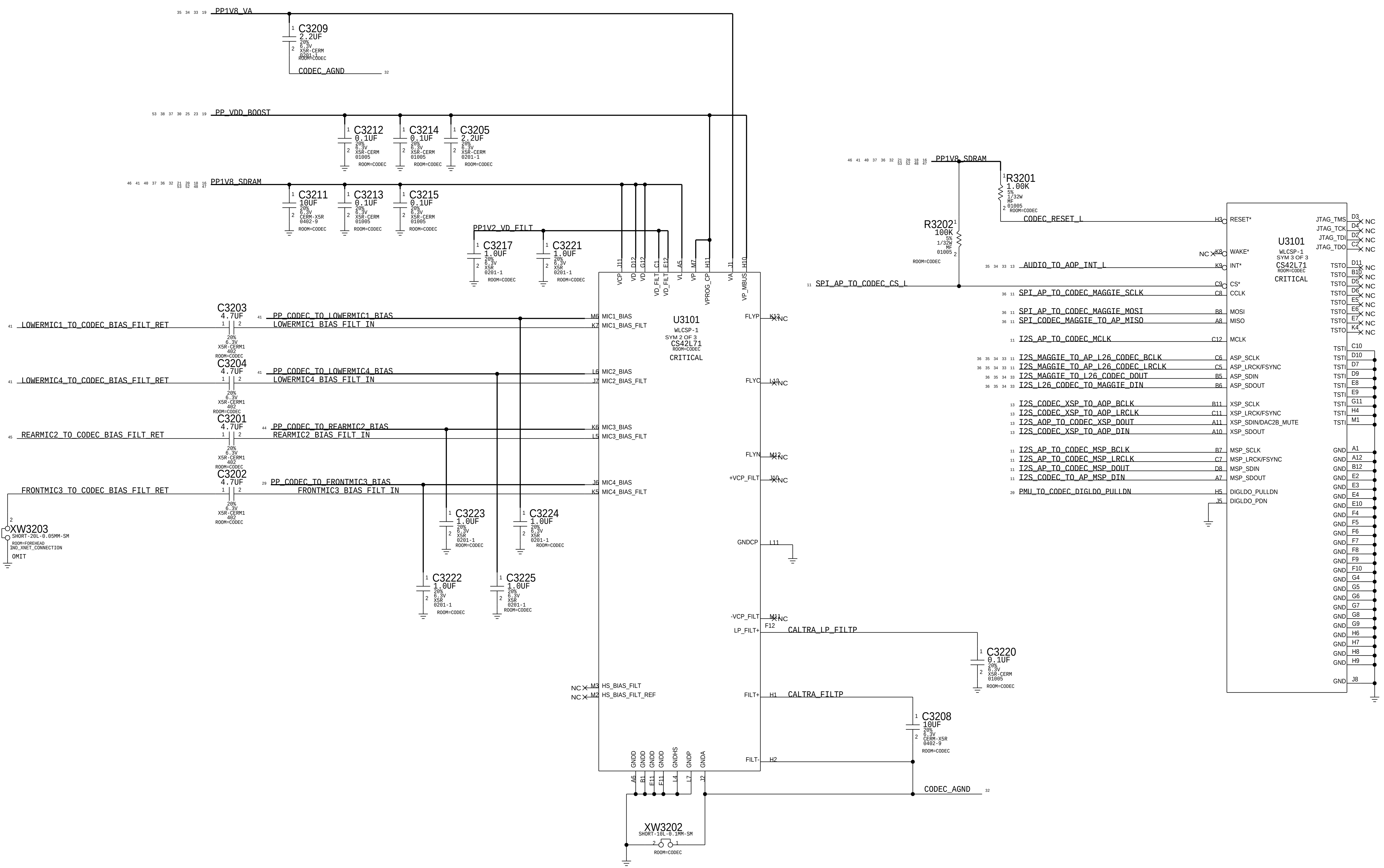
LPDP FILTERS



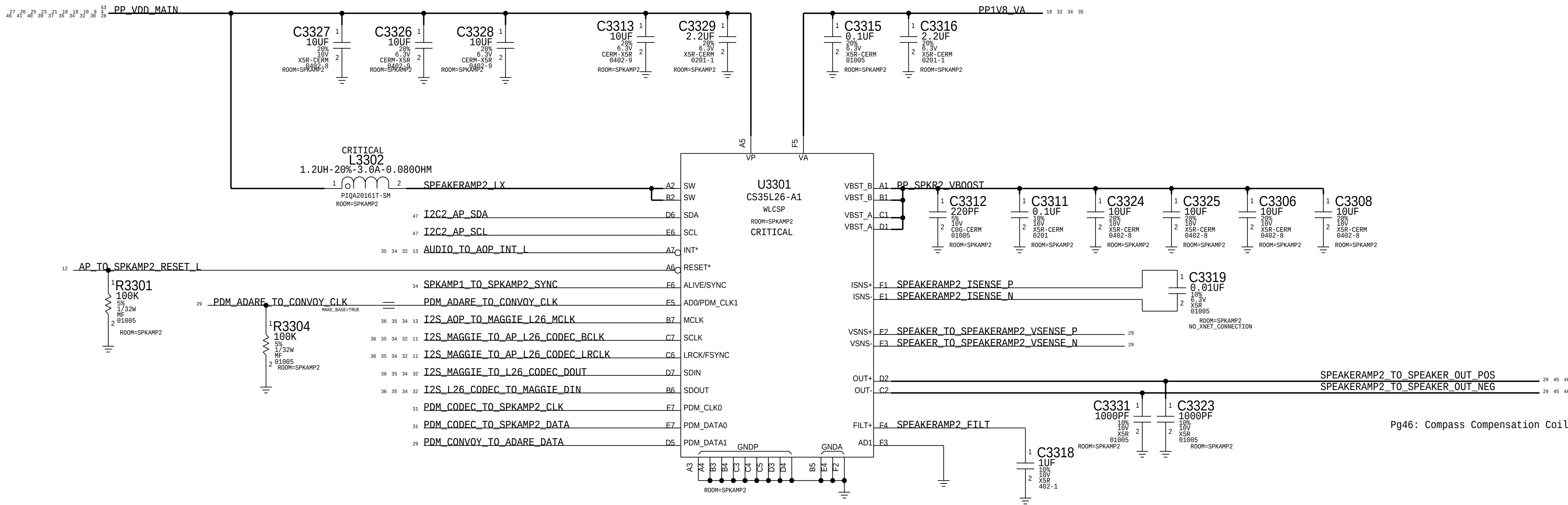
CALTRA AUDIO CODEC (ANALOG INPUTS & OUTPUTS)



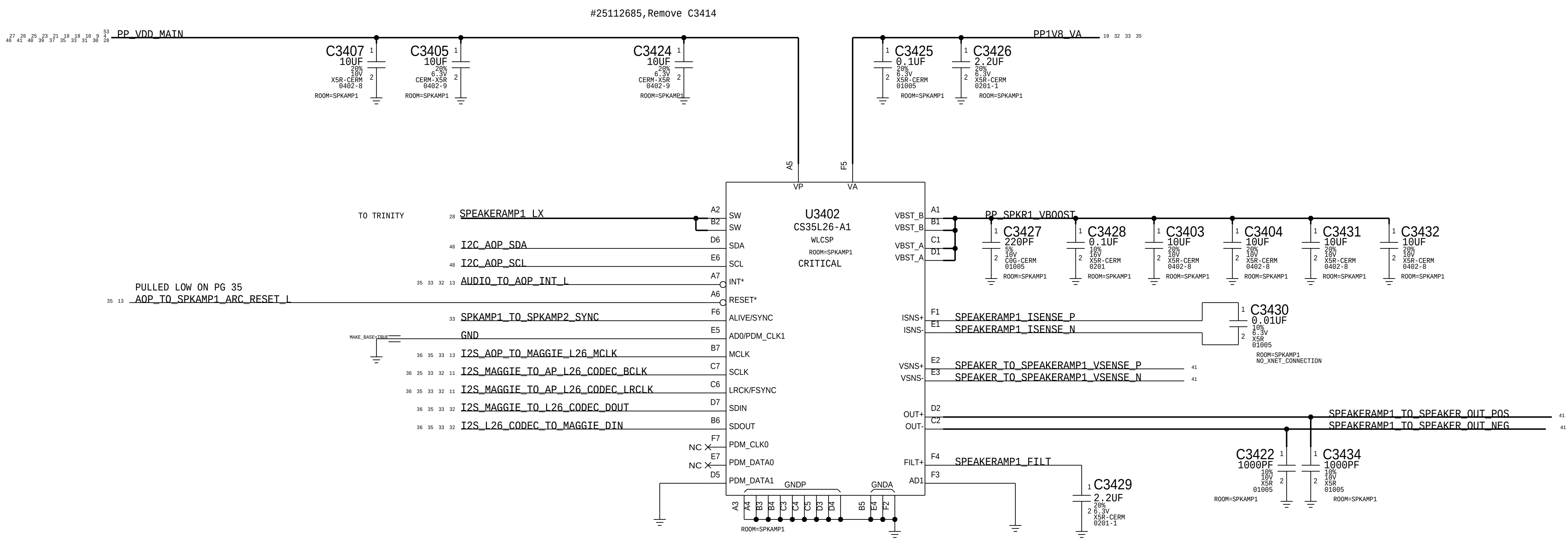
CALTRA AUDIO CODEC (POWER & I/O)



SPEAKER AMPLIFIER 2 (North)



SPEAKER AMPLIFIER 1 (South)



ARC D



D

D

C

C

B

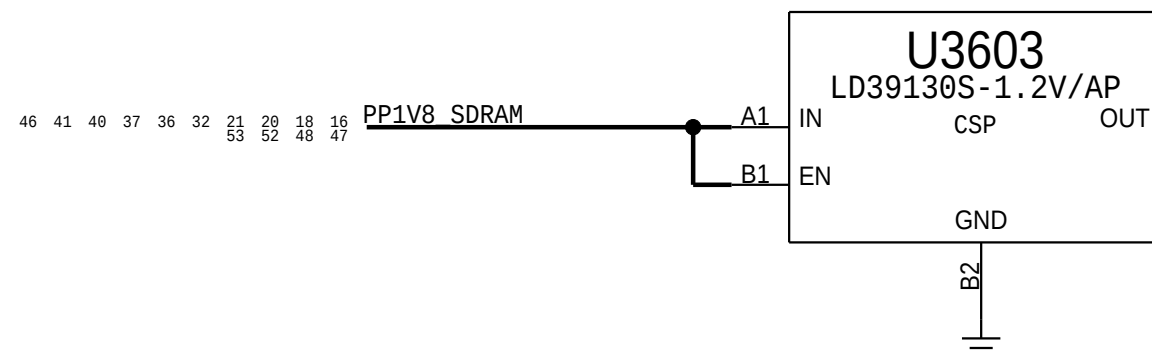
B

A

A

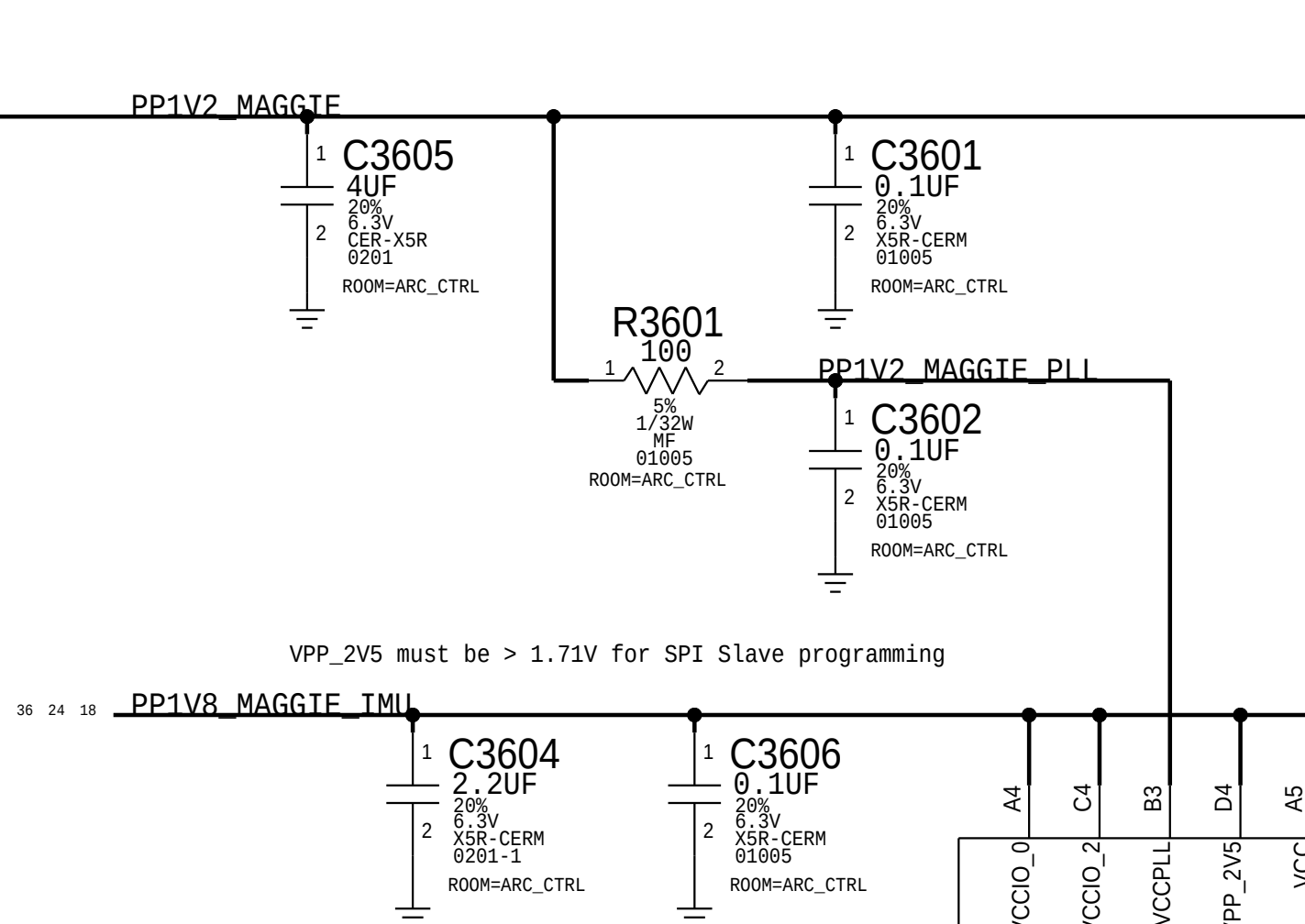
MAGGIE LDO

APN: 353S00842



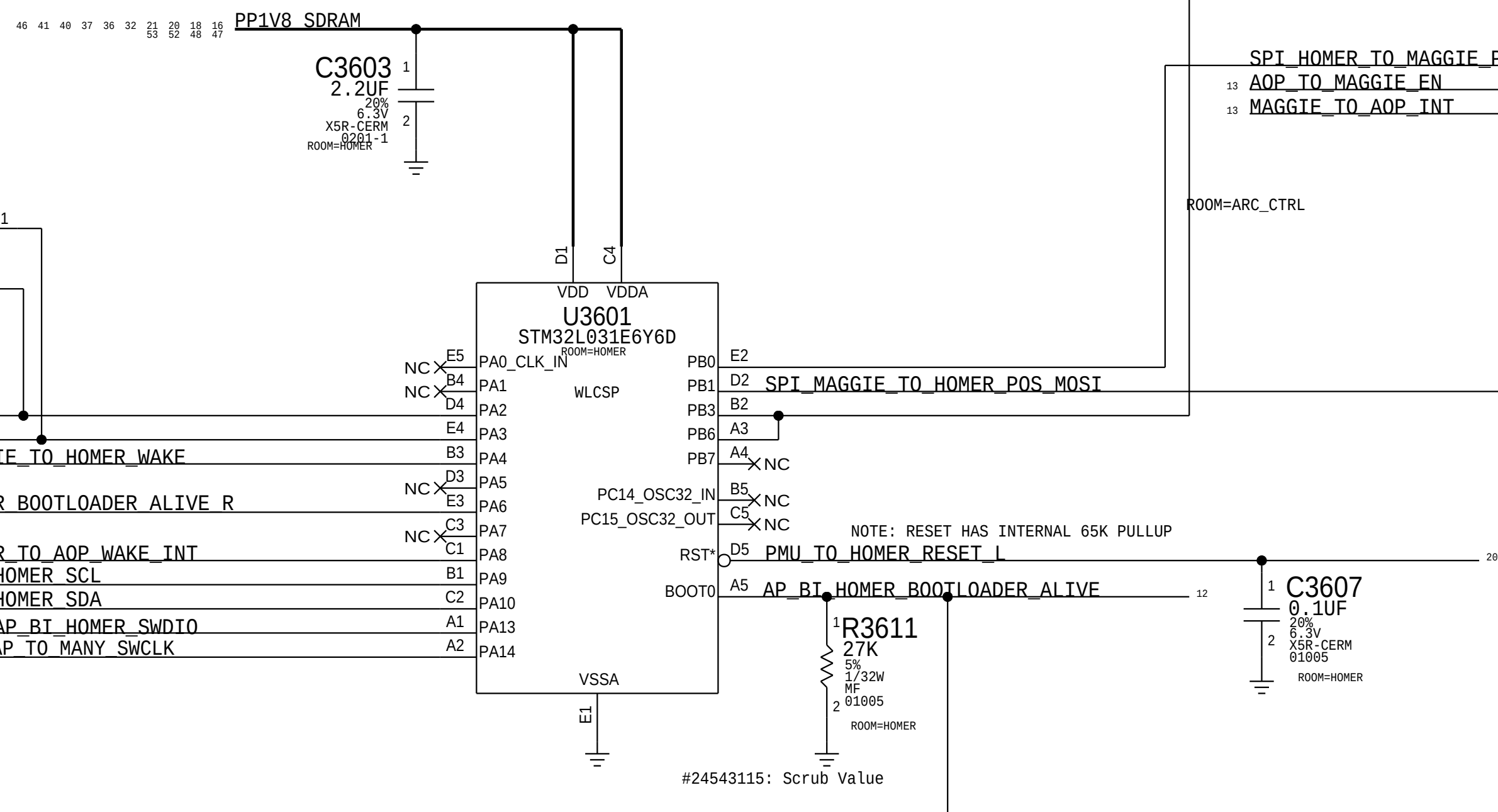
MAGGIE

APN: 336S00020



HOMER STM32L0 MICRO

STM32L03 APN: 337S00231



R3604

33.2

10K

1/32W

01005

I2S_MAGGIE_TO_AP_L26_CODEC_BCLK_R

I2S_MAGGIE_TO_AP_L26_CODEC_BCLK

MAGGIE DRIVES TO ARC, SPKRS, AP, CODEC

MAGGIE <-> AP (SDIN)

MAGGIE <-> ARC, SPKRS, CODEC (SDOUT)

MAGGIE <-> AP, SPKRS, CODEC (SDIN)

MAGGIE <-> AP (SDOUT)

PP1V8_MAGGIE_TMU

18 24 36

1R3605 10K

5% 1/32W

01005

1R3602 10K

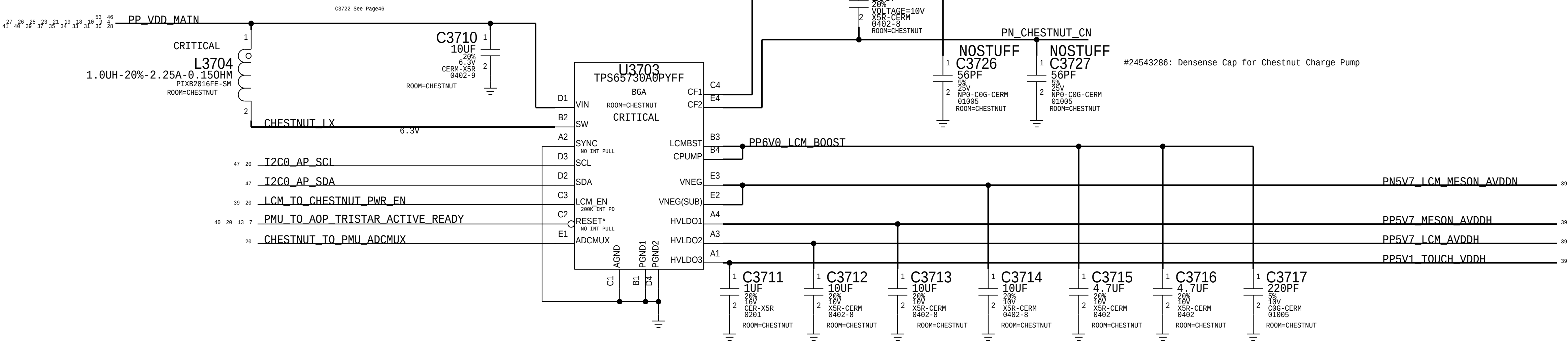
5% 1/32W

01005

DISPLAY & TOUCH - POWER SUPPLIES

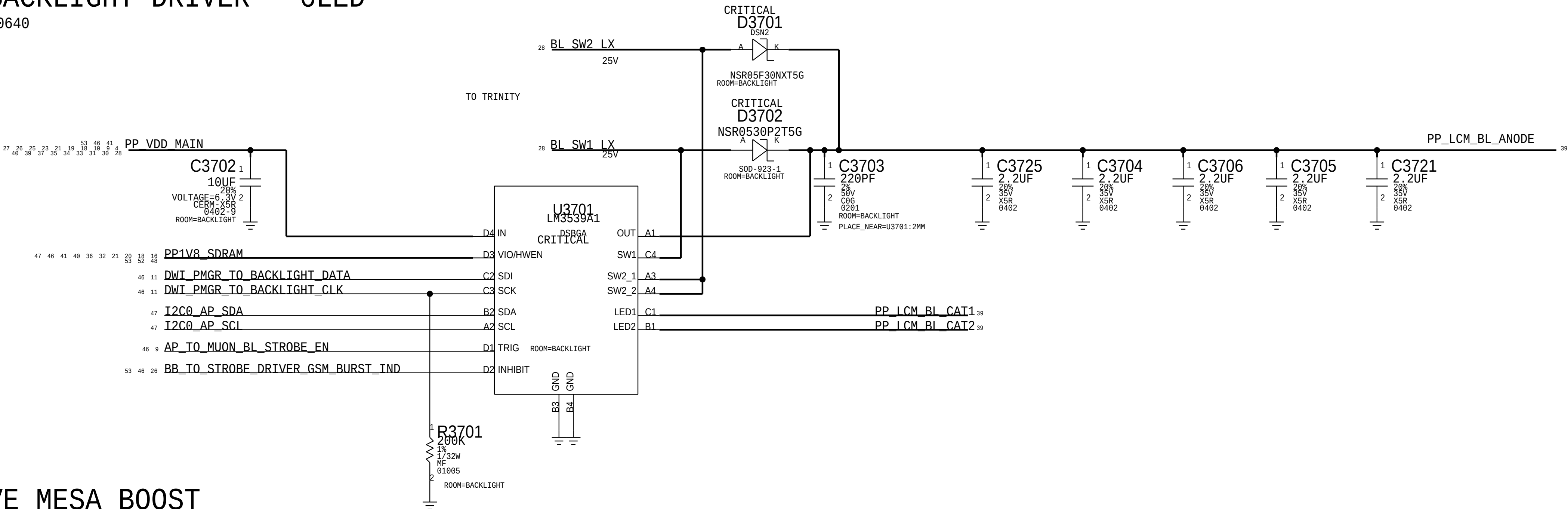
CHESTNUT DISPLAY PMU

APN:338S1172



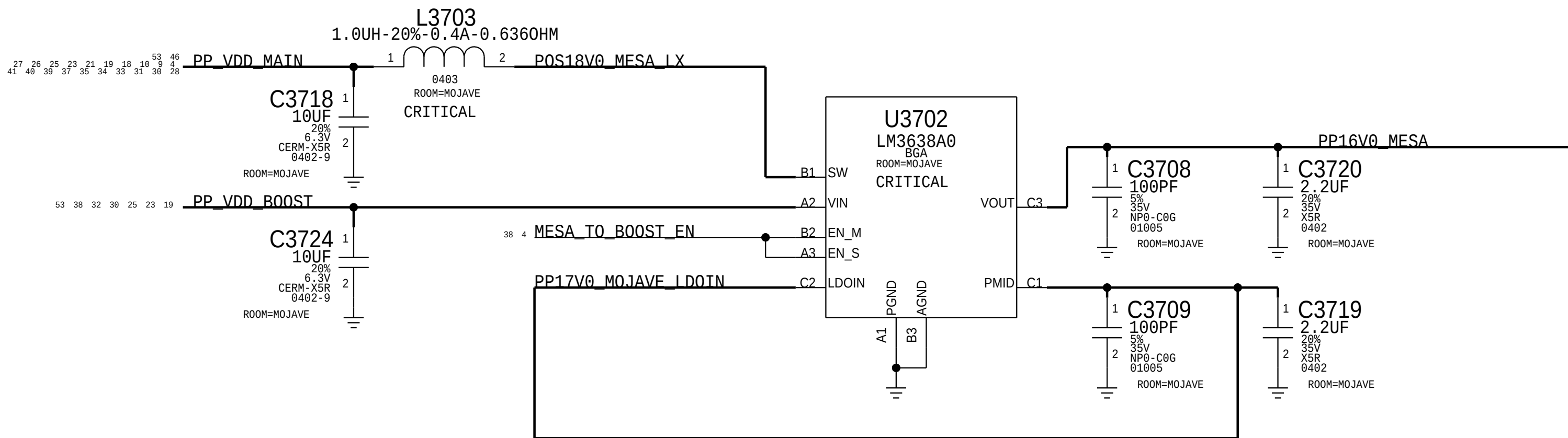
LED BACKLIGHT DRIVER - 6LED

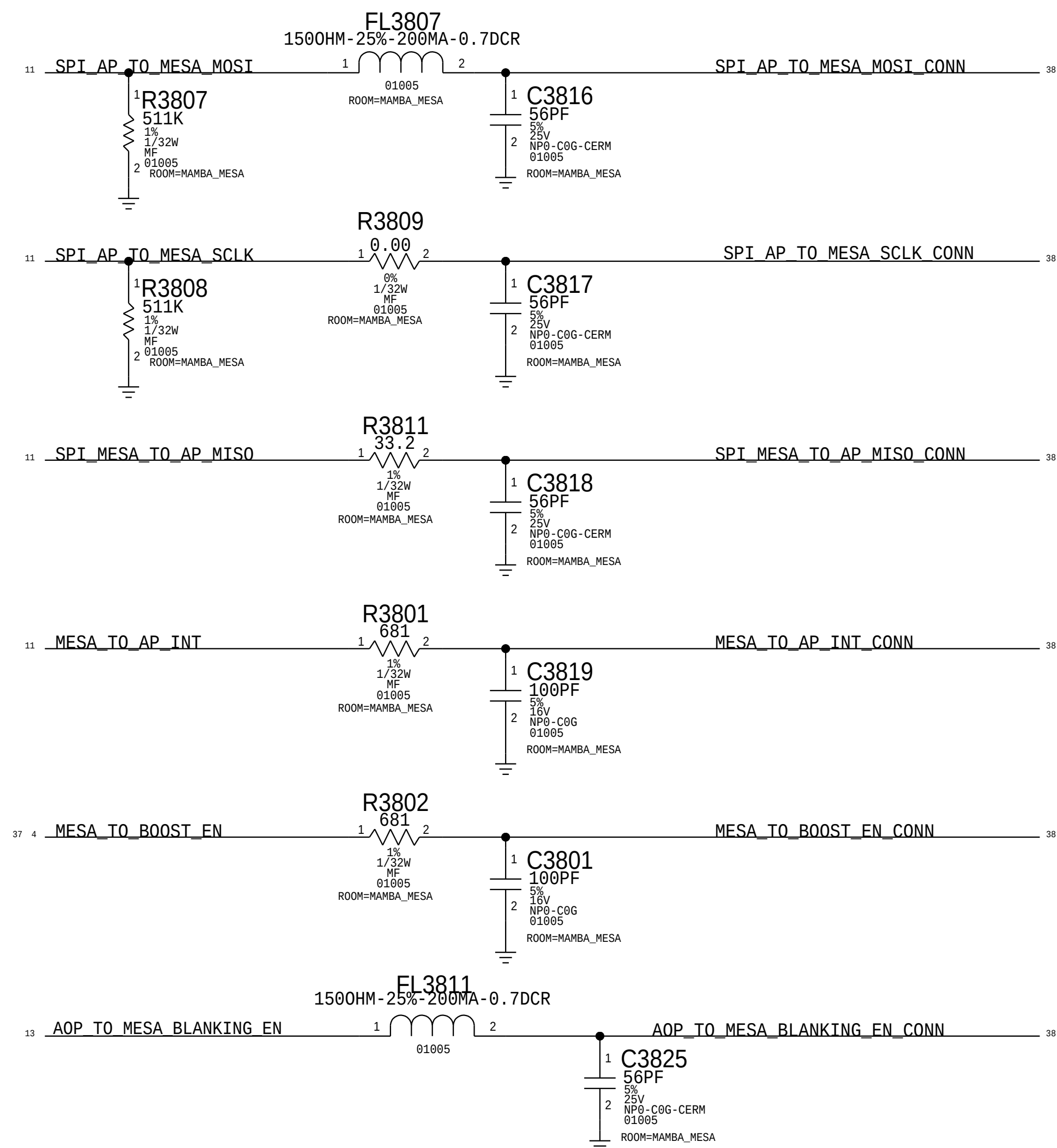
APN:353S00640

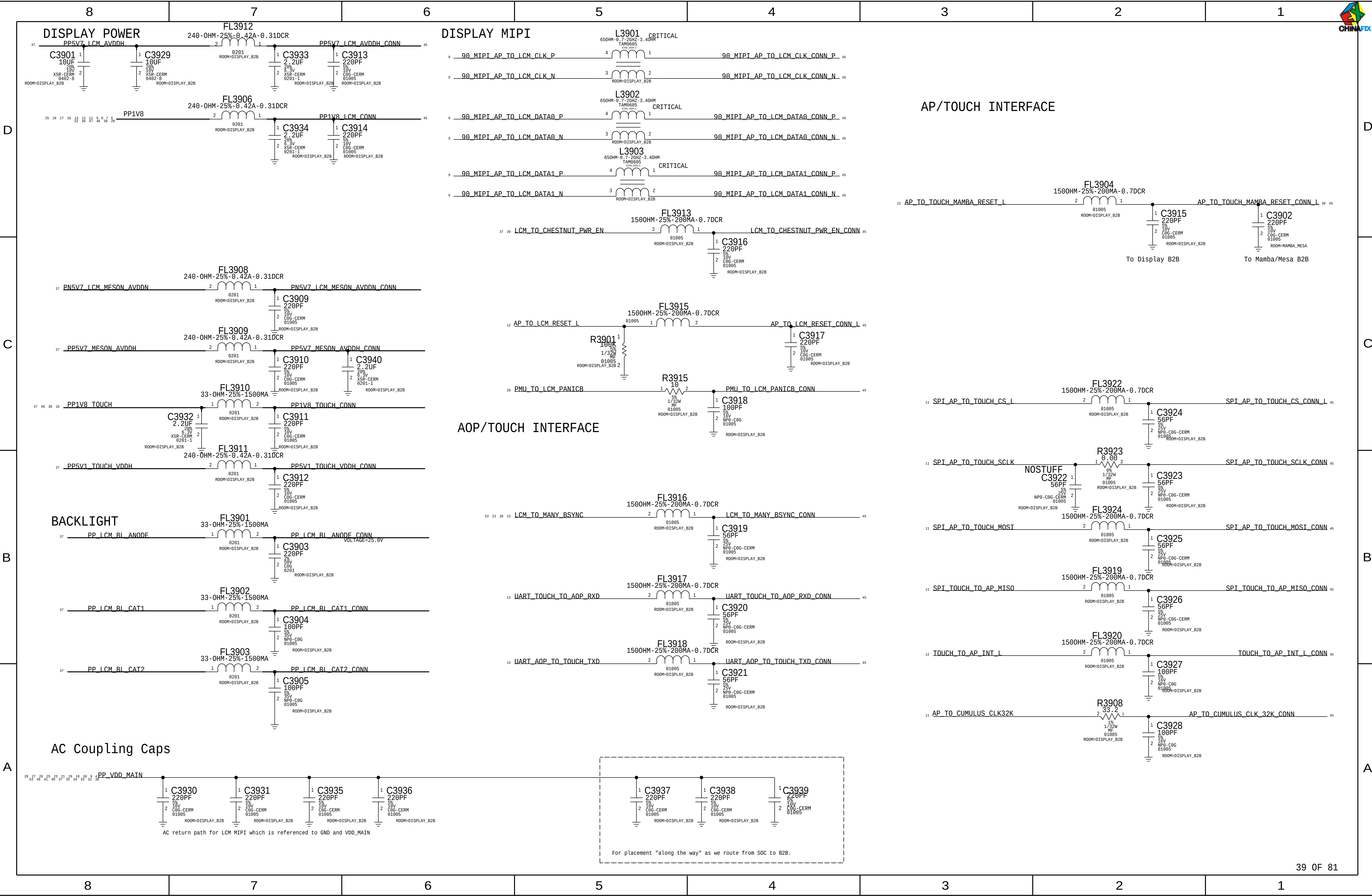


MOJAVE MESA BOOST

APN:353S00671

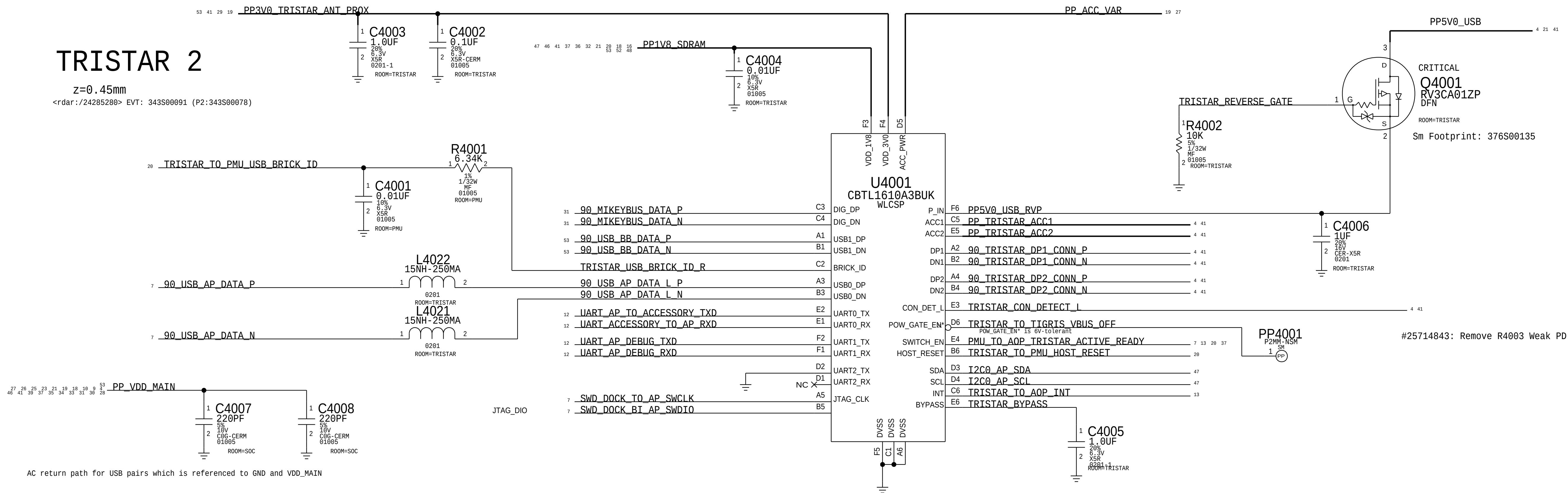






TRISTAR 2

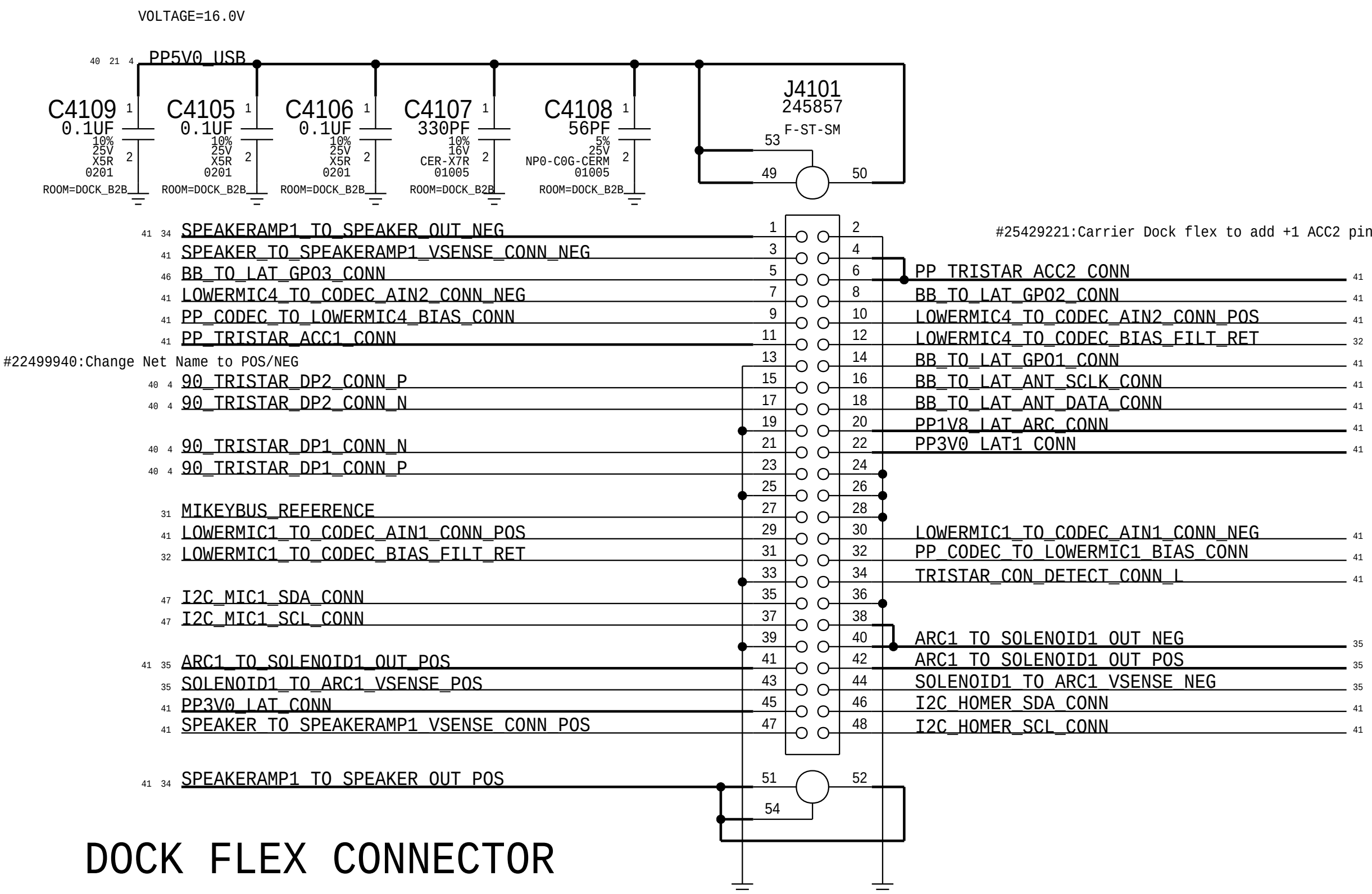
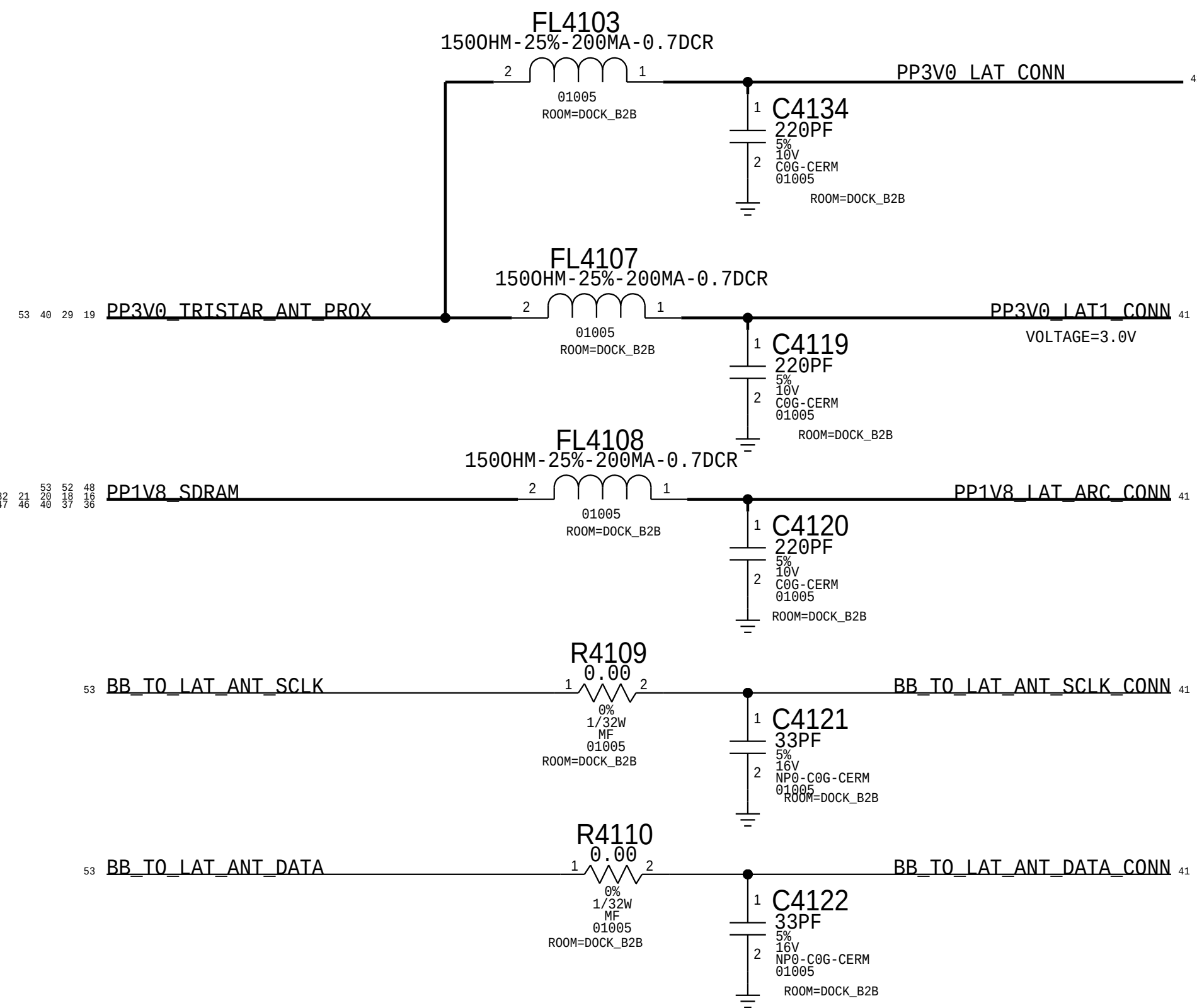
z=0.45mm
<rdar://24285288> EVT: 343S00091 (P2:343S00078)





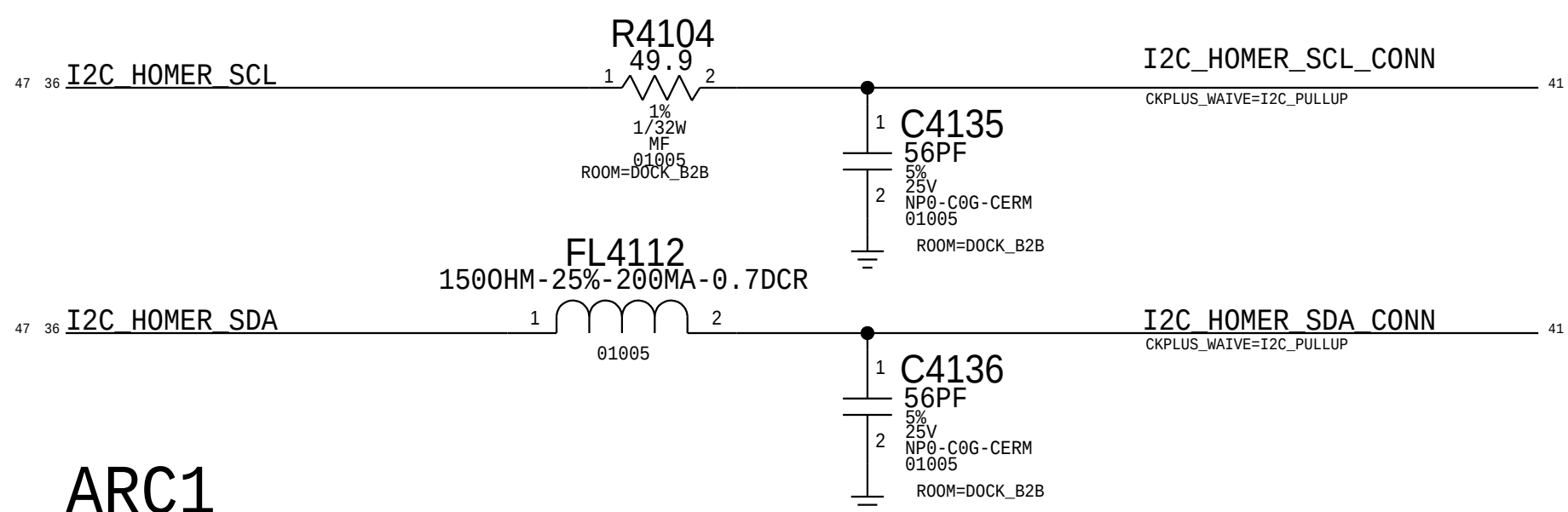
ANTENNA

Please loop in Matt Mow (Antenna Team)
when changing these components!

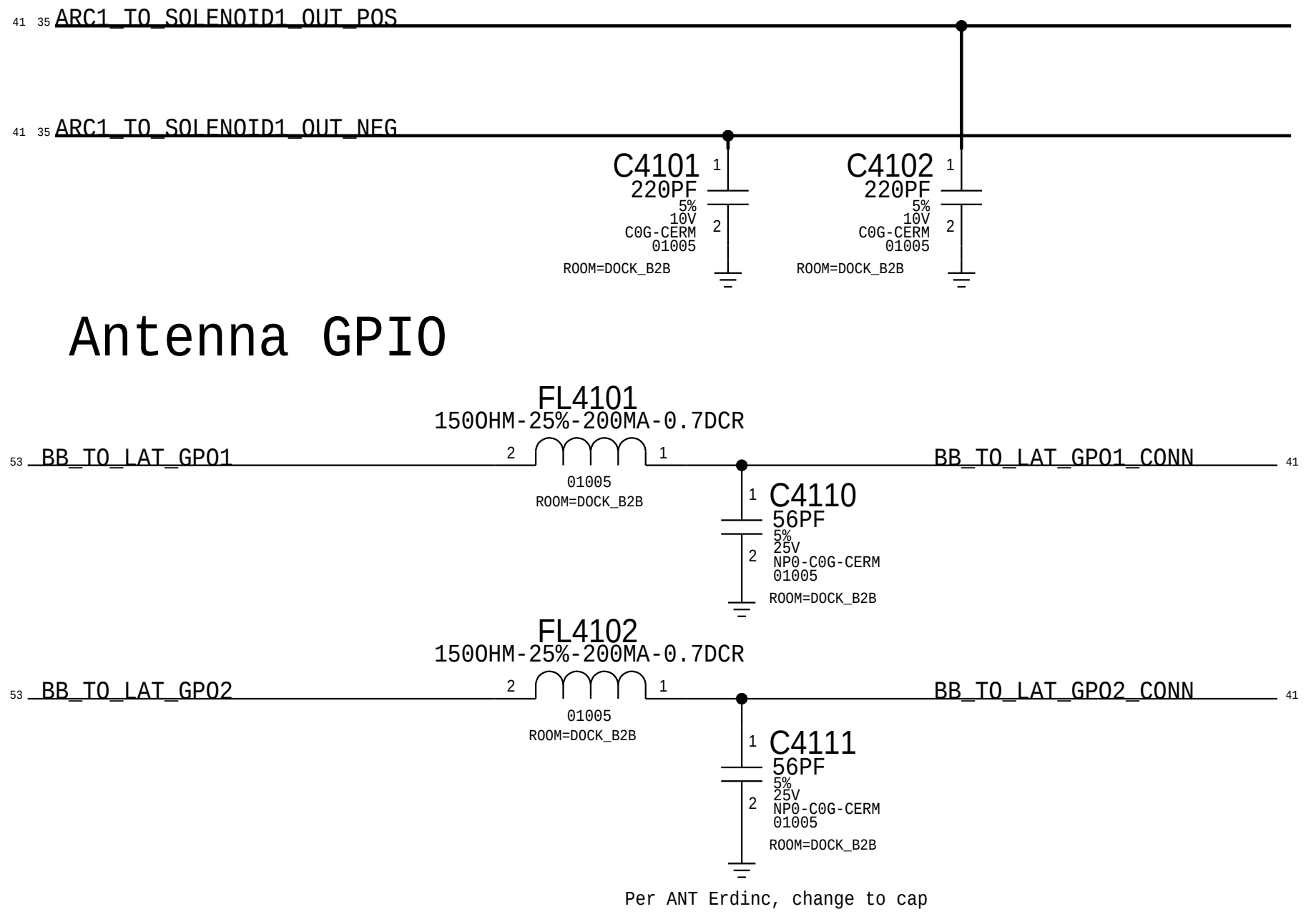


ARC CONTROL

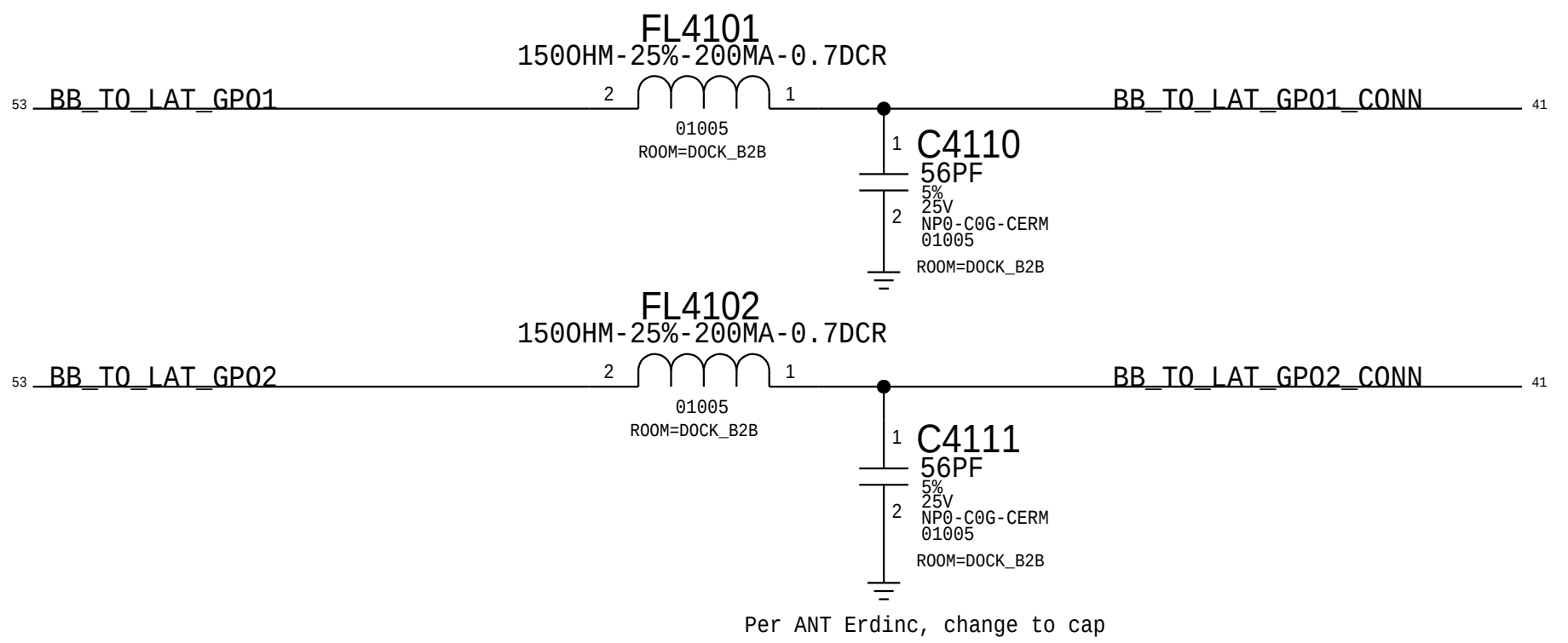
#26118161: Update FL4104 to 49.9ohm



ARC1

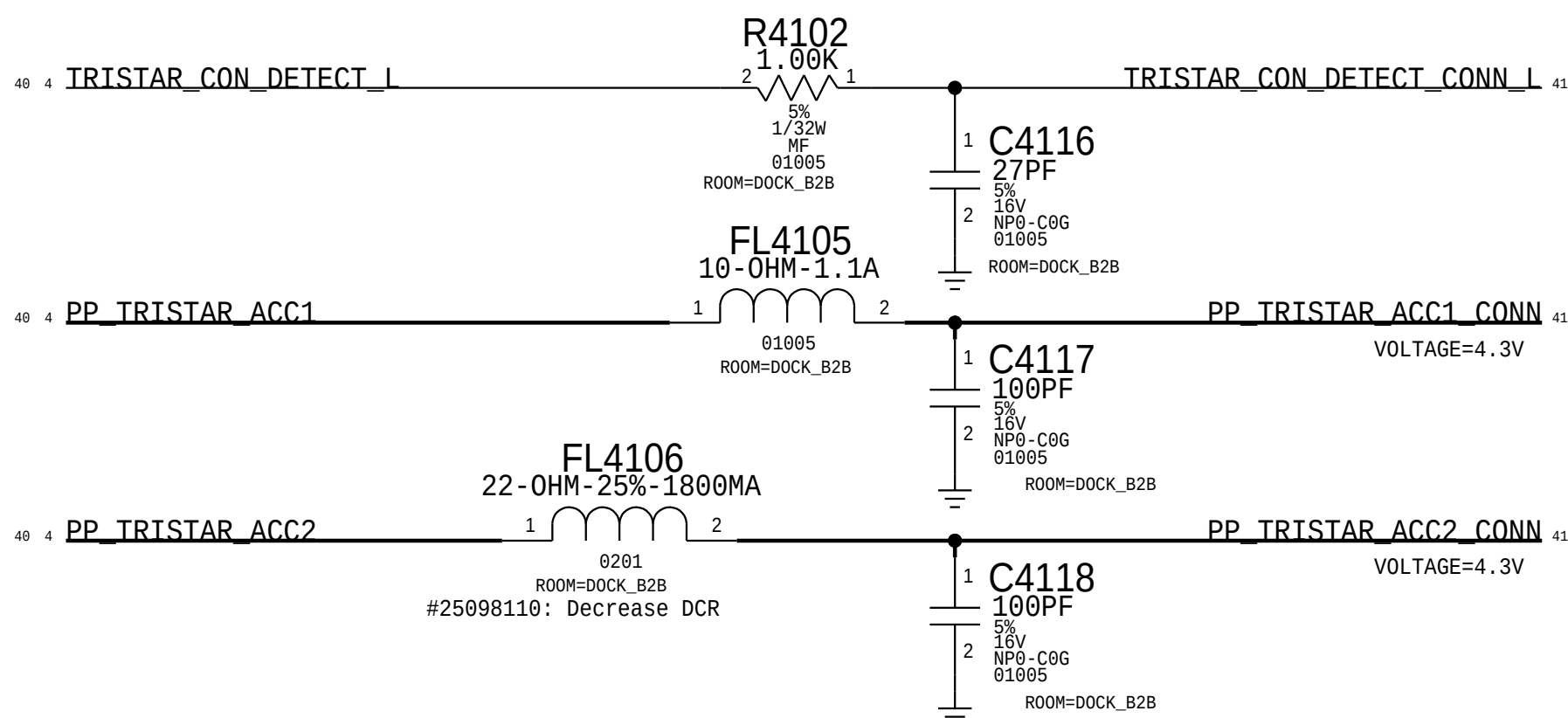


Antenna GPIO

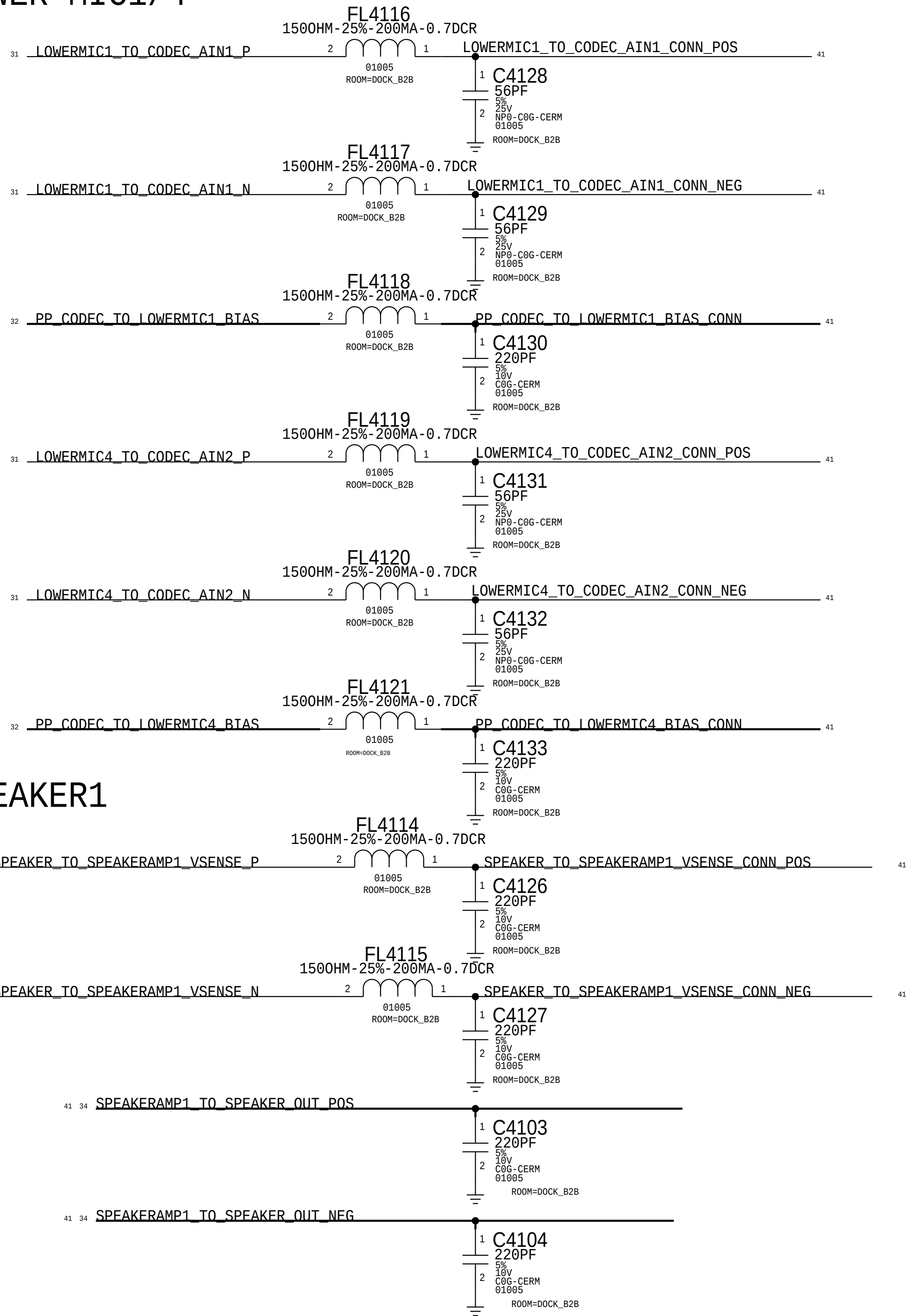


Per ANT Erdinc, change to cap

TRISTAR



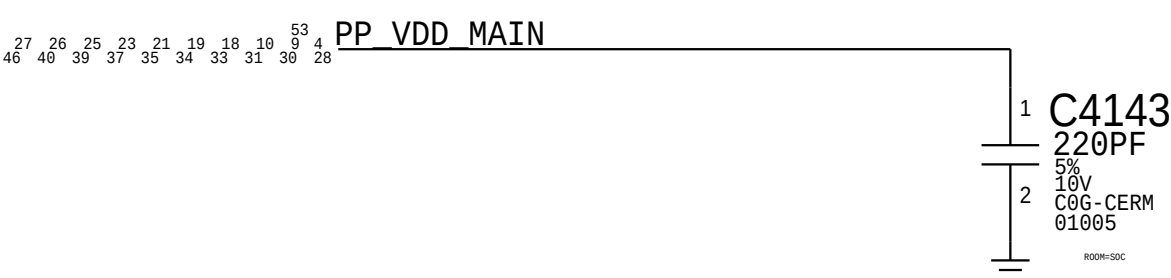
LOWER MIC1/4



SPEAKER1

USB AC Coupling

AC return path for USB pairs which is referenced to GND and VDD_MAIN



DOCK FLEX CONNECTOR



D

C

B

A

1

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7

8

D

C

B

A

1

2

3

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6

7

8



D

C

B

A

1

2

3

4

5

6

7

8

D

C

B

A

1

2

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4

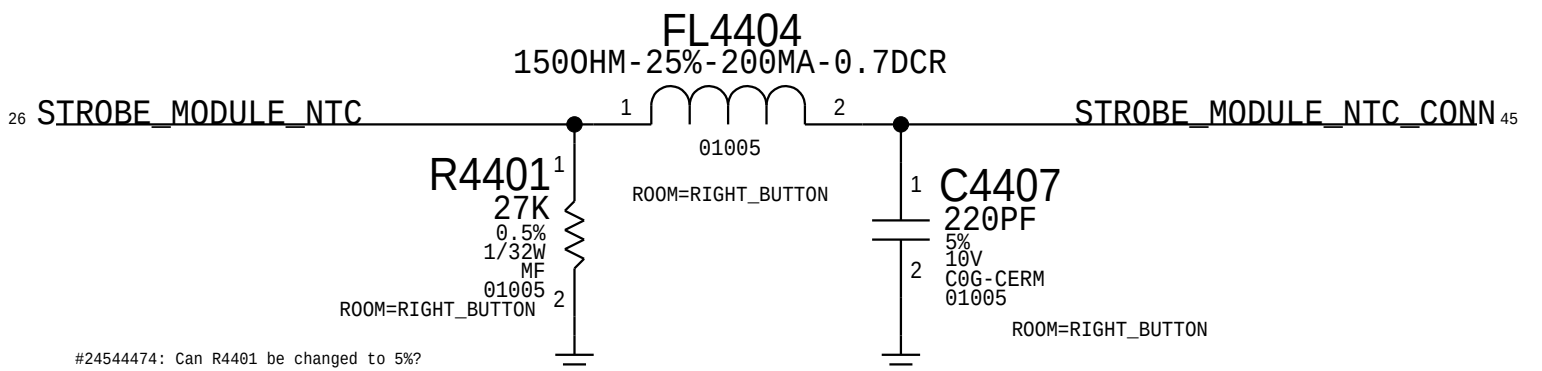
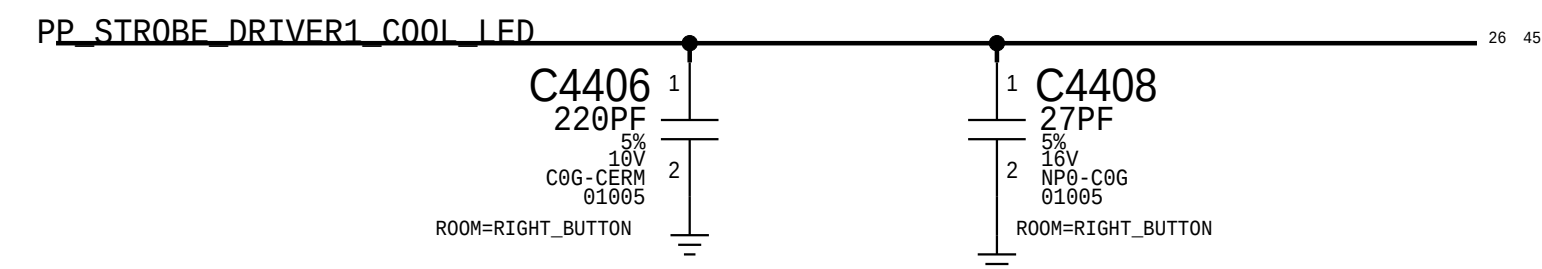
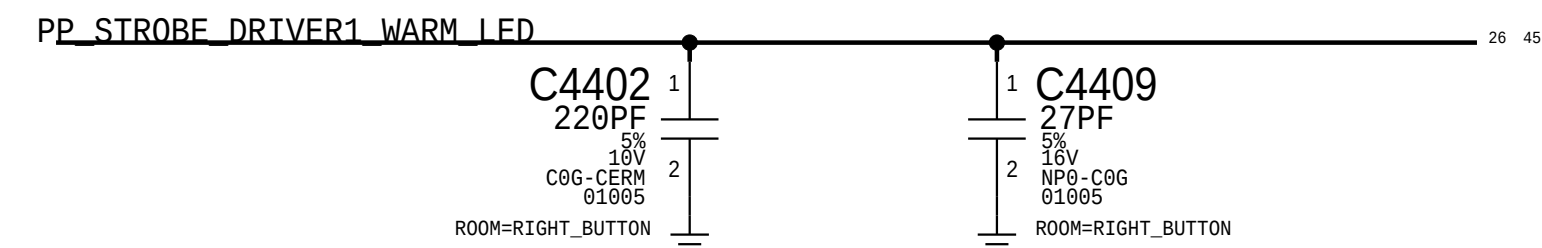
5

6

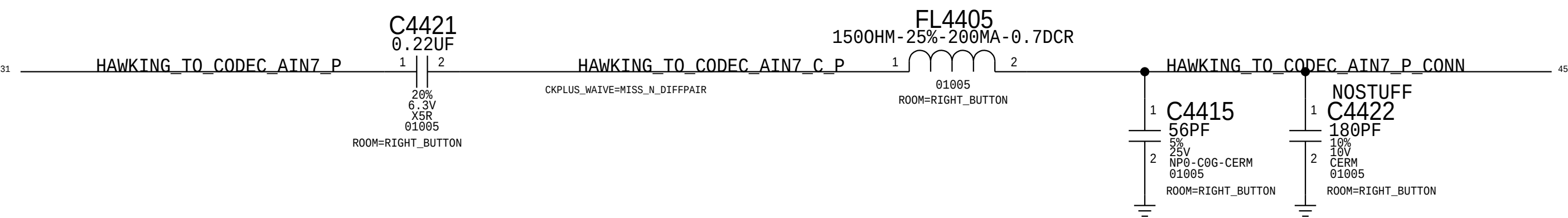
7

8

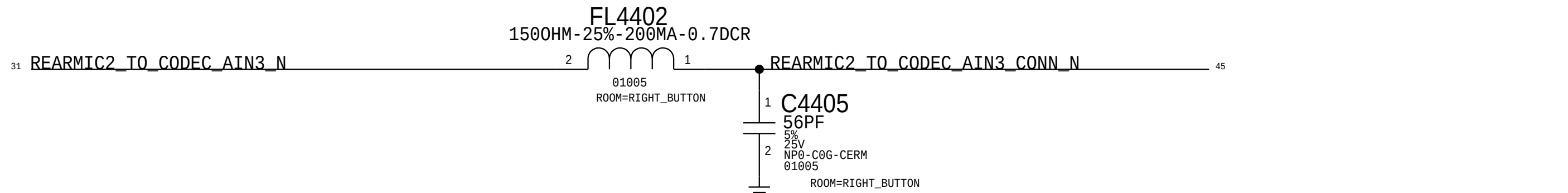
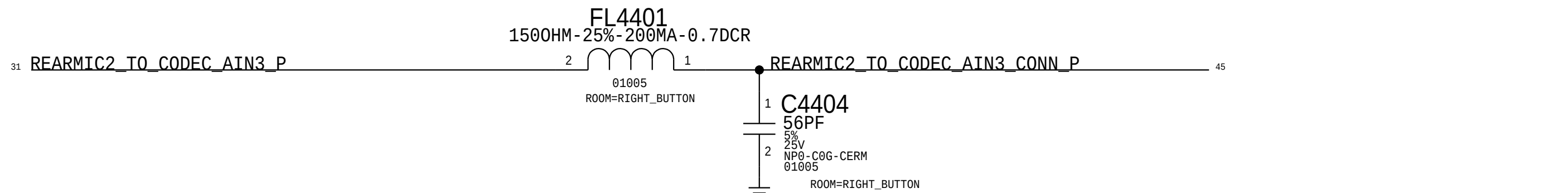
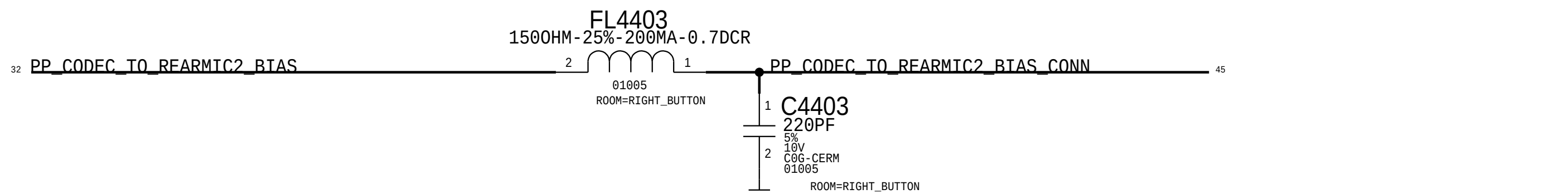
STROBE1



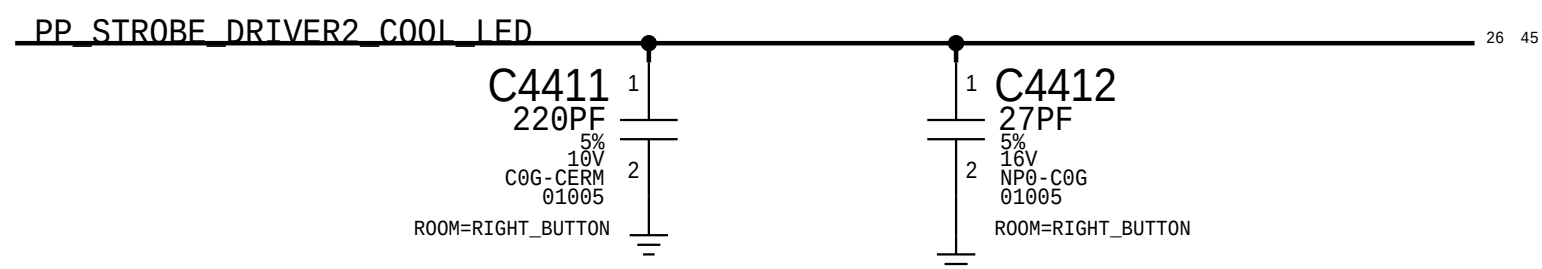
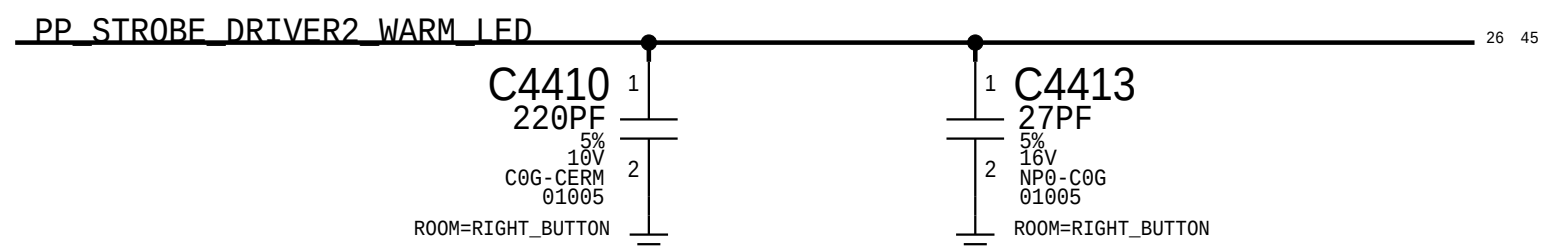
HAWKING



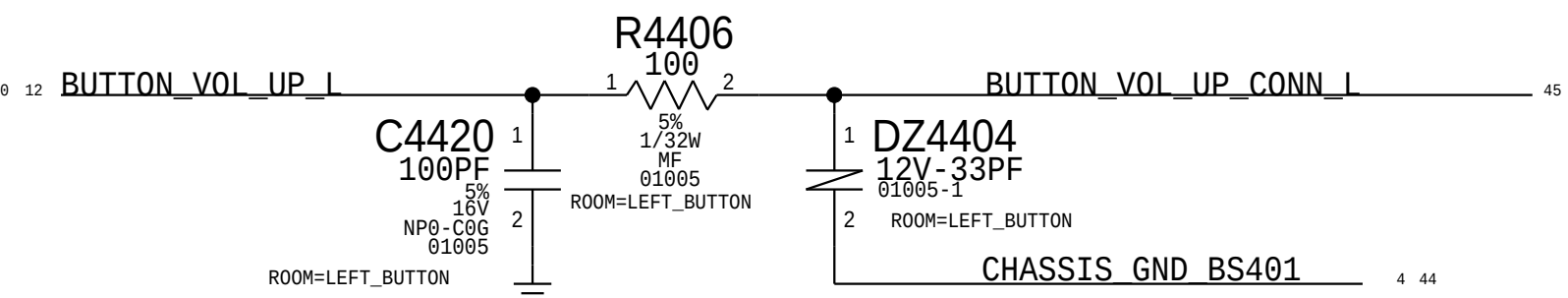
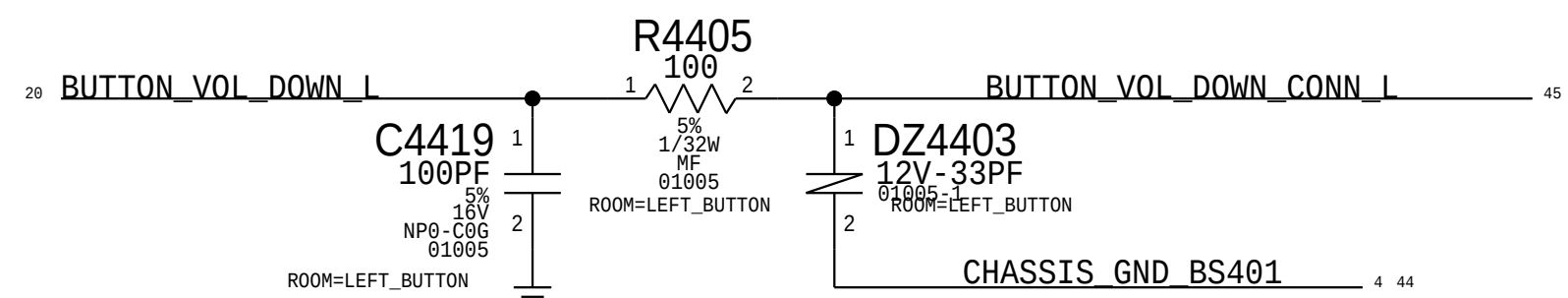
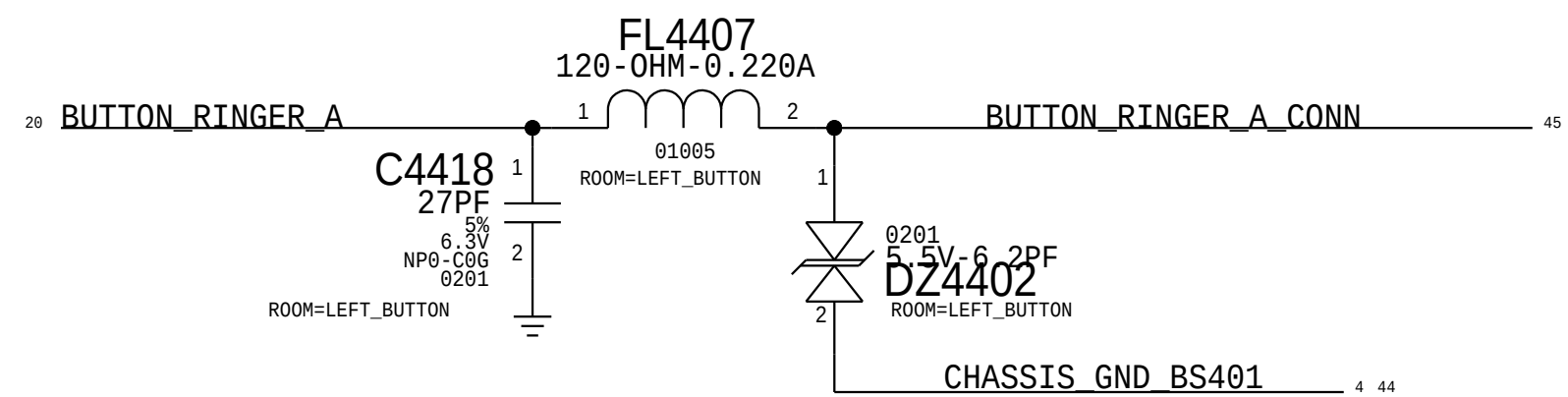
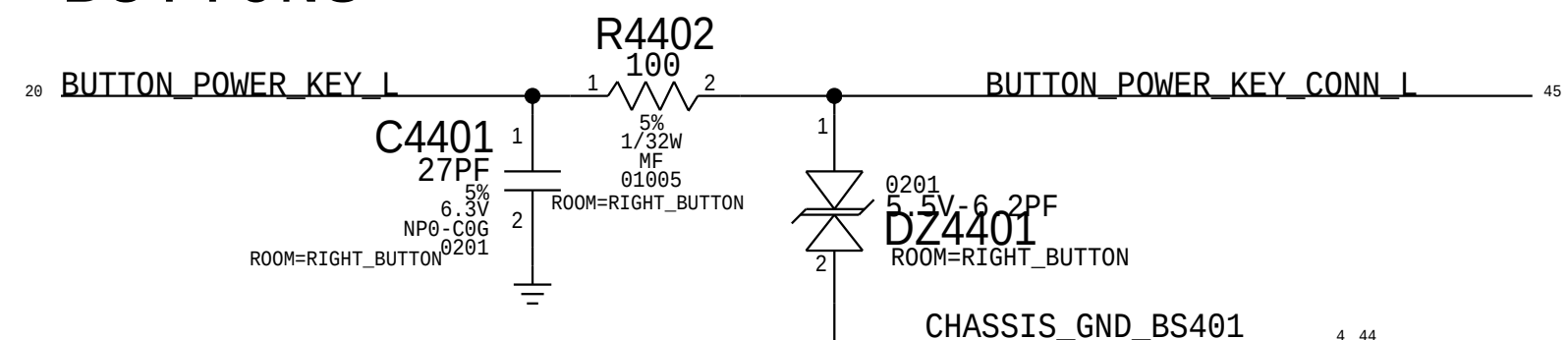
MIC2 (ANC REF)



STROBE2



BUTTONS

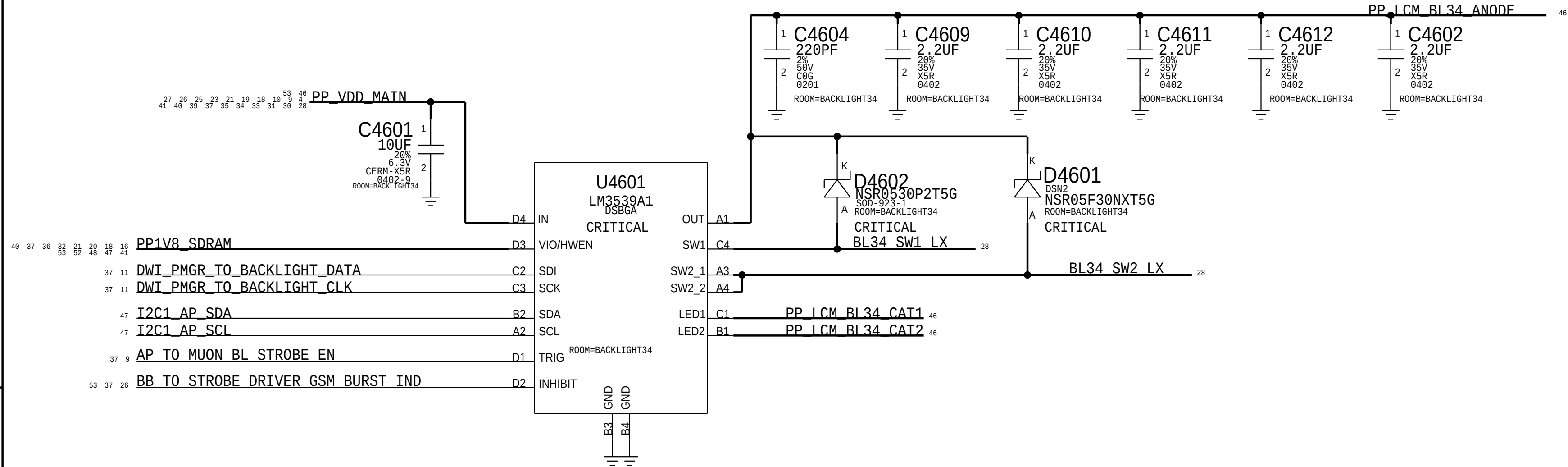






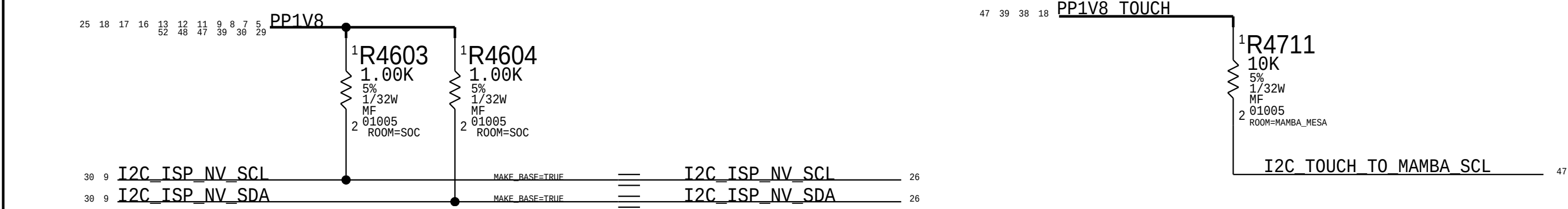
THIS PAGE UNIQUE TO LARGE FORM FACTOR

2ND BACKLIGHT DRIVER - 4LED



ISP I2C1

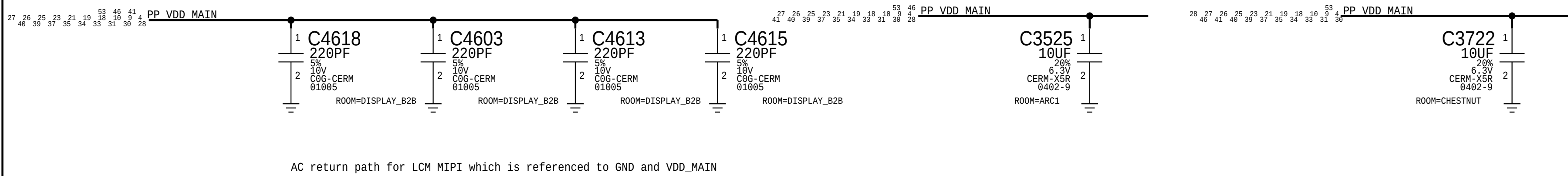
TOUCH I2C



VDD MAIN CAP
AC Coupling Caps

ARC CAP

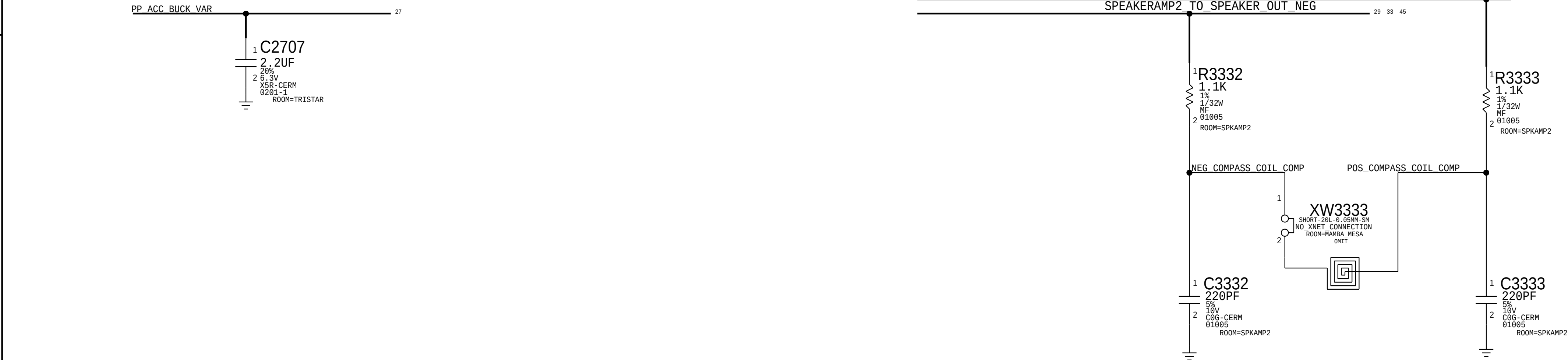
CHESTNUT CAP



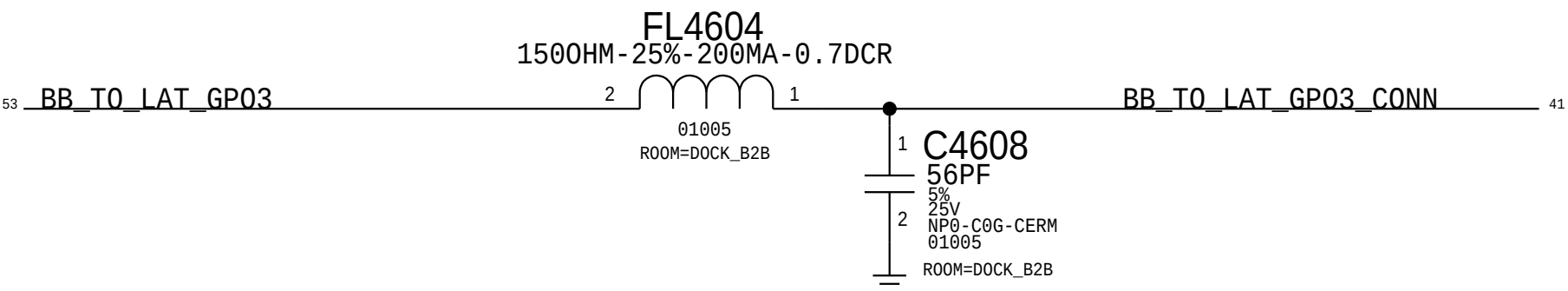
AC return path for LCM MIPI which is referenced to GND and VDD_MAIN

Top Speaker Compass Coil

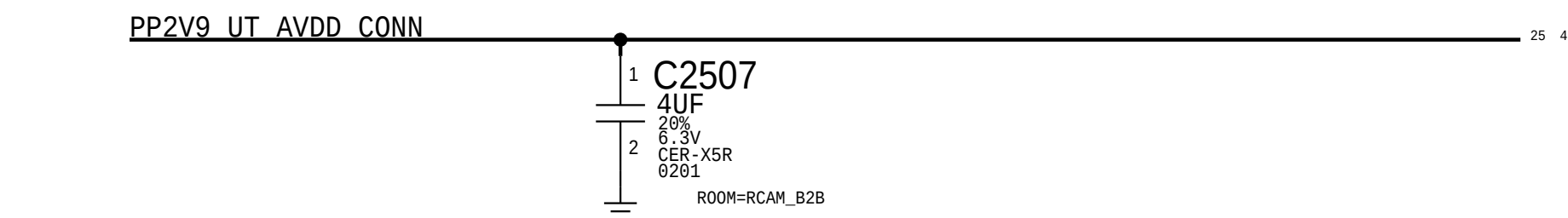
ACC Buck Caps



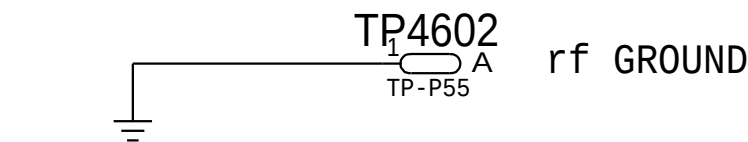
Dock B2B (Pg 41)



UT B2B



Extra RF GND



AP

I2C0

I2C1

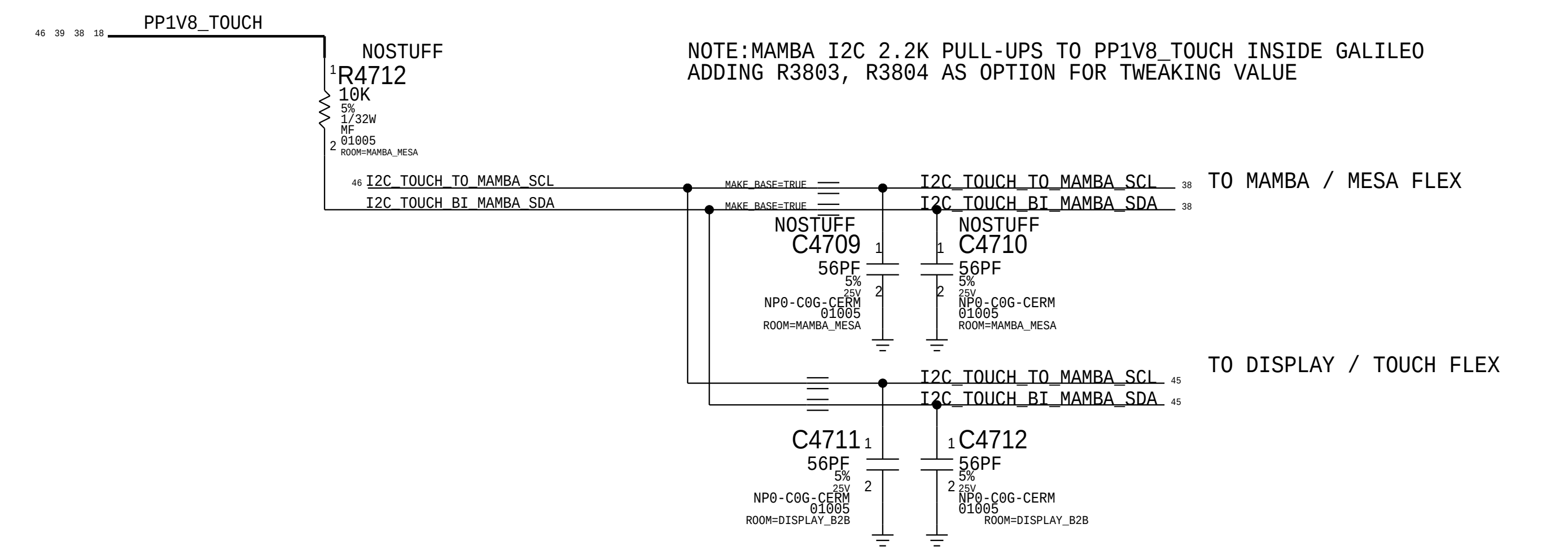
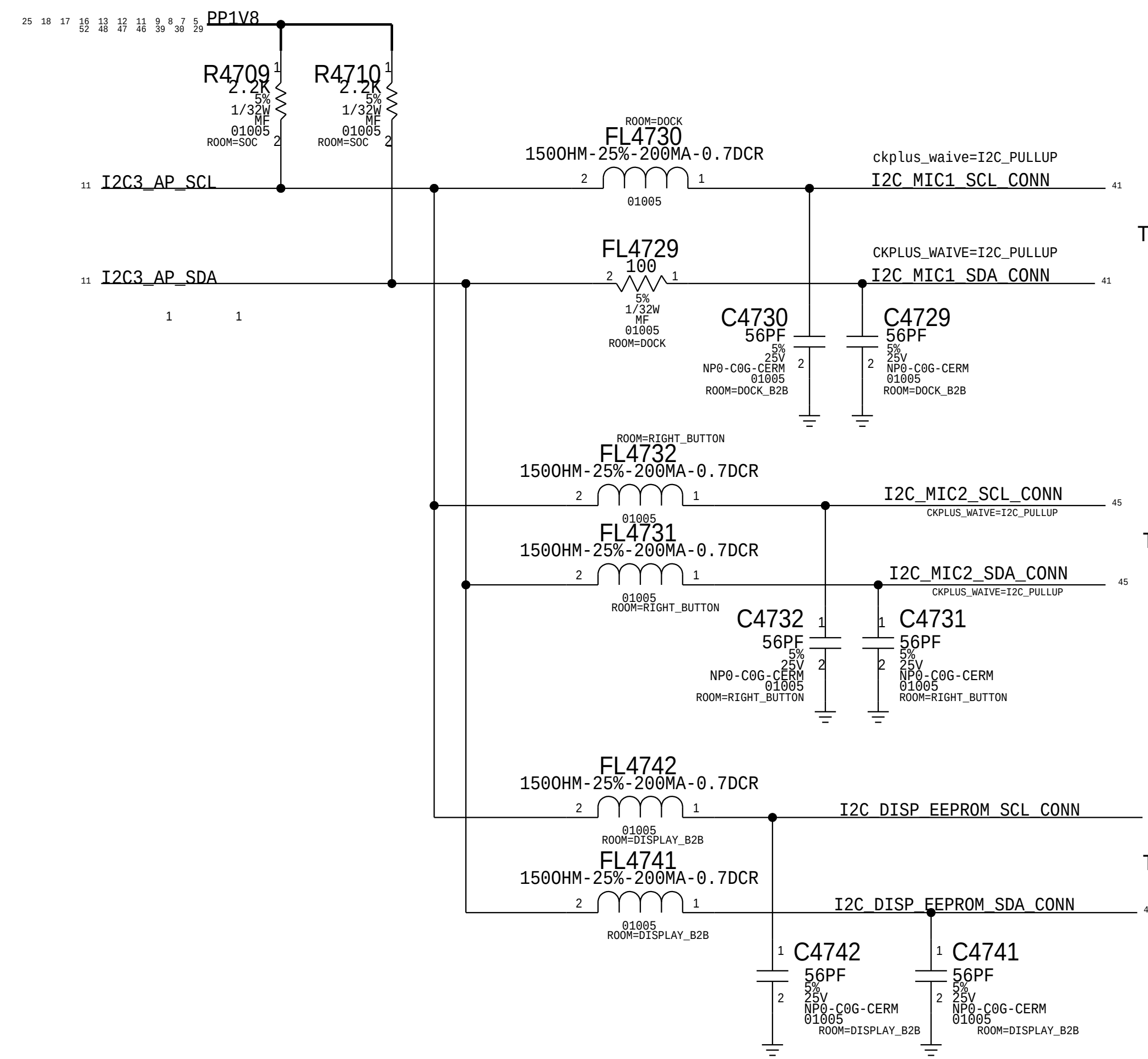
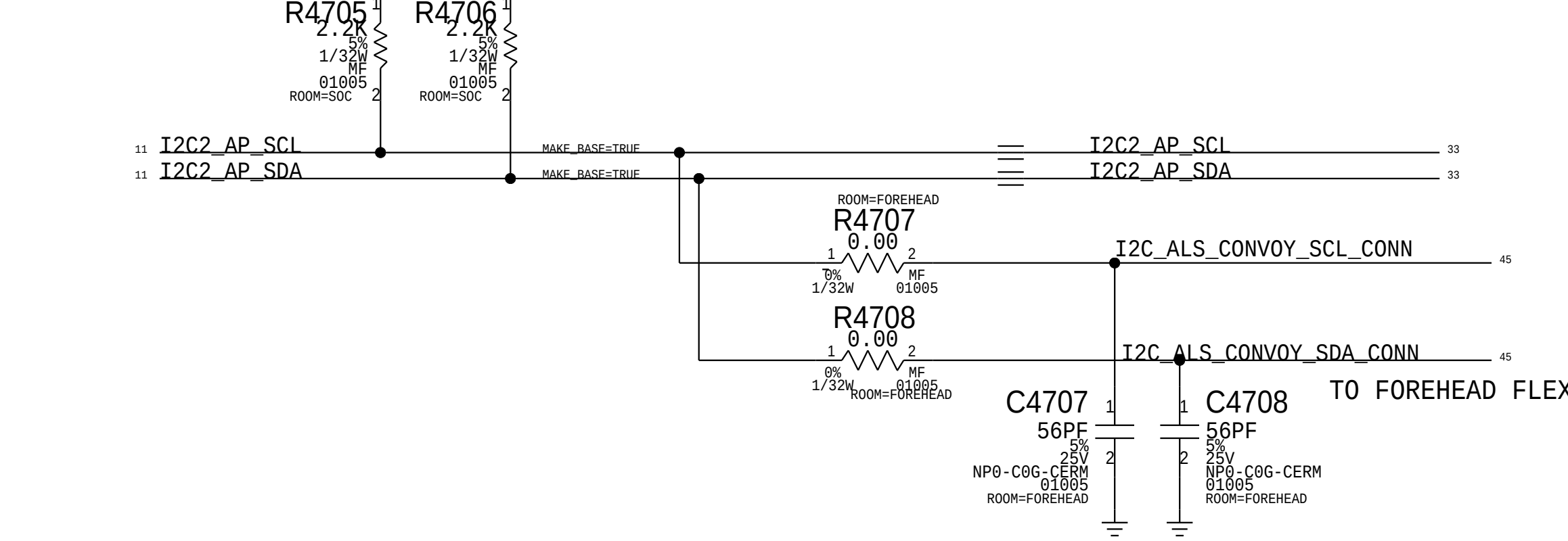
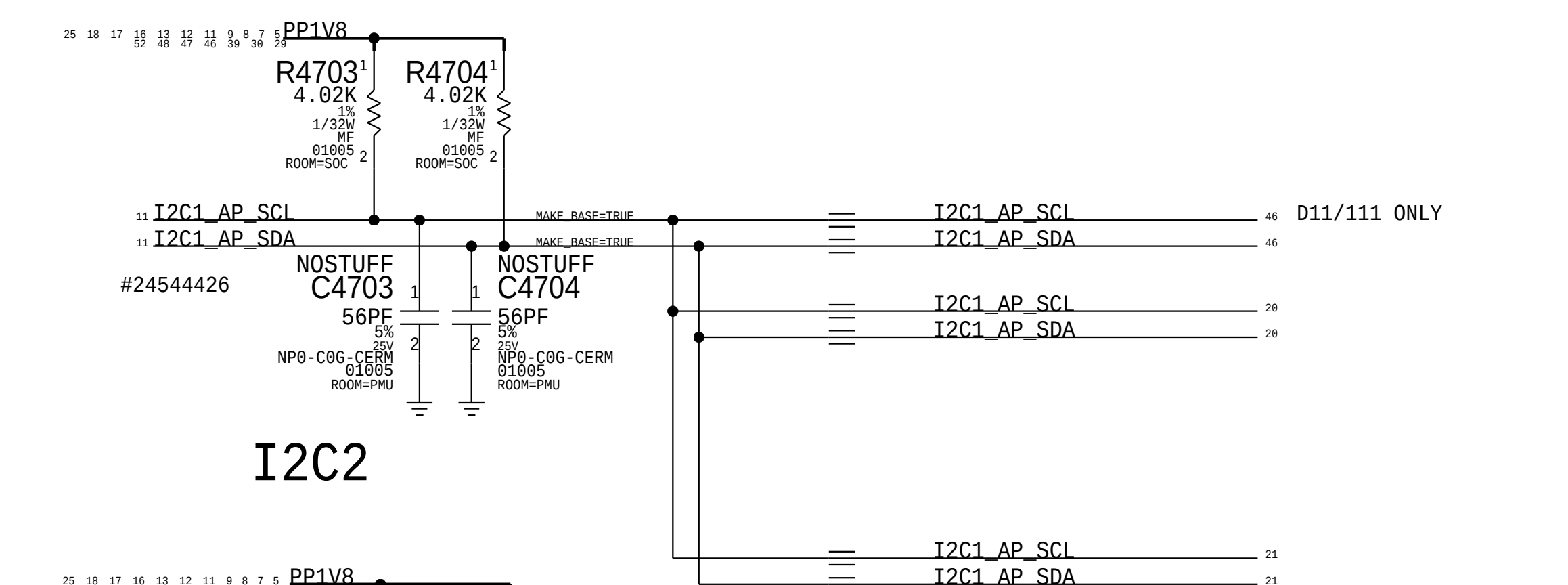
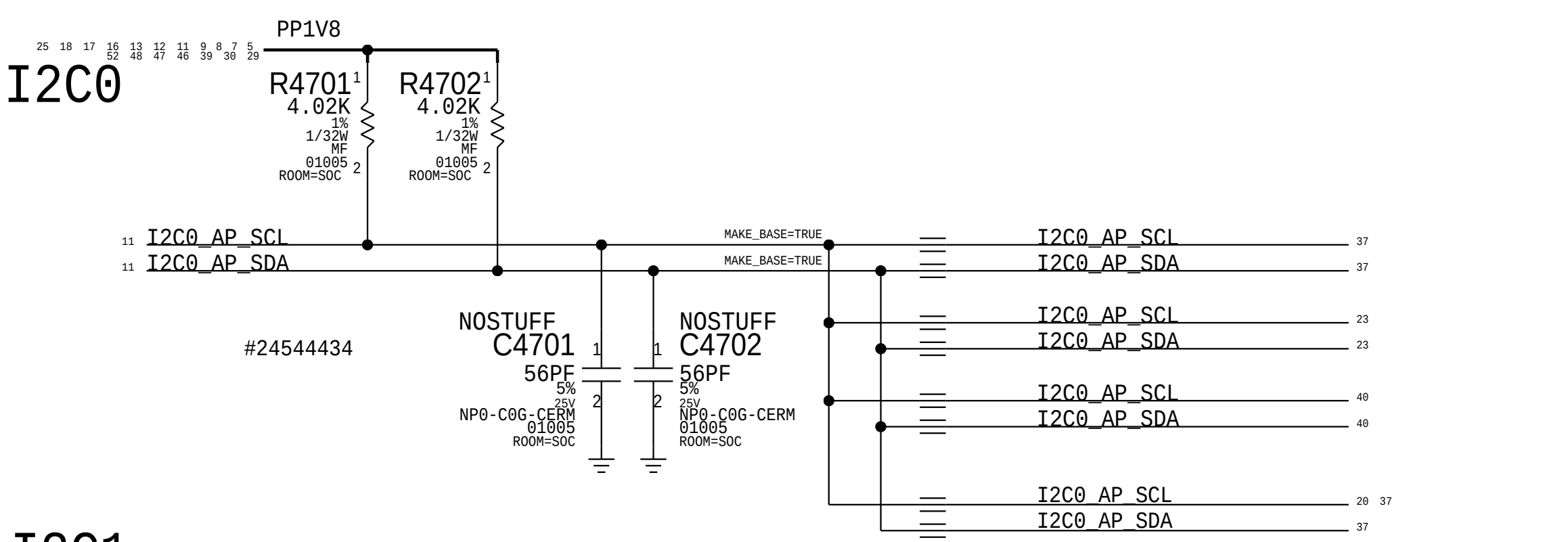
I2C2

I2C3

TOUCH

HOMER

I2C5



NOTE:MAMBA I2C 2.2K PULL-UPS TO PP1V8_TOUCH INSIDE GALILEO
ADDING R3803, R3804 AS OPTION FOR TWEAKING VALUE

AOP

I2C

ISP

I2C0

#24544699: Support 1MHz

#24550735: ISP I2C0 PU

#24958320: Intentional R4815 Change
Reduce undershoot when Prox Driving

Intentional R4815 Change

I2C1
See page 46

I2C2

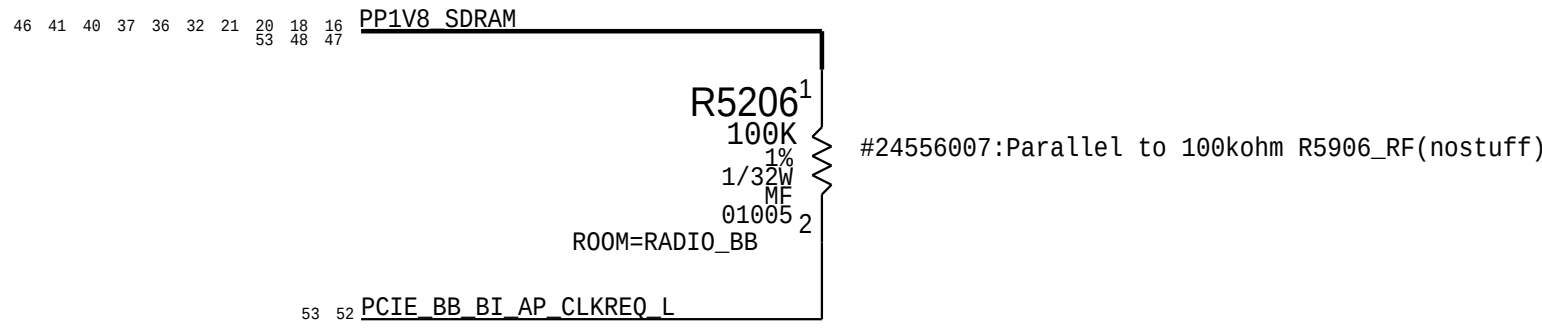
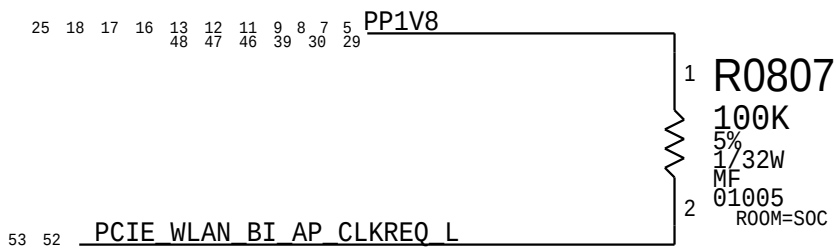
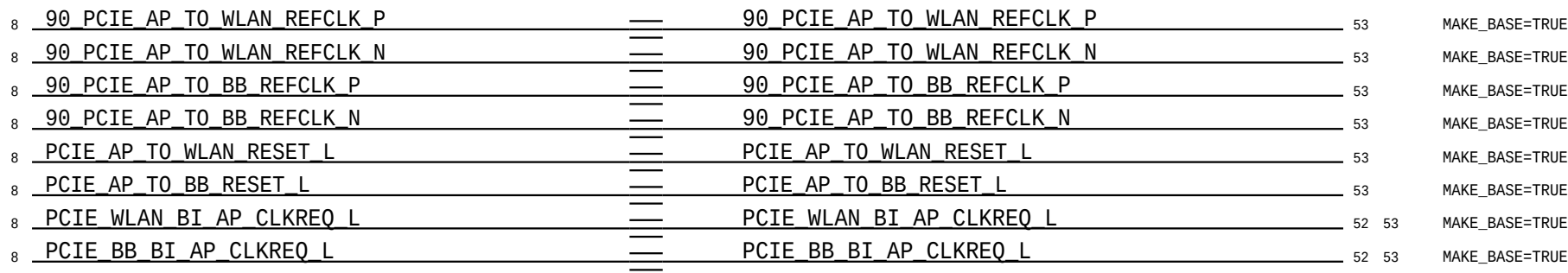
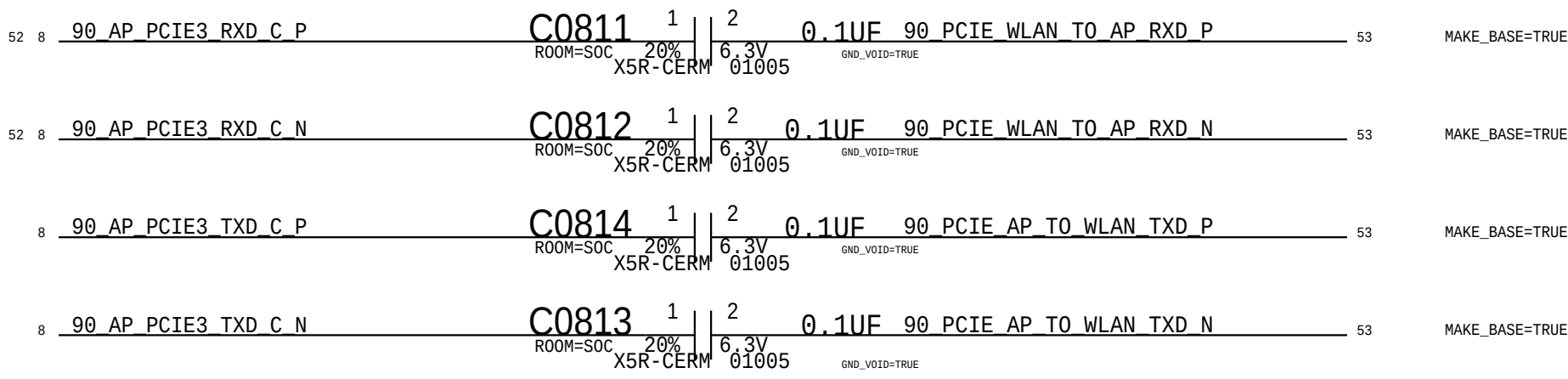
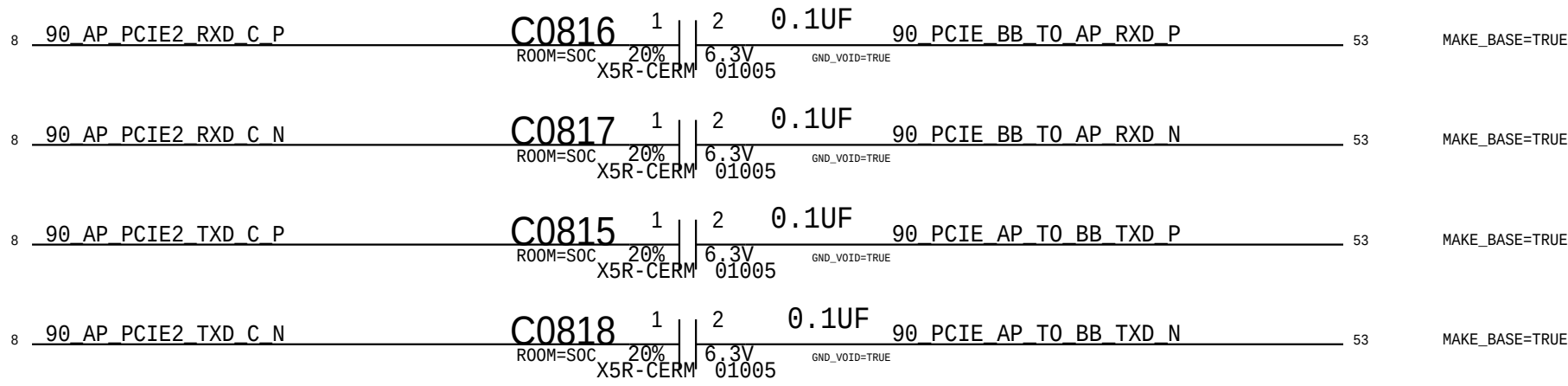
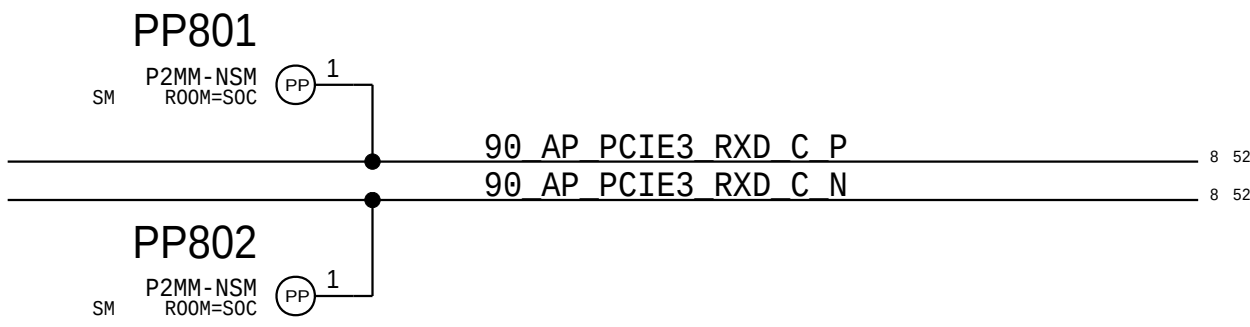
Bus	Device	Binary	7-bit Address	8-bit Address	Max Speed
ISP I2C0 <i>1MHz</i>	SALT LAKE (PRIMARY)	0010000X	0x10	0x20	1 MHz
	SALT LAKE (SECONDARY)	0110000X	0x30	0x60	1 MHz
	GRUNBERG (STANDARD)	0011100X	0x1C	0x38	1 MHz
	GRUNBERG (SPECIAL ABSOLUTE RW)	0011101X	0x1D	0x3A	1 MHz
	GRUNBERG (SPECIAL DELTA READ)	0011110X	0x1E	0x3C	1 MHz
	STROBE 1	1100011X	0x63	0xC6	1 MHz
ISP I2C1 <i>1MHz</i>	LAS VEGAS (PRIMARY)	0100000X	0x20	0x40	1 MHz
	LAS VEGAS (SECONDARY)	0110000X	0x30	0x60	1 MHz
	EDWIN	1100011X	0x63	0xC6	400 kHz
	STROBE 2	0000110X	0x06	0x0C	1 MHz
ISP I2C2 <i>400kHz</i>	CONCORD	0010000X	0x10	0x20	1 MHz
AOP I2C <i>1MHz</i>	ARC DRIVER 1	100000YX	0x41	0x82	1 MHz
	PROX	1011000X	0x58	0xB0	1 MHz
	SPKAMP1 (L26)	100000YX	0x40	0x80	1 MHz
	TURTLE	0101100X	0x2c	0x58	400 kHz
	MESA	1100001X	0x61	0xC2	400 kHz
Touch I2C <i>600kHz</i>	MAMBA	1100000X	0x60	0xC0	1 MHz
	MESON	1000000X	0x40	0x80	510 kHz
EEPROM I2C	EEPROM	1010001X	0x51	0xA2	400 kHz
AP I2C0 <i>400kHz</i>	BOOST	1110101X	0x75	0xEA	3.4 MHz
	TRISTAR	0011010X	0x1A	0x34	400 kHz
	BACKLIGHT 1	1100010X	0x62	0xC4	1 MHz
	CHESTNUT	0100111X	0x27	0x4E	400 kHz
AP I2C1 <i>400kHz</i>	ADELYN	1110100X	0x74	0xE8	400 kHz
	TIGRIS	1110101X	0x75	0xEA	400 kHz
	BACKLIGHT 2	1100010X	0x62	0xC4	1 MHz
AP I2C2 <i>400kHz</i>	ALS	0101001X	0x29	0x52	400 kHz
	CONVOY	0100001X	0x21	0x42	3.4 MHz
		0100010X	0x22	0x44	400 kHz
	SPKAMP 2 (L26)	10000ZYX	0x40	0x80	400 MHz
AP I2C3 <i>100kHz</i>	DOCK FLEX (MIC1)	1010100X	0x54	0xA8	1 MHz
	COMBINED BUTTON FLEX (MIC2)	1010100X	0x56	0xAC	1 MHz
	DISP EEPROM	1010001X	0x51	0xA2	400kHz
Homer I2C <i>1MHz</i>	SCHRODINGER (STANDARD)	0001110X	0xE	0x1C	1 MHz
	SCHRODINGER (CAL REG'S)	0001111X	0xF	0x1D	1 MHz
	EEPROM (STANDARD)	1010000X	0x50	0xA0	
	EEPROM (CONFIG)	1010000X	0x50	0xA0	

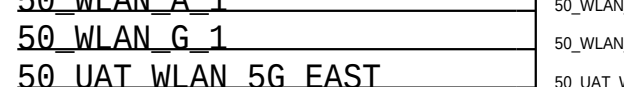
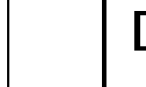




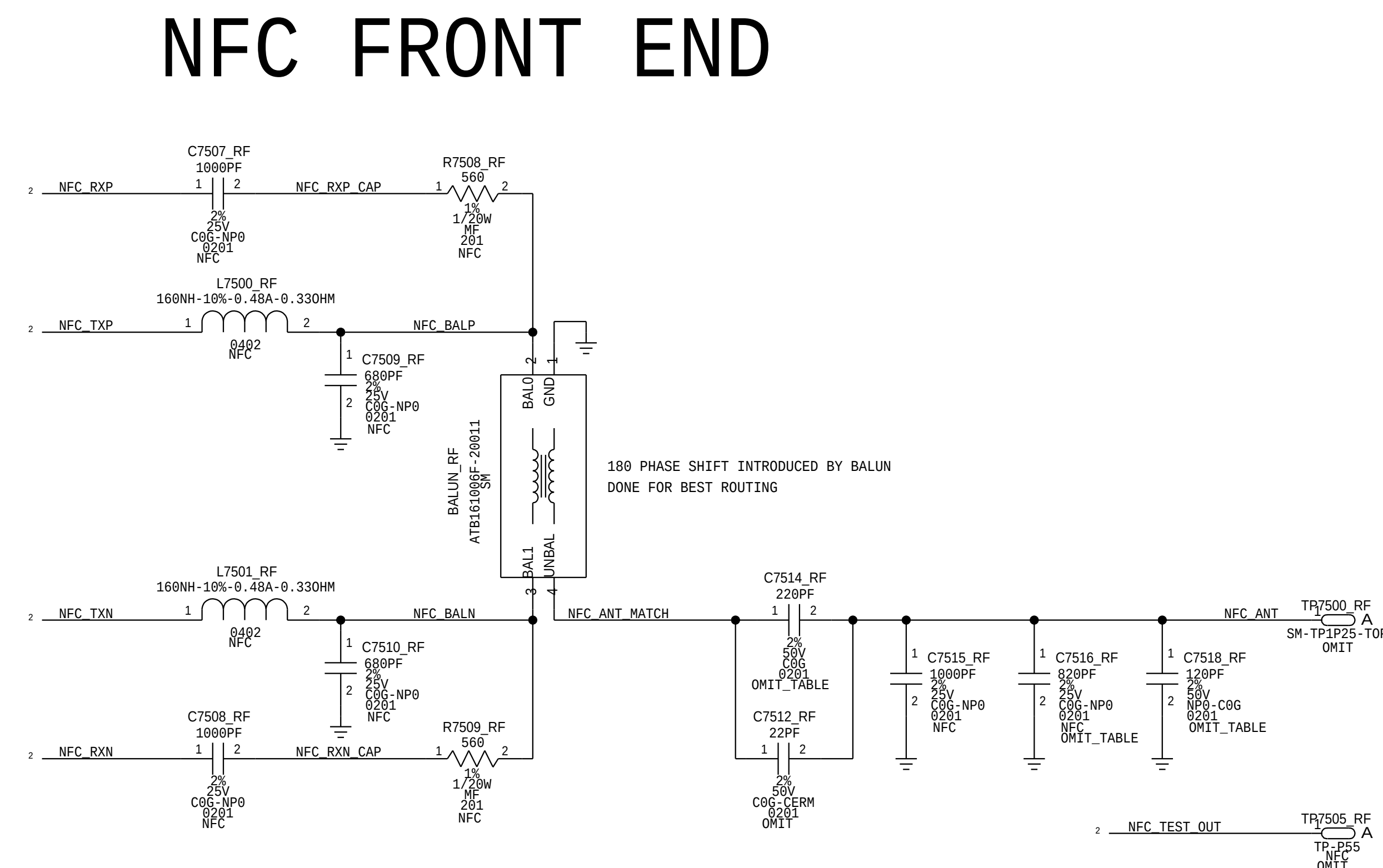
This page controls elements different accross all MLBs

PCIe Lanes





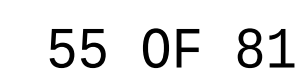
NFC CONTROLLER



```

graph LR
    A2[PP_VDD_MAIN] --> OMIT_TABLE[NFC5W_RF  
FPF1204UCX  
WLCSP-COMBO]
    B2[PP1V8_SDRAM] --> OMIT_TABLE
    OMIT_TABLE --> A1[PP_VDD_MAIN_NFC]
    A1 --> R7520_1[R7520 10k]
    R7520_1 --> GND1[GND]
    A1 --> R7520_2[R7520 10k]
    R7520_2 --> GND2[GND]
    A1 --> R7520_3[R7520 10k]
    R7520_3 --> GND3[GND]

```



METROCIRC

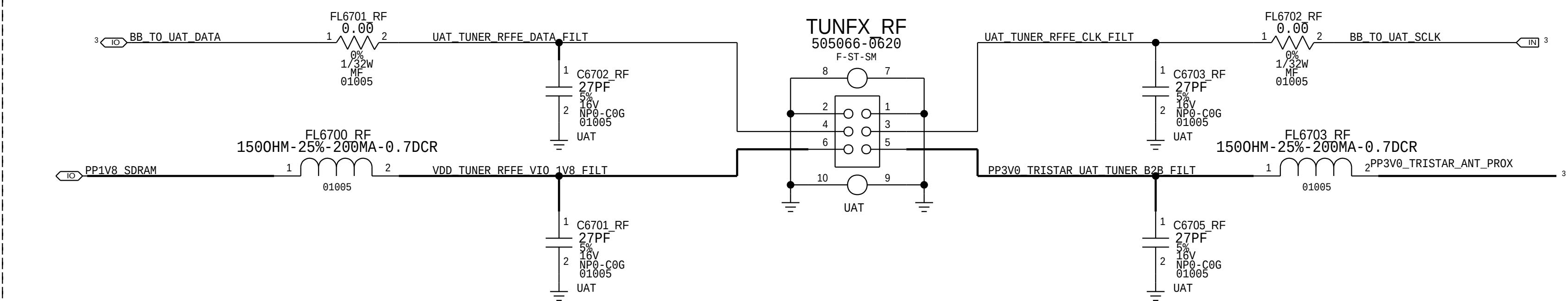
D11 NORTH-SOUTH METROCIRC 339S00147

D11 EAST-WEST METROCIRC 339S00146

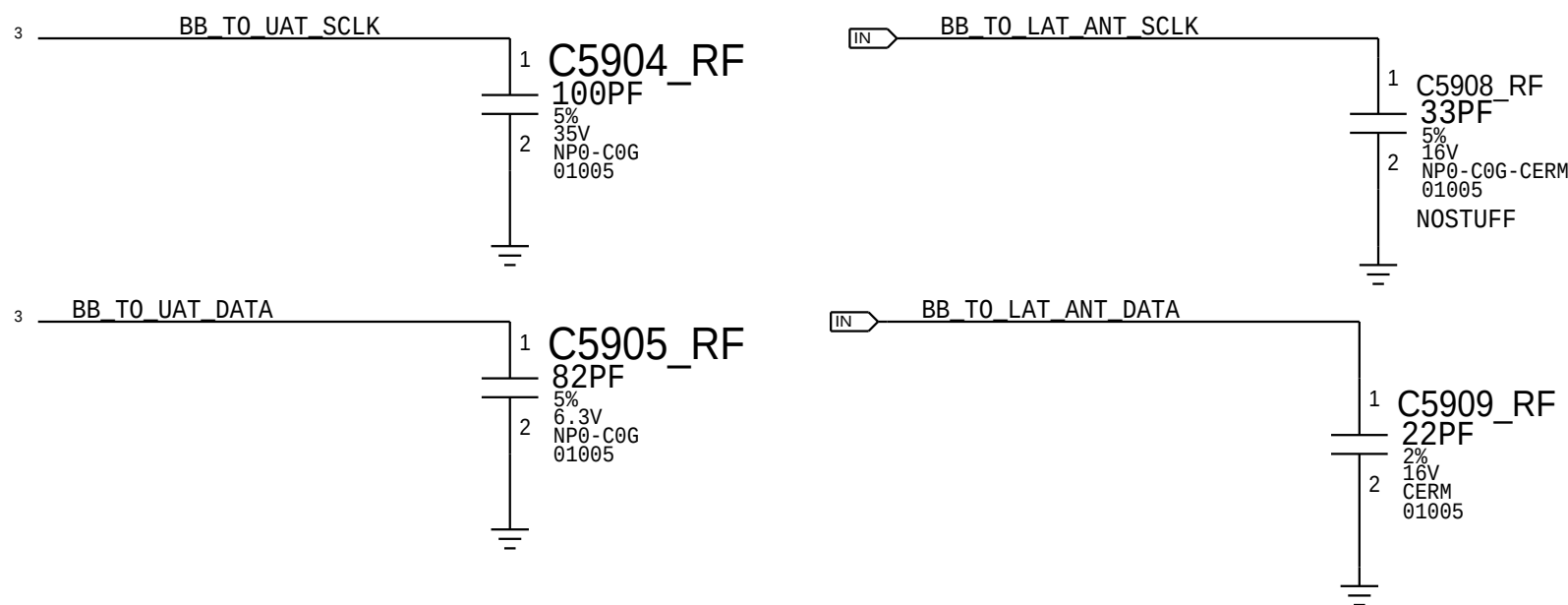


UAT TUNER FLEX

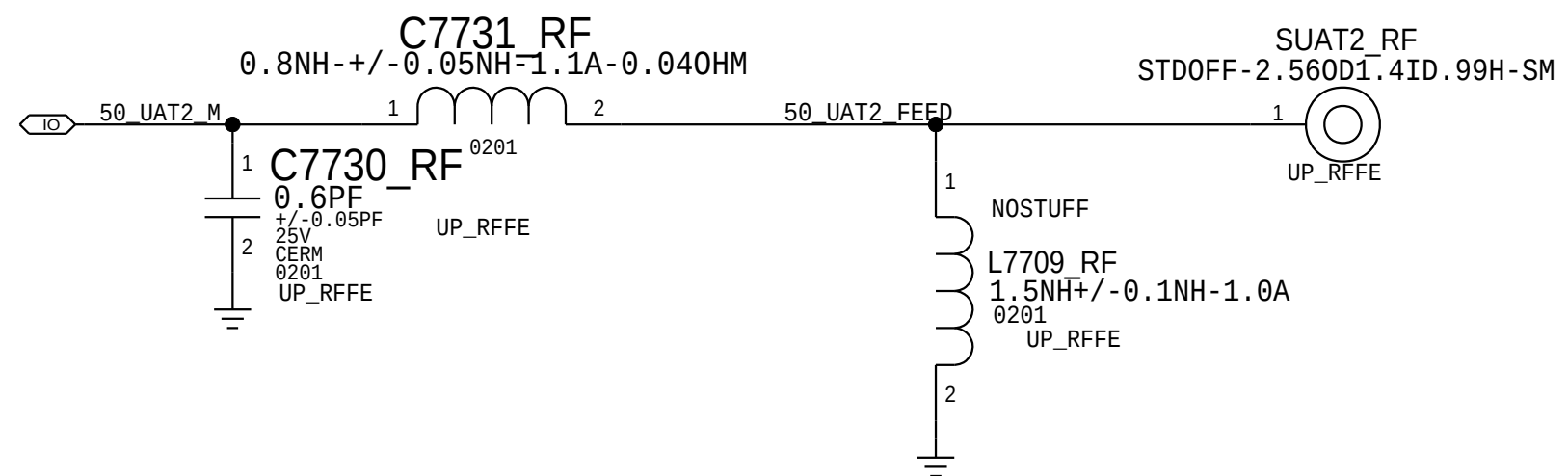
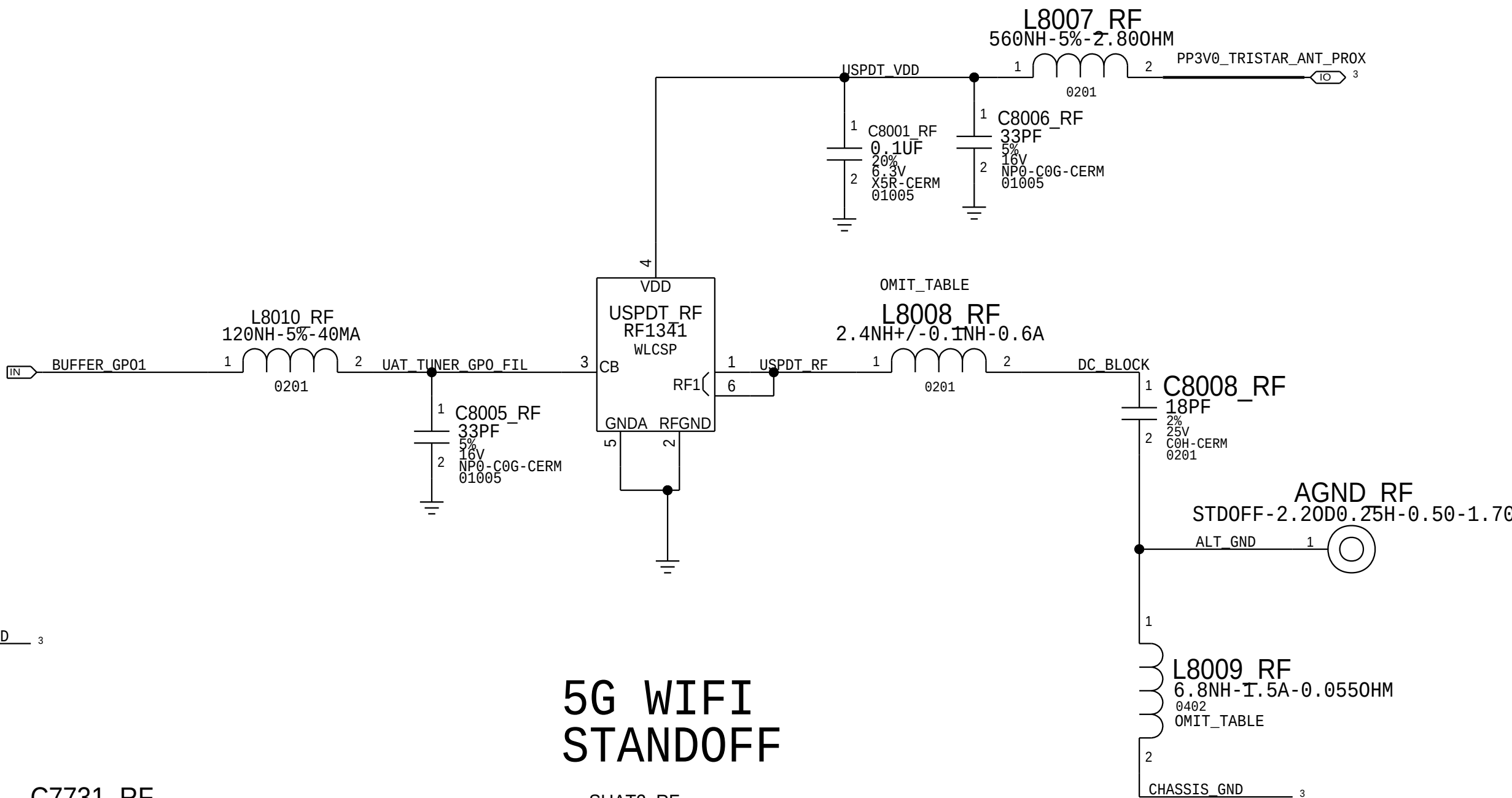
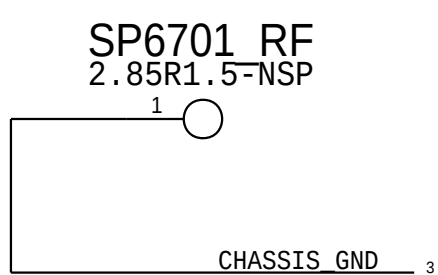
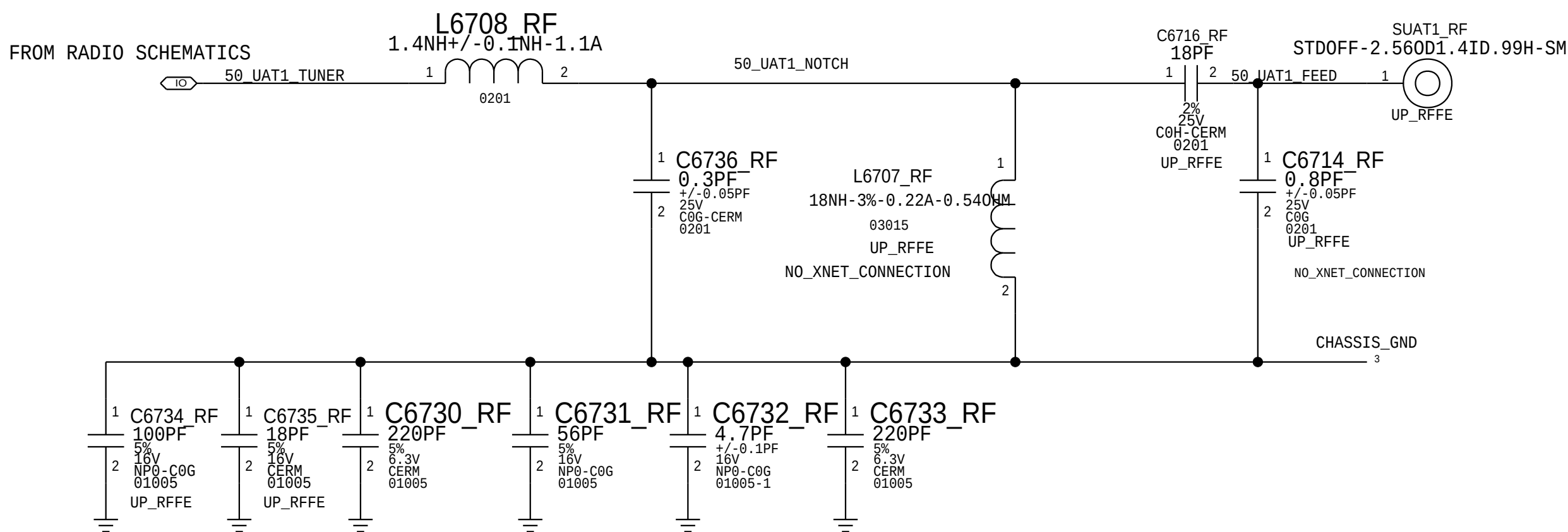
PLEASE CONTACT ANTENNA (MATT MOW)
FOR ANY COMPONENT CHANGE.



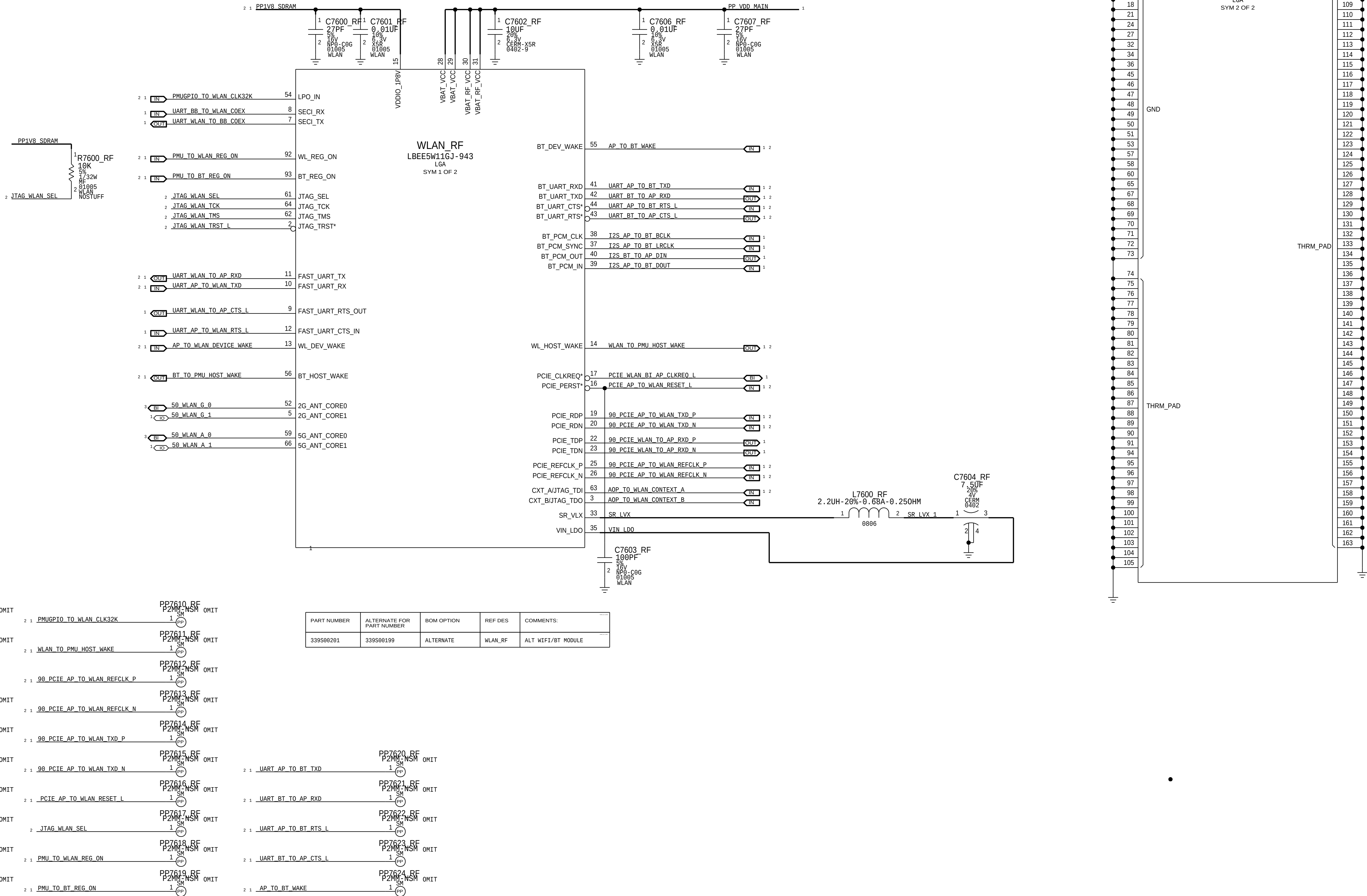
ALT UAT1 GND



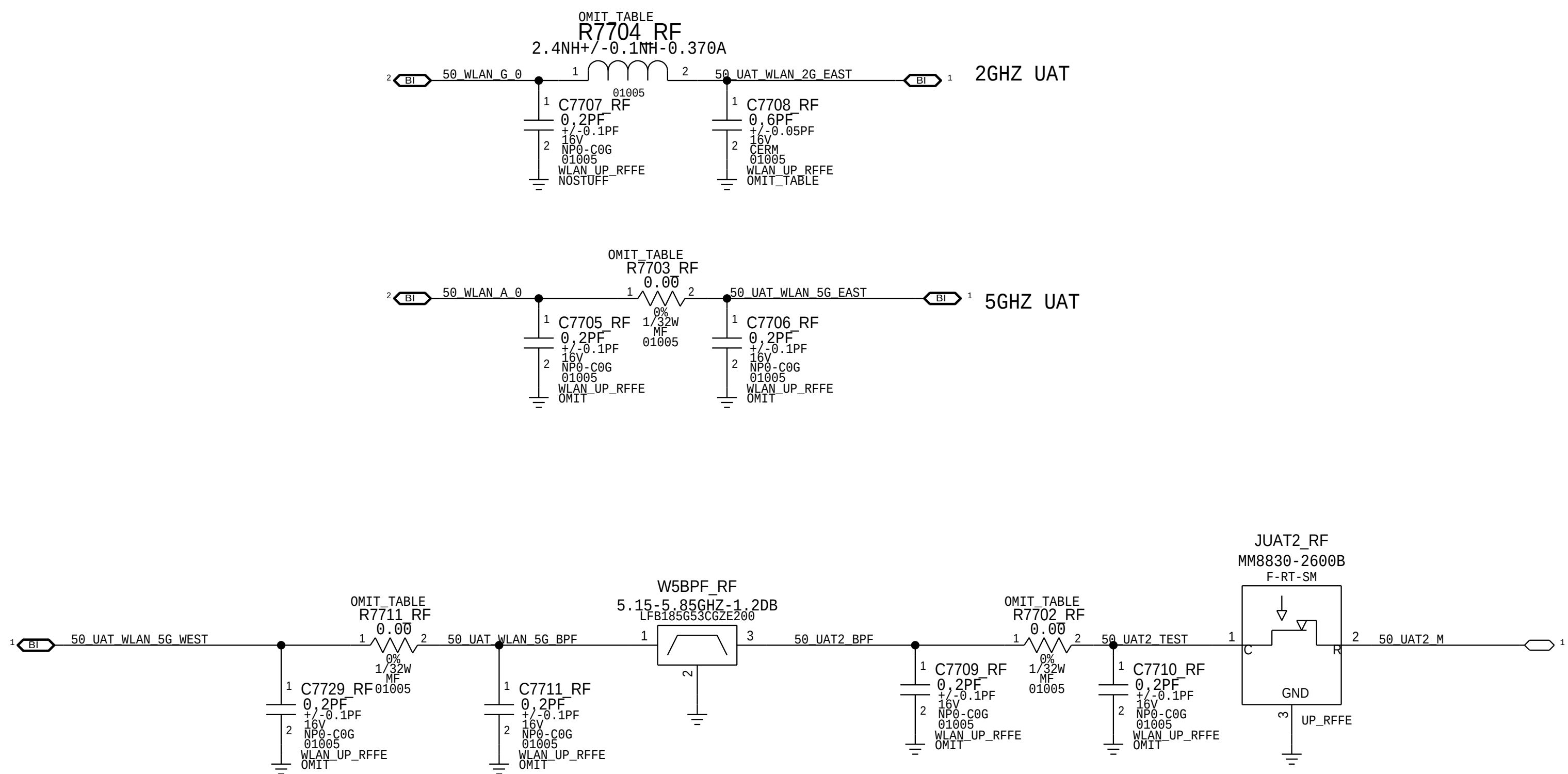
LB/MLB/GNSS/MB/HB STANDOFF



WIFI/BT

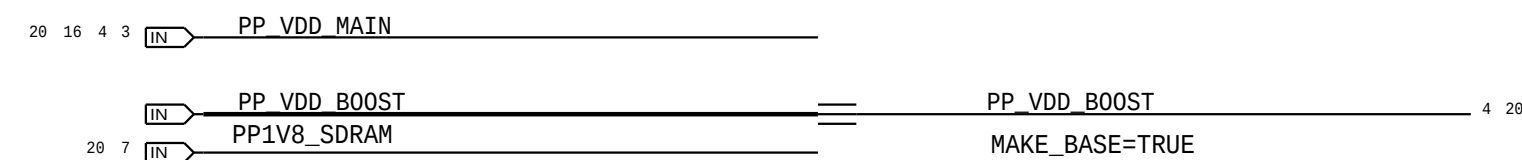


WIFI UPPER ANTENNA FEEDS

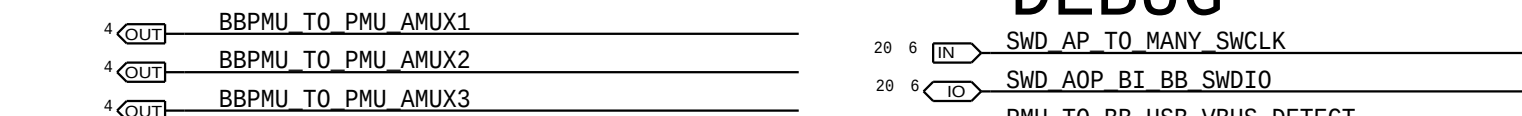


AP CONNECTIONS

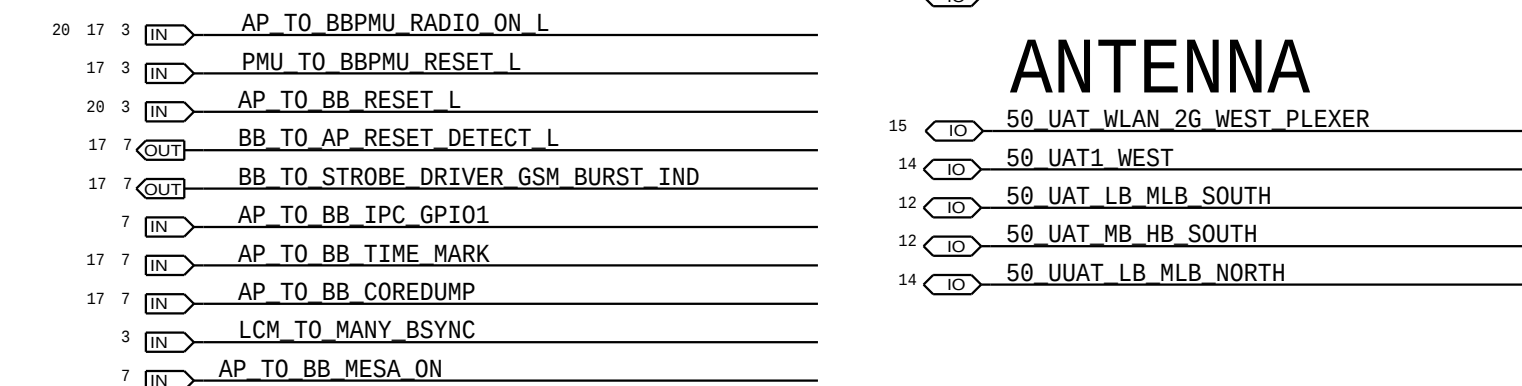
POWER



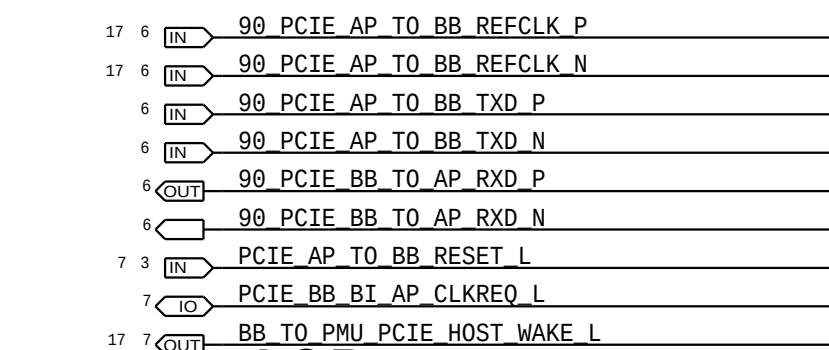
AMUX



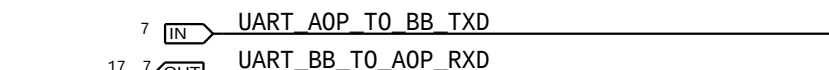
BB CONTROL



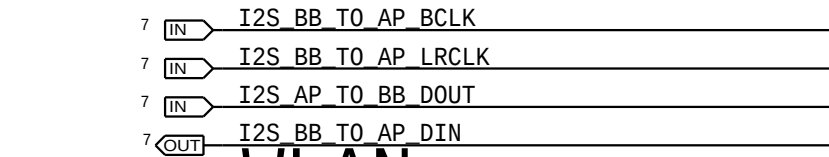
PCIE



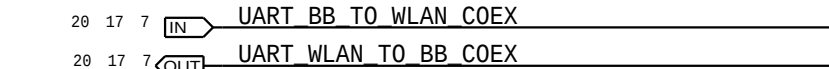
AOP



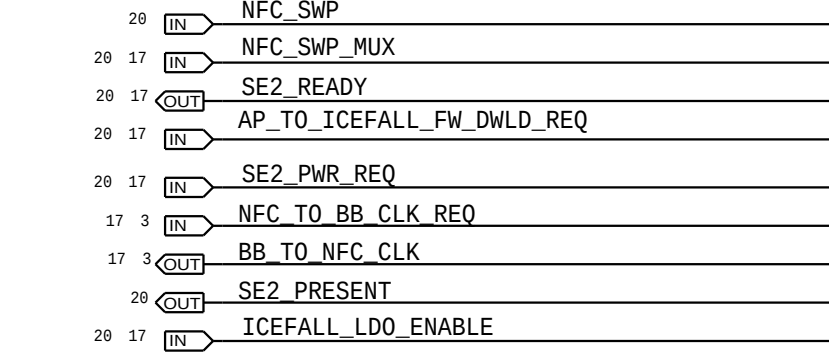
AUDIO



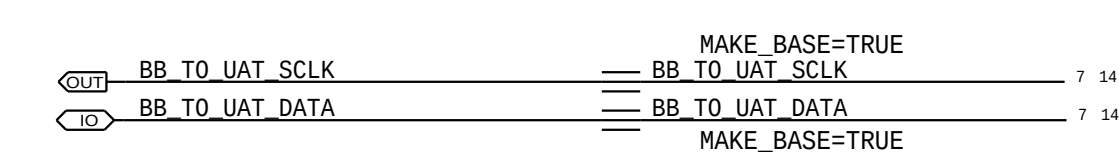
WLAN



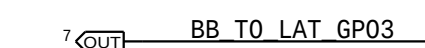
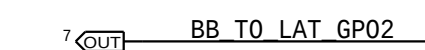
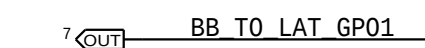
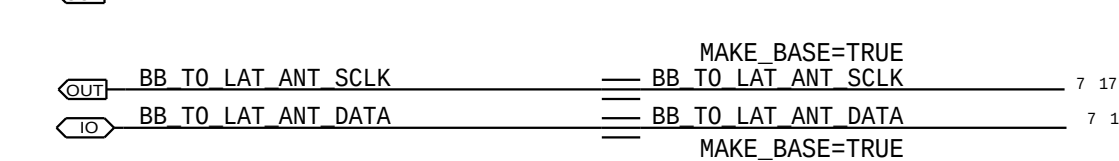
NFC



TUNER



DOCK



62 0F 81



BOM OPTIONS:

LMBRF:

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S00723	1	MLB PAD	MLBPA_RF	CRITICAL	LMBRF
117S00002	1	MLB PAD LAT OUTPUT MATCH	R7105_RF	CRITICAL	LMBRF
152S2042	1	MLB PAD UAT OUTPUT MATCH	R7106_RF	CRITICAL	LMBRF
353S00627	1	MLB LNA	MLBLN_RF	CRITICAL	LMBRF
117S00002	1	MLB LNA OUTPUT MATCH	R6710_RF	CRITICAL	LMBRF
155S00139	1	PENTAPLEXER	PPLXR_RF	CRITICAL	LMBRF
118S0043	1	RES,WF,4.99K OHM,01005	R7506_RF	CRITICAL	LMBRF
155S00193	1	FLTR,DIPLEXER,LB-MB-HB,DPX,SHIELD,0005	UATDI_RF	CRITICAL	LMBRF
155S00158	1	FLTR,DIPLEXER,LB-MB/HB,MIRROR,0005	UPPDI_RF	CRITICAL	LMBRF
337S00284	1	1C,SECURE ELEMENT,BOM20211,MLBGA25	SE2_RF	CRITICAL	LMBRF
138S0739	1	CAP,1UF,0201	C7201_RF	CRITICAL	LMBRF
138S0739	1	CAP,1UF,0201	C7528_RF	CRITICAL	LMBRF
132S0316	1	CAP,0.1UF,01005	C7501_RF	CRITICAL	LMBRF
138S0739	1	CAP,1UF,0201	C7523_RF	CRITICAL	LMBRF
138S0739	1	CAP,1UF,0201	C7524_RF	CRITICAL	LMBRF
130S00032	1	CAP,2.2UF,0201	C7529_RF	CRITICAL	LMBRF
130S00032	1	CAP,2.2UF,0201	C7530_RF	CRITICAL	LMBRF
131S0217	1	CAP,100PF,01005	C7531_RF	CRITICAL	LMBRF
118S0627	1	RES,10KOHM,01005	R7512_RF	CRITICAL	LMBRF
353S00026	1	LDO,05A,2X2	SE2LDO_RF	CRITICAL	LMBRF
131S0630	1	CAP,CER,NPO/C0G,27PF,2%,16V,01005	C6345_RF	CRITICAL	LMBRF
131S0341	1	CAP,CER,C0G,3.0PF,+/-0.05,25V,0201,HI-Q	C6348_RF	CRITICAL	LMBRF
152S2006	1	IND,FILM,6.2NH,3%,400MA,UH-Q,0201	L6322_RF	CRITICAL	LMBRF
131S0630	1	CAP,CER,NPO/C0G,27PF,2%,16V,01005	C6315_RF	CRITICAL	LMBRF
152S2006	1	IND,FILM,6.2NH,3%,400MA,UH-Q,0201	L6305_RF	CRITICAL	LMBRF
131S0341	1	CAP,CER,C0G,3.0PF,+/-0.05PF,25V,0201,HI-Q	C6306_RF	CRITICAL	LMBRF
131S0630	1	CAP,CER,NPO/C0G,27PF,2%,16V,01005	C6106_RF	CRITICAL	LMBRF
152S2051	1	IND,1.3NH,1.1A,0201	R6404_RF	CRITICAL	LMBRF
152S2000	1	IND,2.0NH,600MA,0201	R7104_RF	CRITICAL	LMBRF
131S00001	1	CAP,CER,0.1PF,25V,0201	C7119_RF	CRITICAL	LMBRF
118S0724	1	RES,WF,600M,1/20W,0201	R6703_RF	CRITICAL	LMBRF
118S0724	1	RES,WF,600M,1/20W,0201	R6700_RF	CRITICAL	LMBRF
131S0363	1	CAP,CER,C0G,HQ,0.6PF,+/-0.05PF,25V,0201	C7123_RF	CRITICAL	LMBRF
152S2002	1	IND,2.7NH,+/-0.1NH,600MA,0201,UH-Q	R6605_RF	CRITICAL	LMBRF
131S0337	1	CAP,1.5PF,+/-0.05PF,25V,0201,HQ	C6610_RF	CRITICAL	LMBRF
118S0724	1	RES,WF,600M,1/20W,0201	R6006_RF	CRITICAL	LMBRF
131S0275	1	CAP,CER,C0G,0.3PF,+/-0.1PF,25V,0201,HQ	C6416_RF	CRITICAL	LMBRF
118S0008	1	RES,WF,1K OHM,1%,1/32W,01005	R5911_RF	CRITICAL	LMBRF
152S1356	1	IND,10NH,3%,250MA,HI-Q,0201	C6613_RF	CRITICAL	LMBRF

NOLMBRF:

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
131S0444	1	DCS/PCS RX FILTER MATCH	R6301_RF	CRITICAL	NOLMBRF
152S2020	1	DCS/PCS RX FILTER MATCH	L6320_RF	CRITICAL	NOLMBRF
155S00149	1	DCS/PCS RX FILTER	GSMRX_RF	CRITICAL	NOLMBRF
152S2005	1	DCS/PCS RX MATCH (DCS)	L6310_RF	CRITICAL	NOLMBRF
131S0319	1	DCS/PCS RX MATCH (DCS)	C6332_RF	CRITICAL	NOLMBRF
131S0630	1	DCS/PCS RX MATCH (DCS)	C6340_RF	CRITICAL	NOLMBRF
152S2005	1	DCS/PCS RX MATCH (DCS)	L6319_RF	CRITICAL	NOLMBRF
131S0305	1	DCS/PCS RX MATCH (DCS)	C6333_RF	CRITICAL	NOLMBRF
131S0630	1	DCS/PCS RX MATCH (DCS)	C6341_RF	CRITICAL	NOLMBRF
155S00163	1	QUADPLEXER	PPLXR_RF	CRITICAL	NOLMBRF
155S00199	1	FLTR,DPX,PASS THRU,LB-MB-HB,SHLD,0005	UATDI_RF	CRITICAL	NOLMBRF
155S00234	1	FLTR,DPX,PASS THRU,LB-MB-HB,UNSHLD,0005	UPPDI_RF	CRITICAL	NOLMBRF
118S0724	1	RES,WF,600M,1/20W,0201	R6404_RF	CRITICAL	NOLMBRF
152S2044	1	IND,2.2NH,+/-0.1NH,600MA,0201	R7104_RF	CRITICAL	NOLMBRF
152S00157	1	1.2NH,+/-0.05NH,1.1A,UH-Q,0201	R6703_RF	CRITICAL	NOLMBRF
131S0549	1	10PF,0201,25V	R6700_RF	CRITICAL	NOLMBRF
131S0445	1	CAP,CER,12PF,5%,25V,0201,HI-Q	R6606_RF	CRITICAL	NOLMBRF

D10 SPECIFIC:

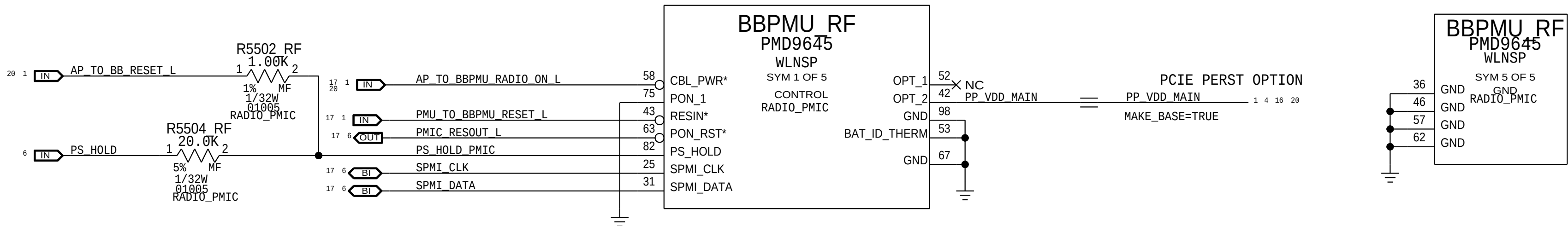
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
131S0278	1	CAP,CER,C0G,1PF,+/-0.1PF,25V,0201,HI-Q	C7120_RF	CRITICAL	D10

D11 SPECIFIC:

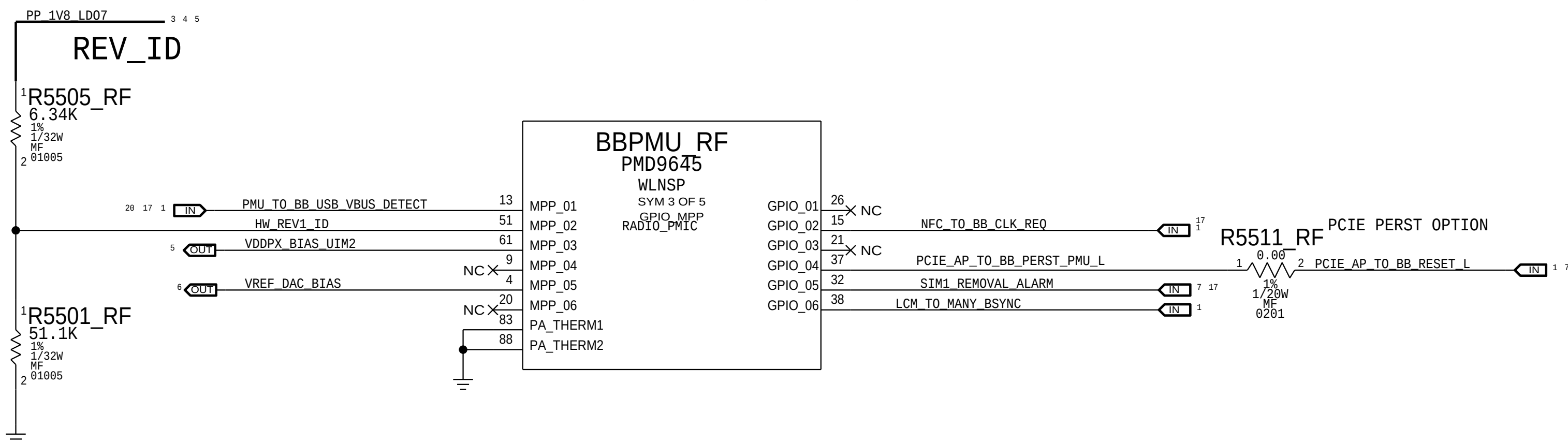
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
131S0425	1	CAP,CER,C0G,0.5PF,+/-0.05PF,25V,0201,HI-Q	C7120_RF	CRITICAL	D11

PMU: CONTROL AND CLOCKS

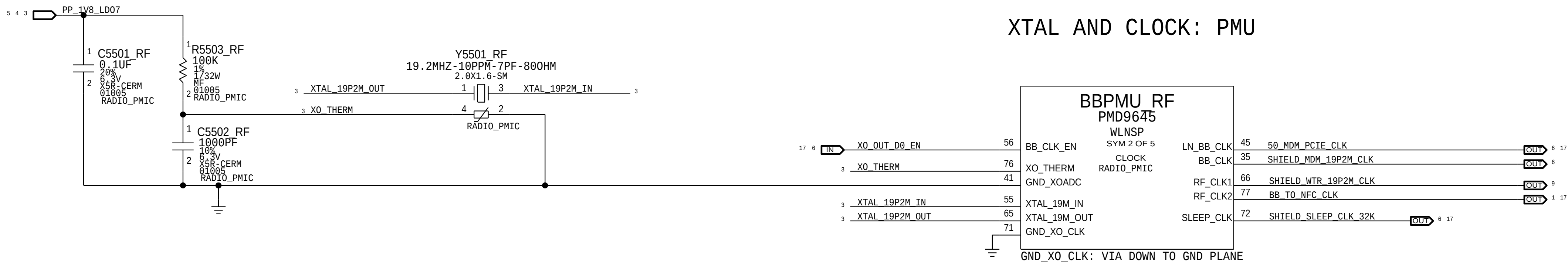
RESET AND CONTROL: PMU



MPPS AND GPIOs: PMU

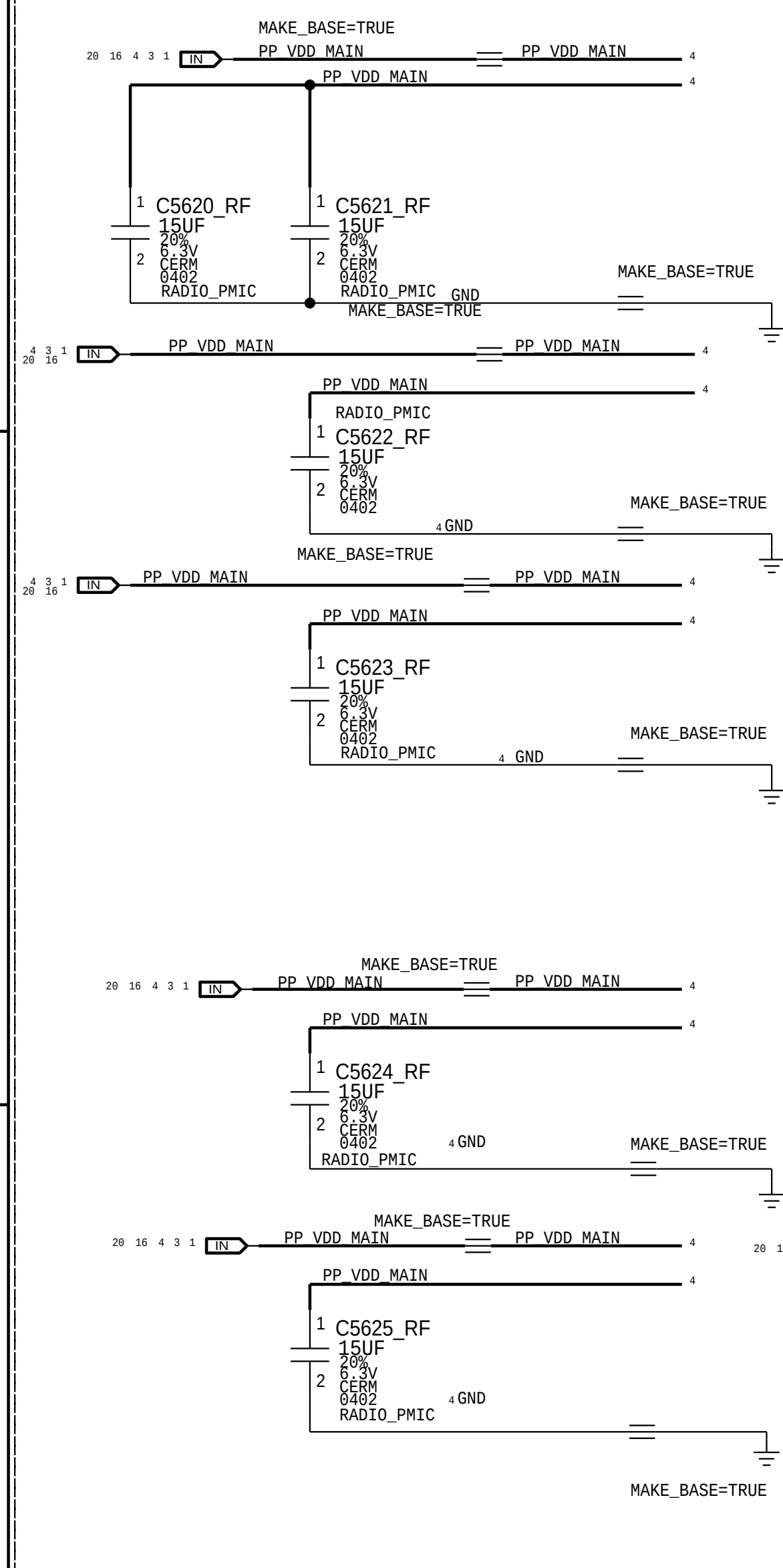


XTAL AND CLOCK: PMU

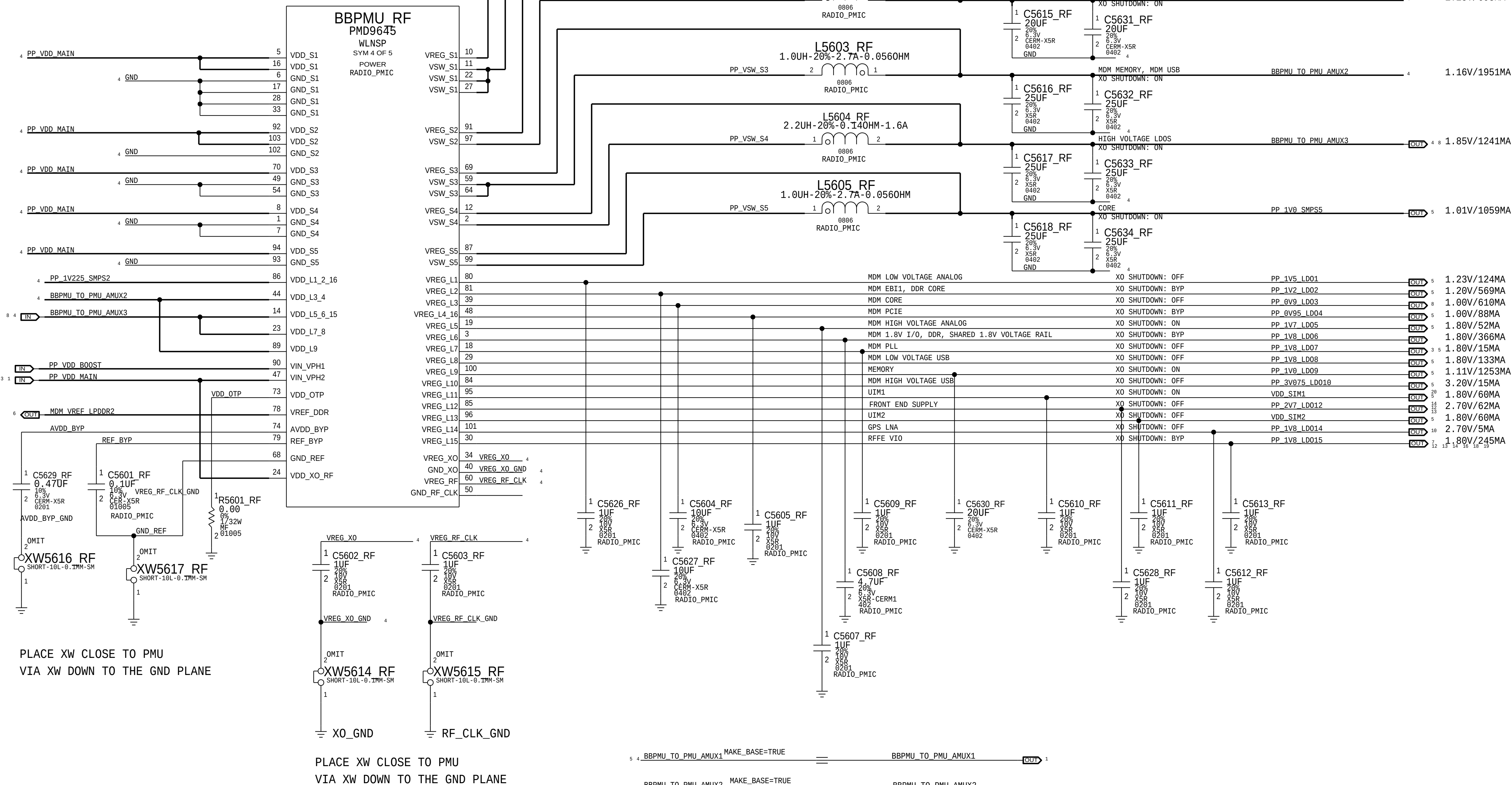
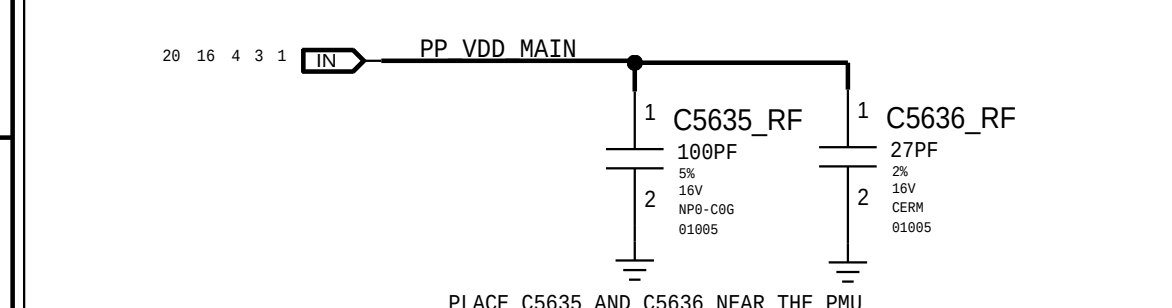


HW_REV_ID	R5505	R5501	REVISION
0.10V	887K	51.1K	DEV1
0.20V	422K	51.1K	DEV2
0.30V	255K	51.1K	DEV 2.1
0.40V	180K	51.1K	DEV 3
0.50V	124K	51.1K	T181
0.60V	102K	51.1K	PP/P1
0.70V	82.5K	51.1K	DEV4
0.80V	63.4K	51.1K	P2
0.90V	51.1K	51.1K	DEV5
1.00V	51.1K	63.4K	EVT
1.10V	51.1K	82.5K	EVT DOE
1.20V	50.0K	100K	EVT ALT/CARBON
1.30V	39.0K	100K	DEV6
1.40V	14.7K	51.1K	CARRIER
1.50V	40.2K	200K	DVT
1.60V	6.34K	51.1K	PVT

SWITCHERS BULK CAPS



DESENSE CAPS



BASEBAND: POWER

D

C

B

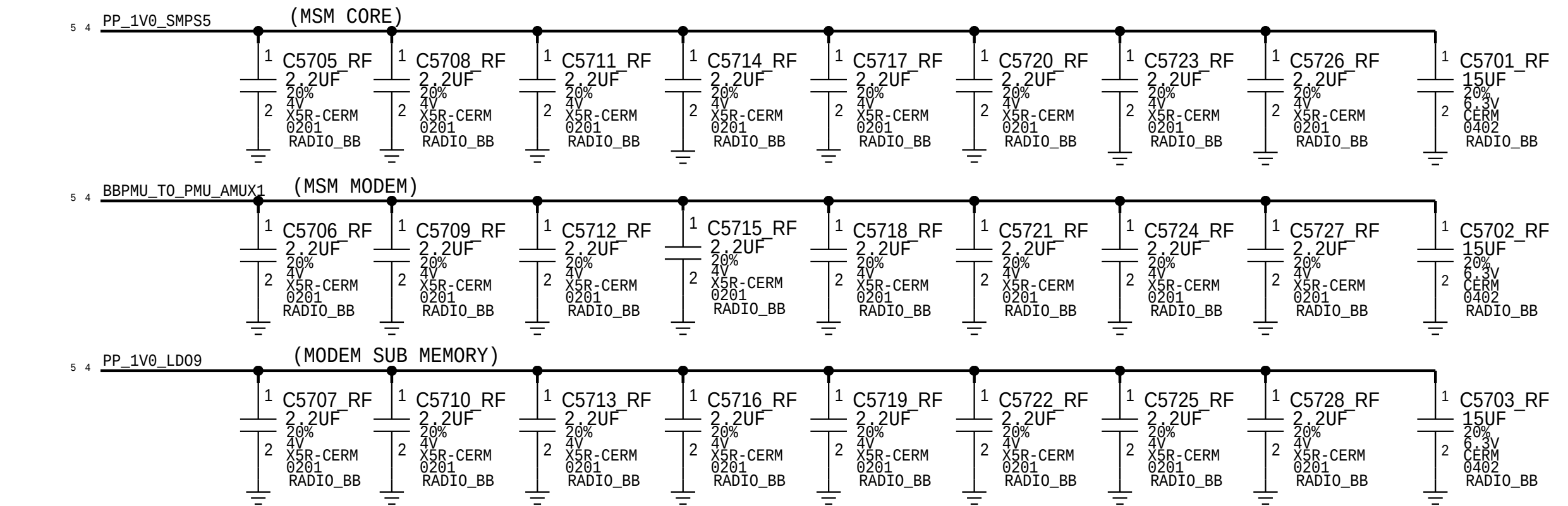
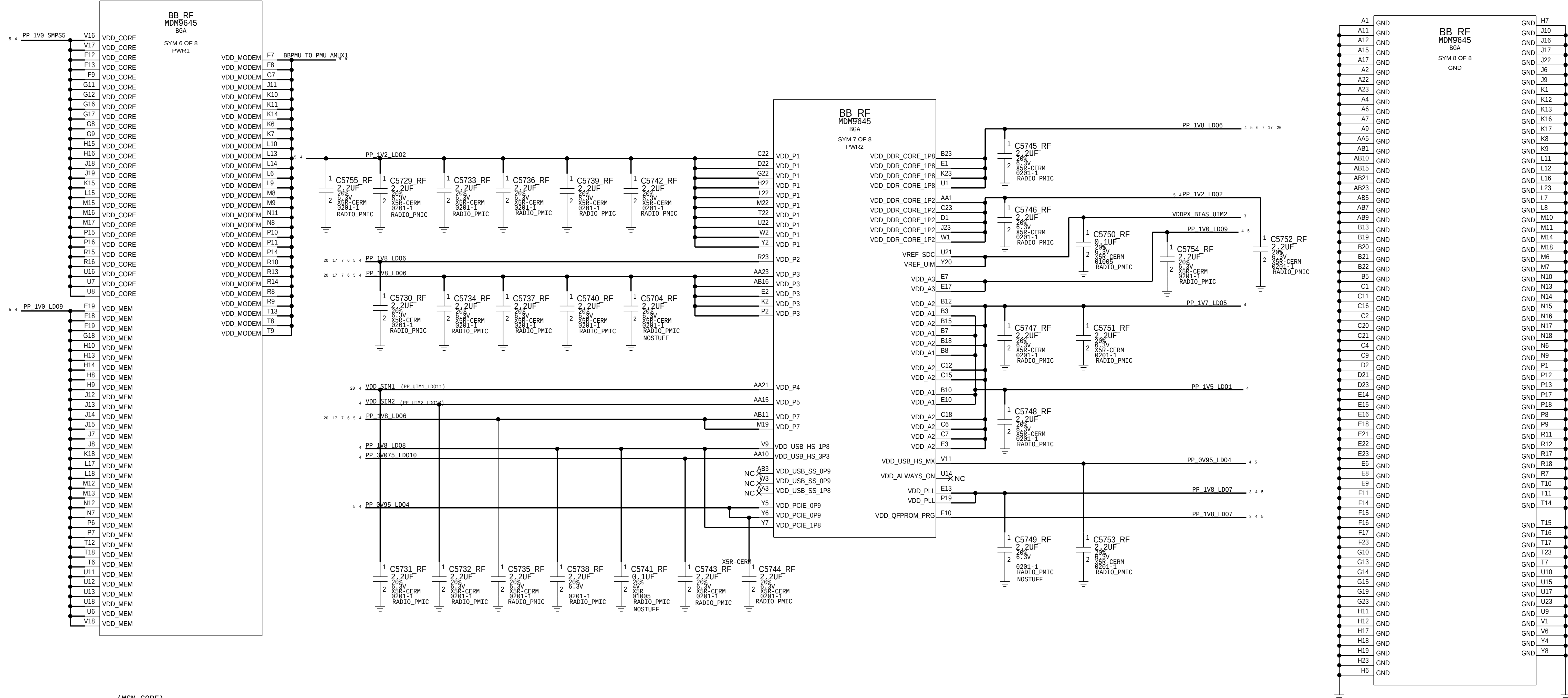
A

D

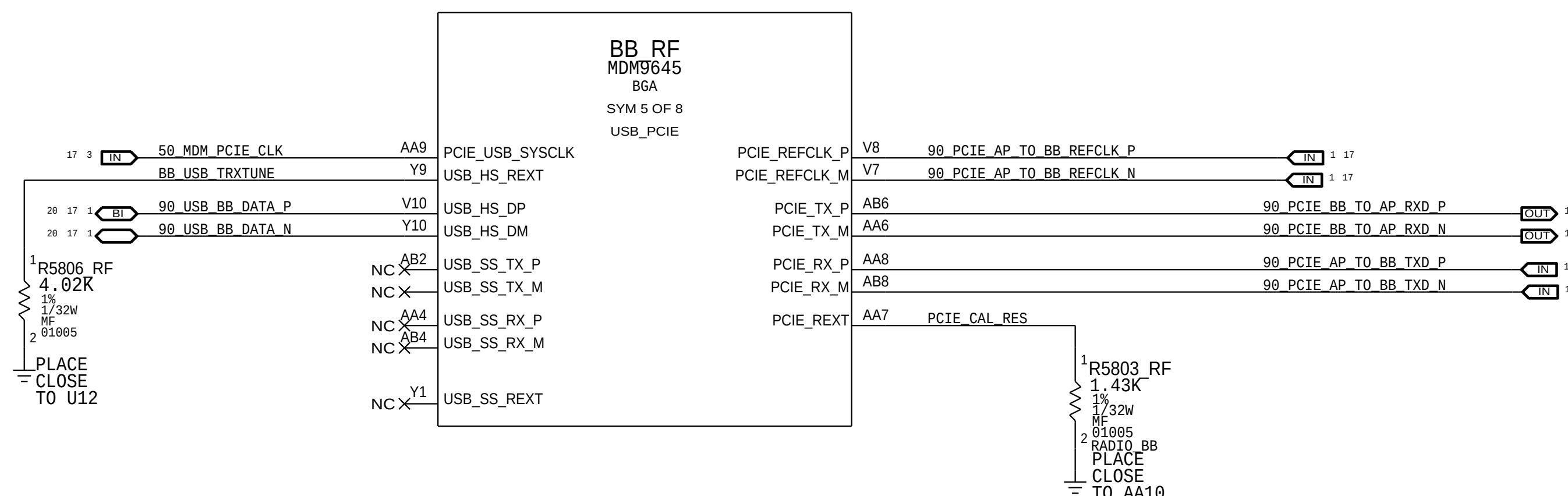
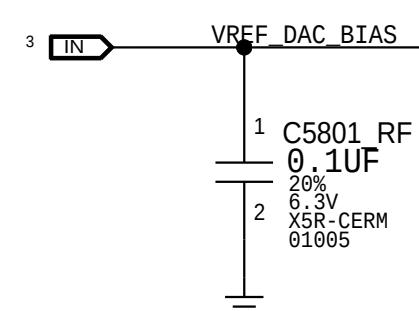
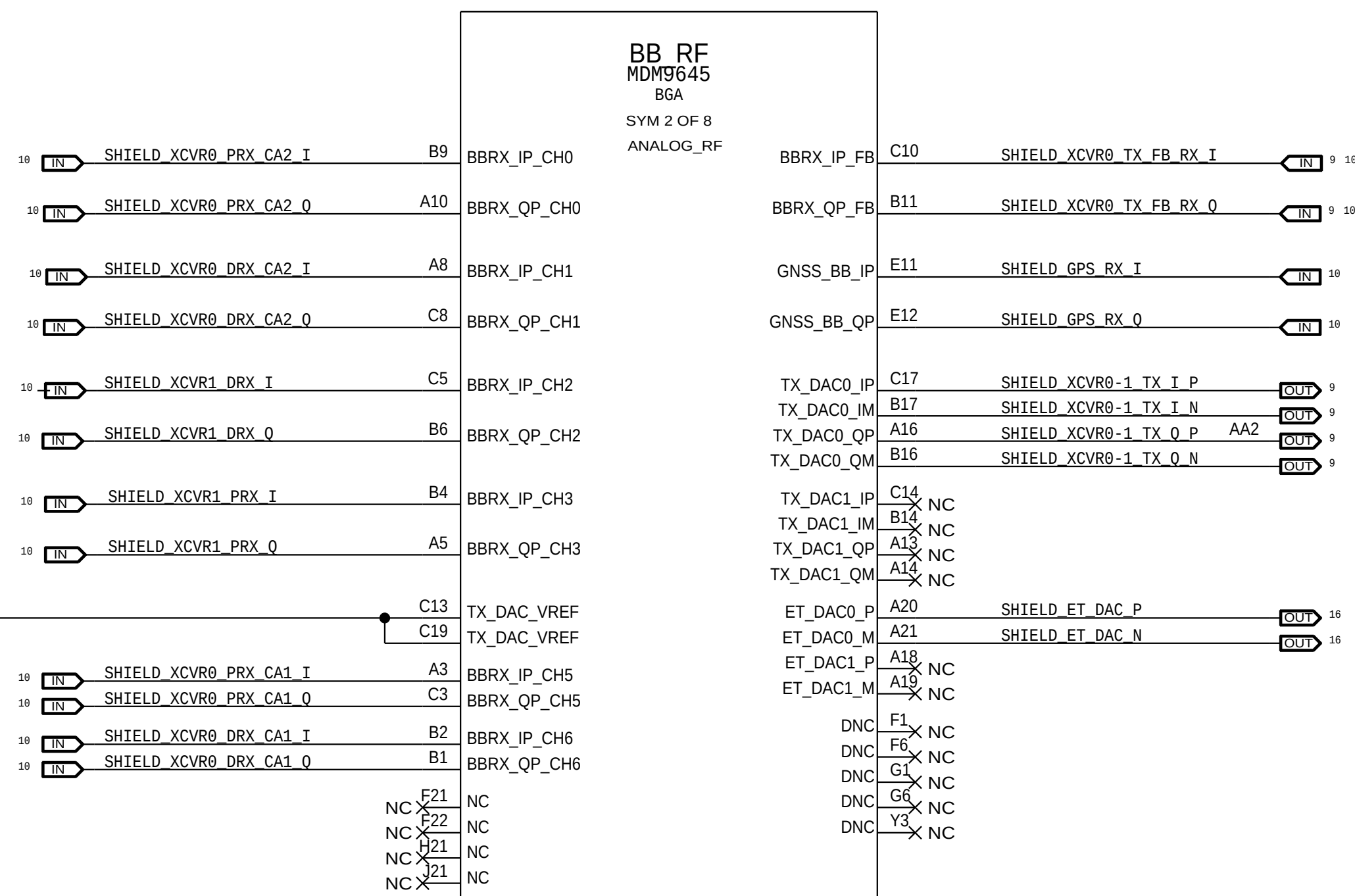
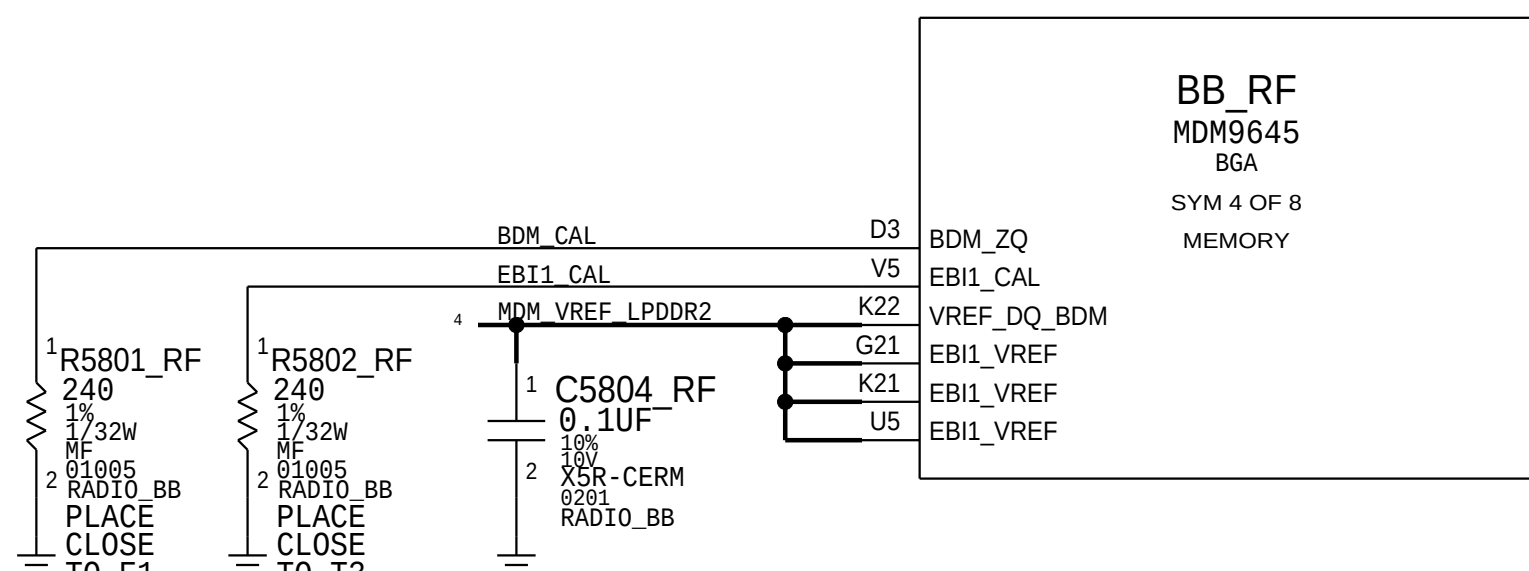
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B

A



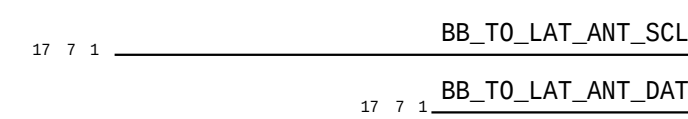
67 0F 81



1

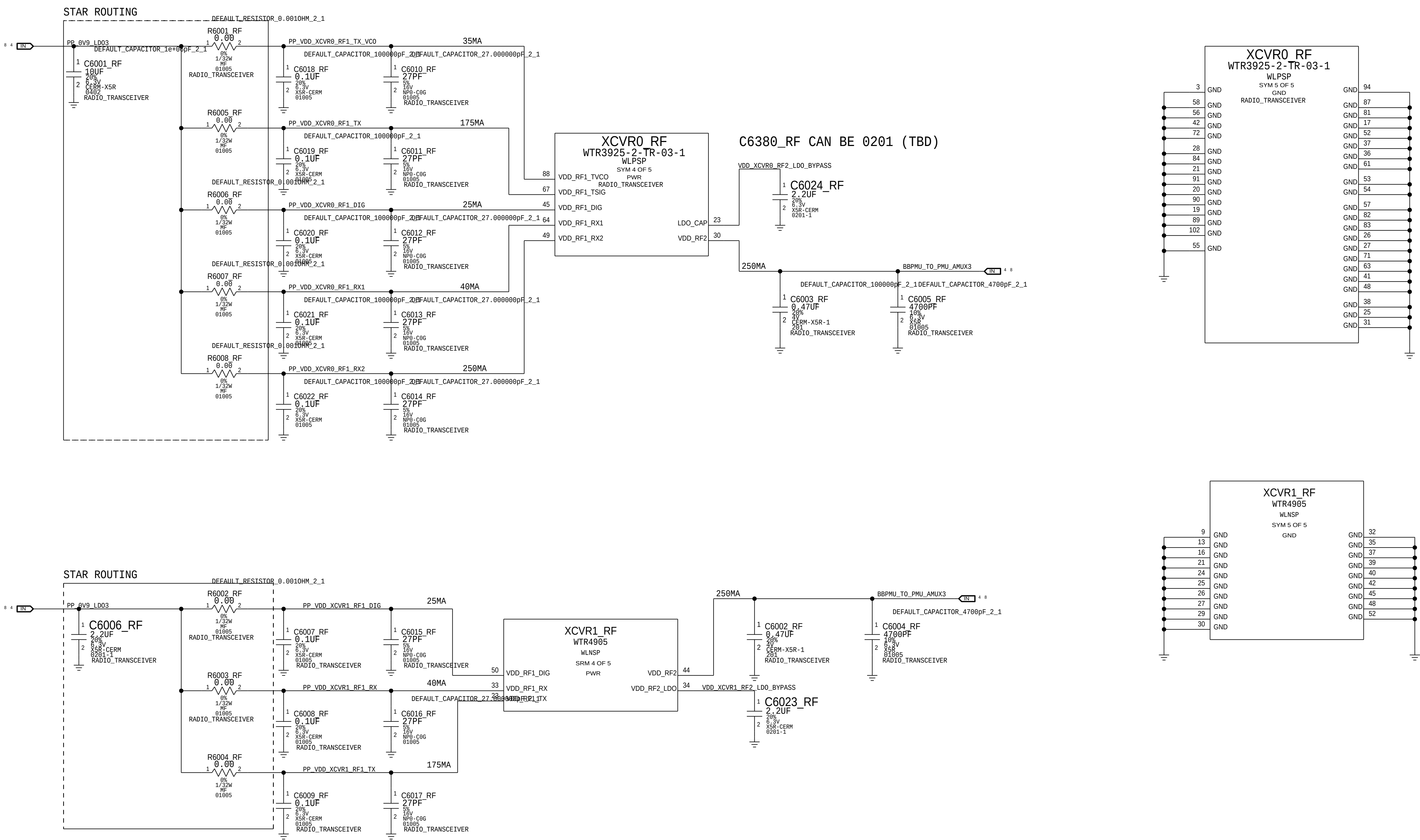


RFFE7 LB PA, COUPLERS

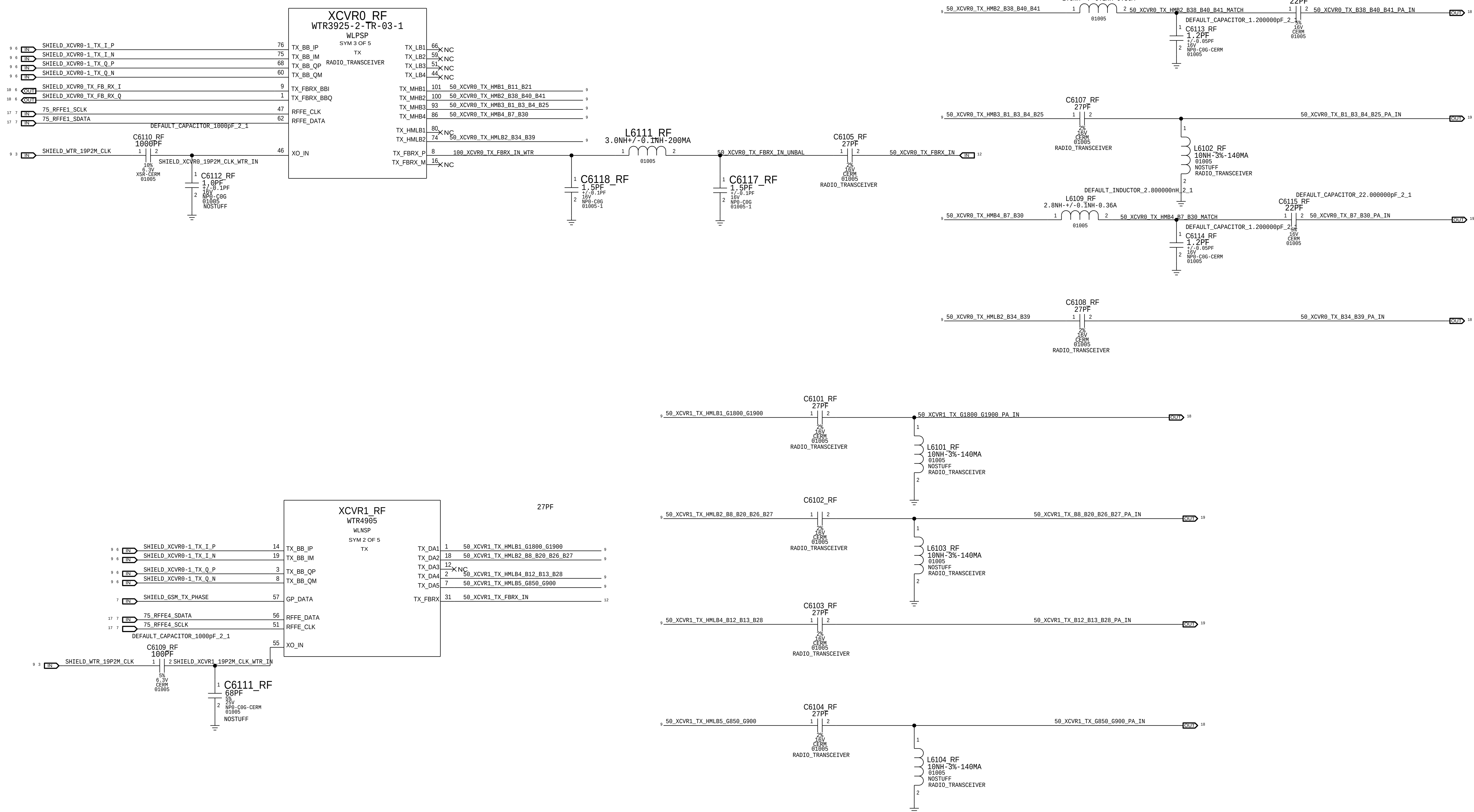


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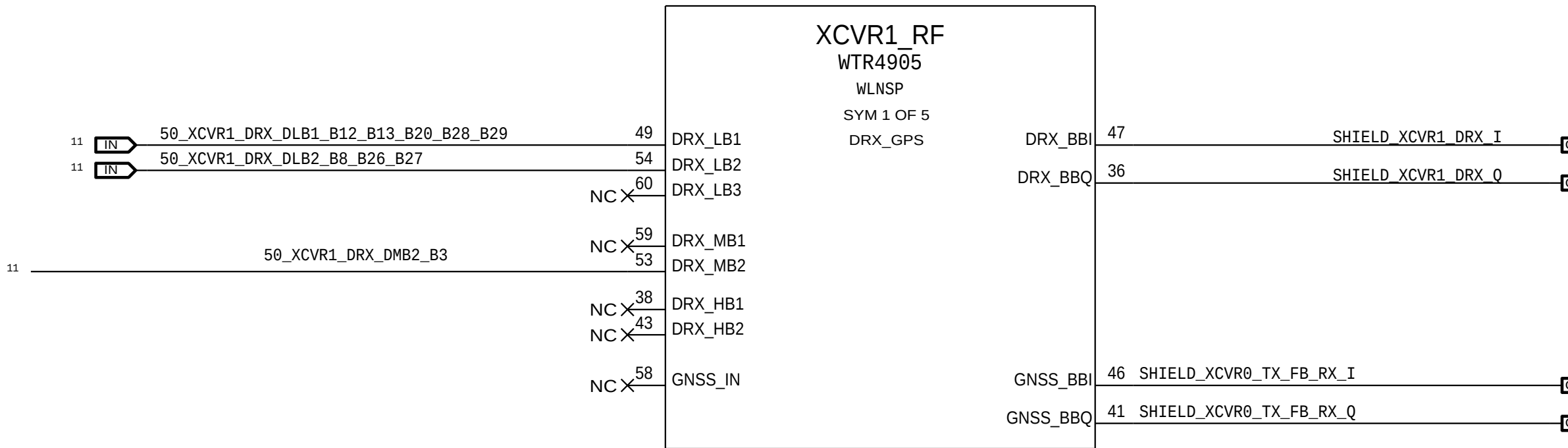
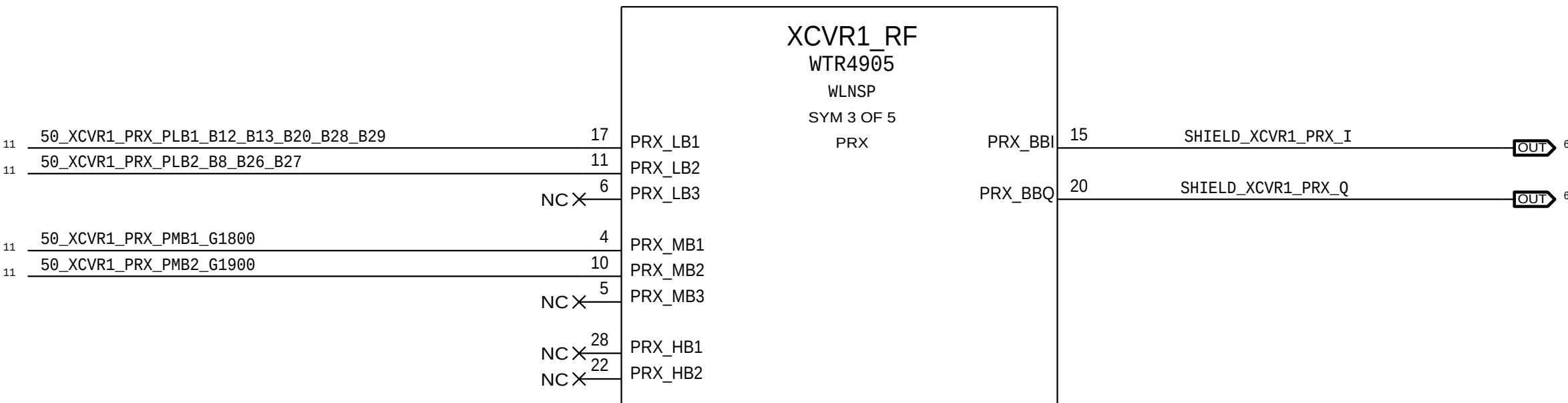
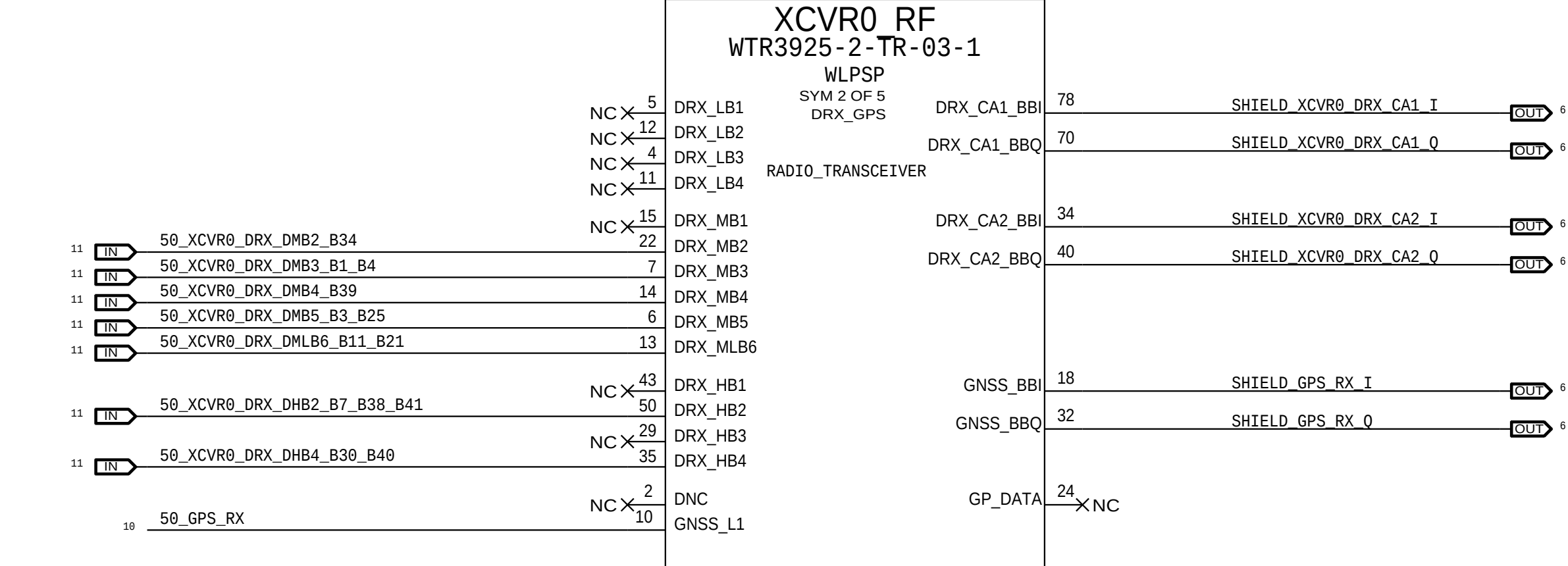
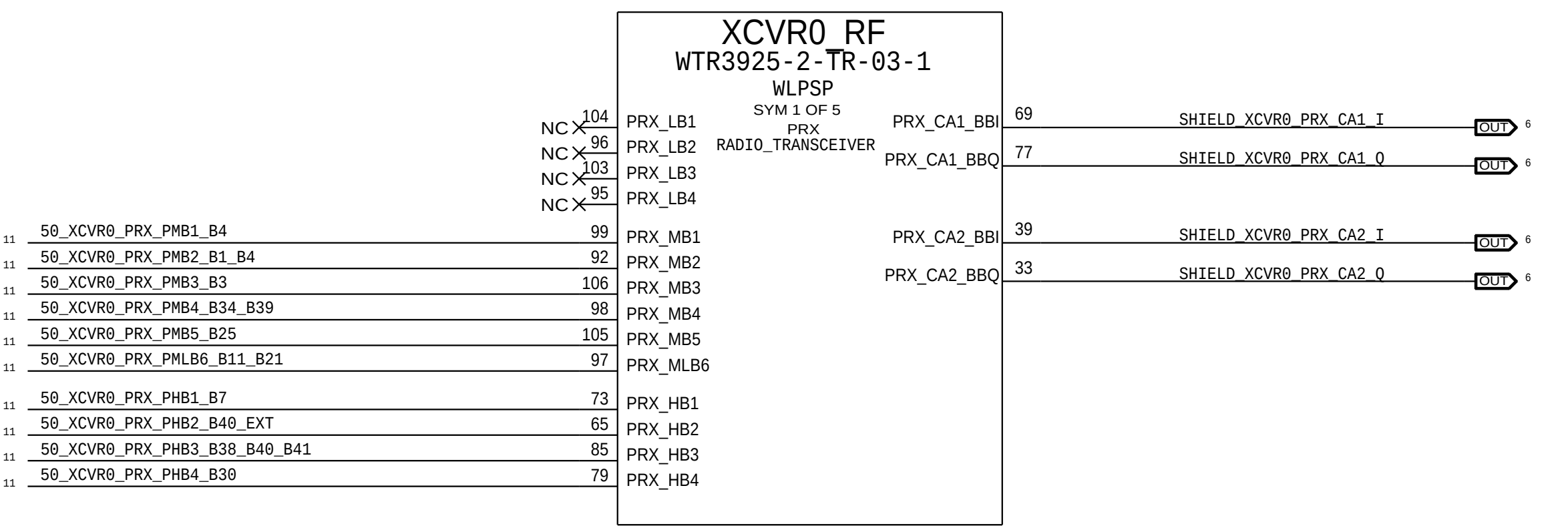
TRANSCEIVER: POWER



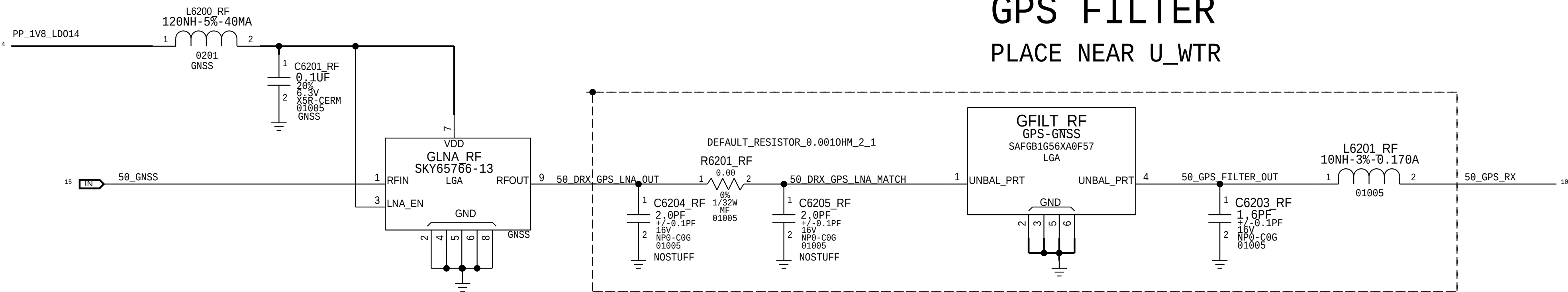
70 0F 81



TRANSCEIVER: PRX, DRX, & GPS PORTS



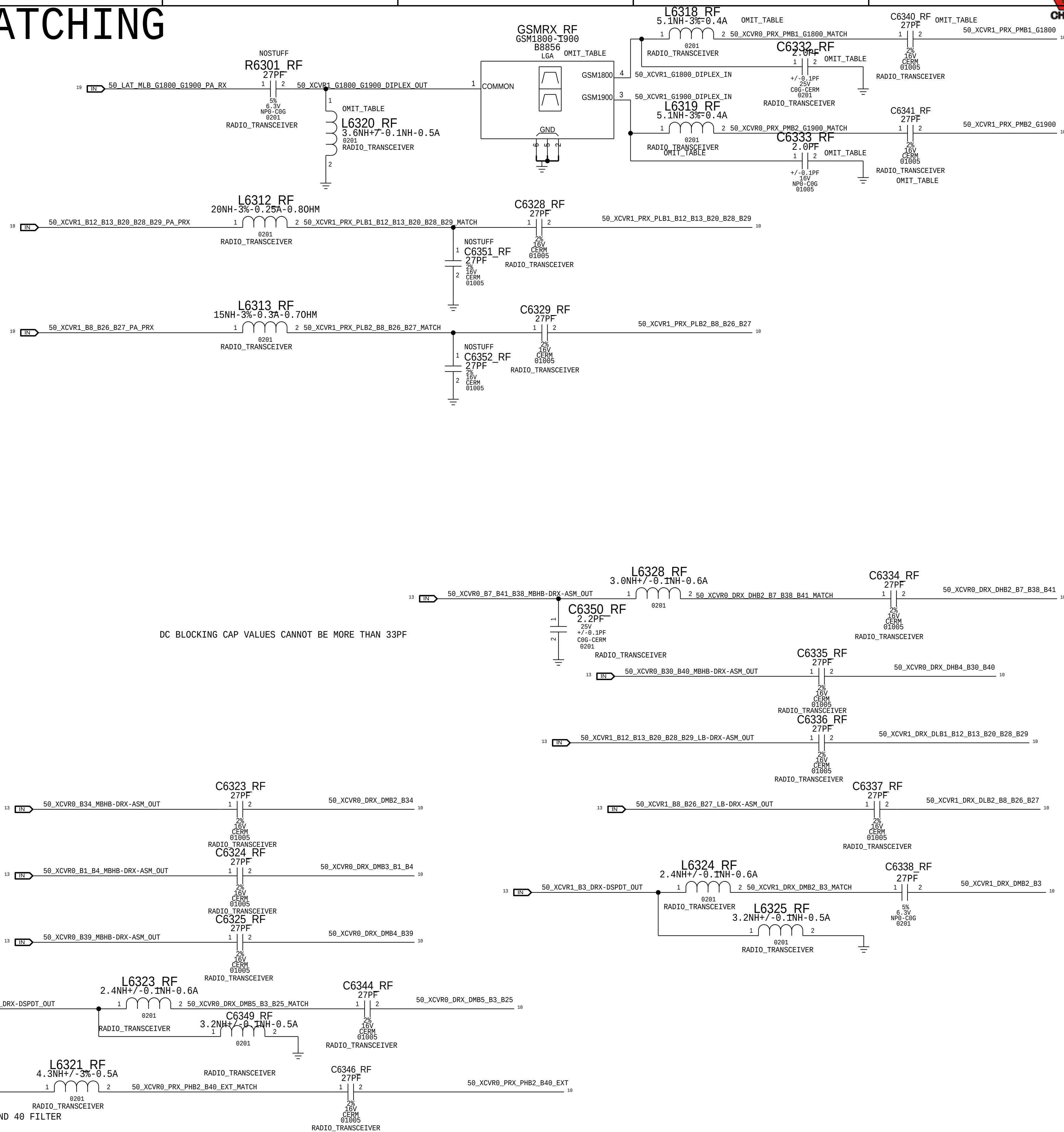
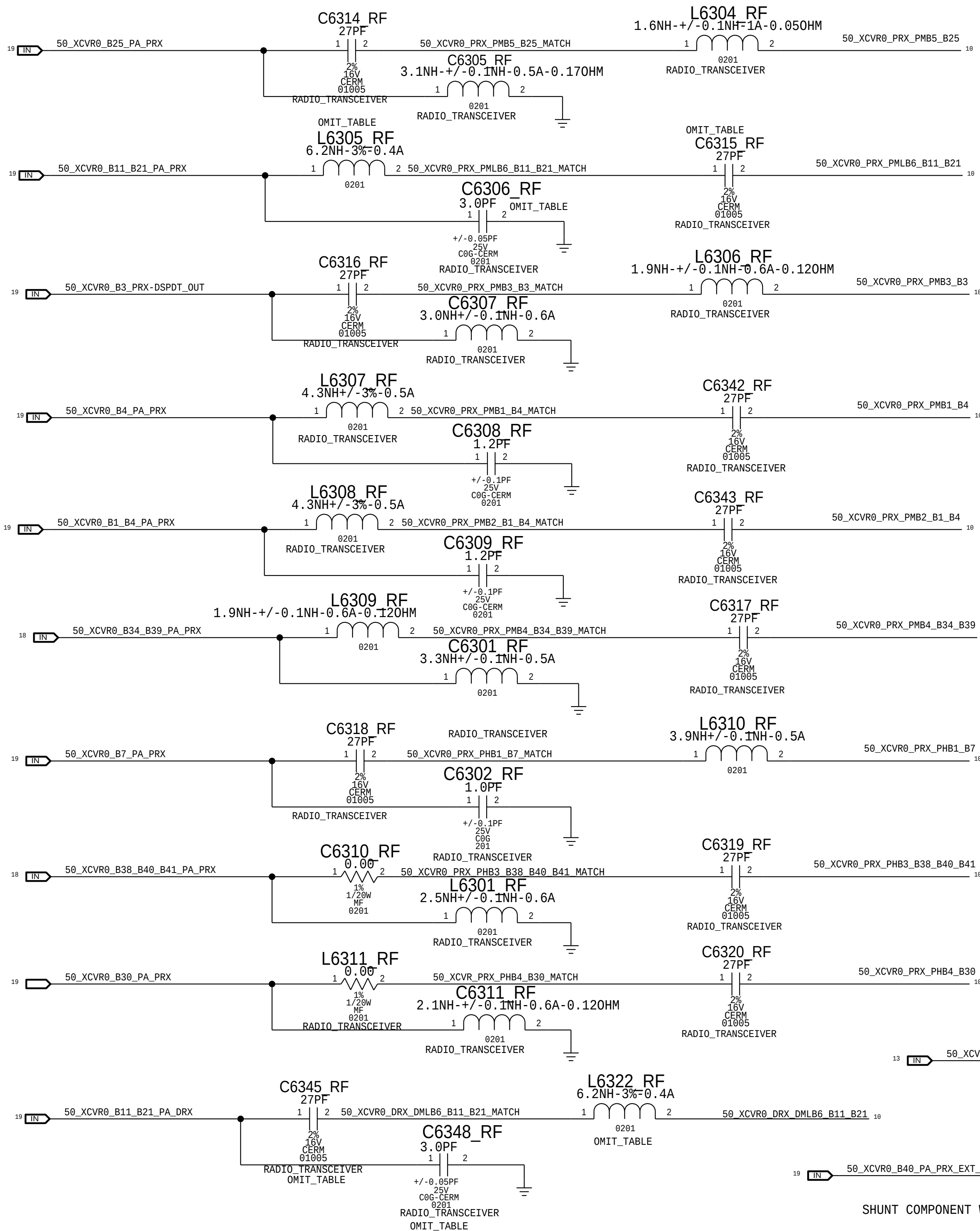
GPS FILTER PLACE NEAR U_WTR





PRIMARY & DIVERSITY RECEIVE MATCHING

DC BLOCKING CAP VALUES CANNOT BE MORE THAN 33PF



CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSE ONLY - NOT A CHANGE REQUEST

LOWER ANTENNA AND COUPLER

D

D

C

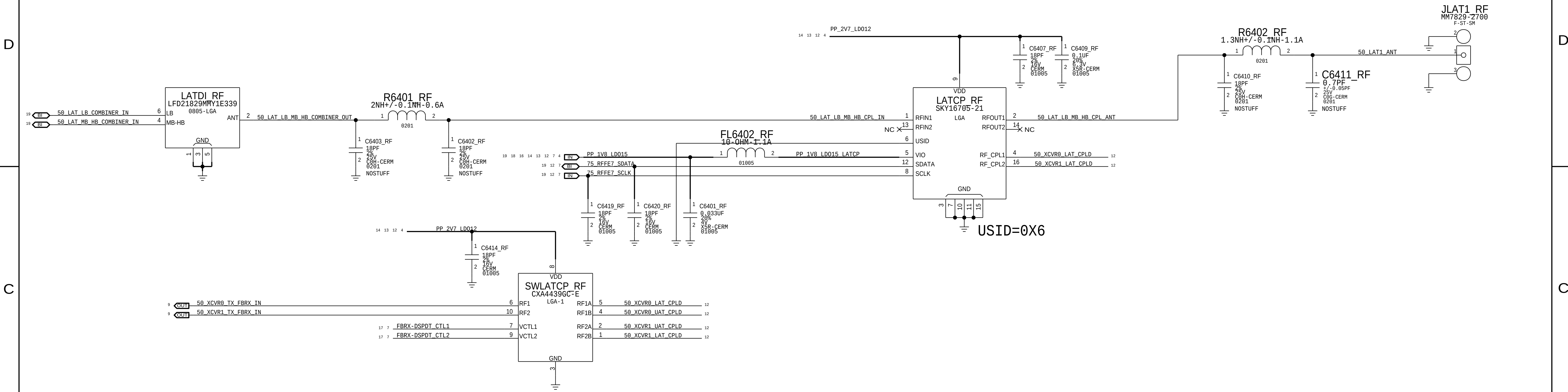
C

B

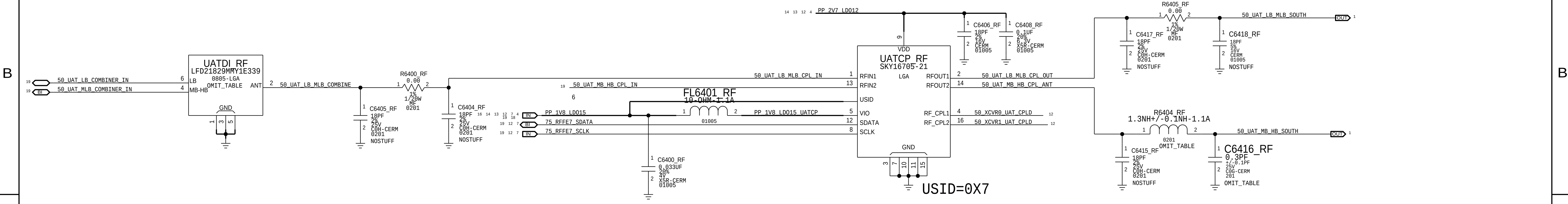
B

A

A

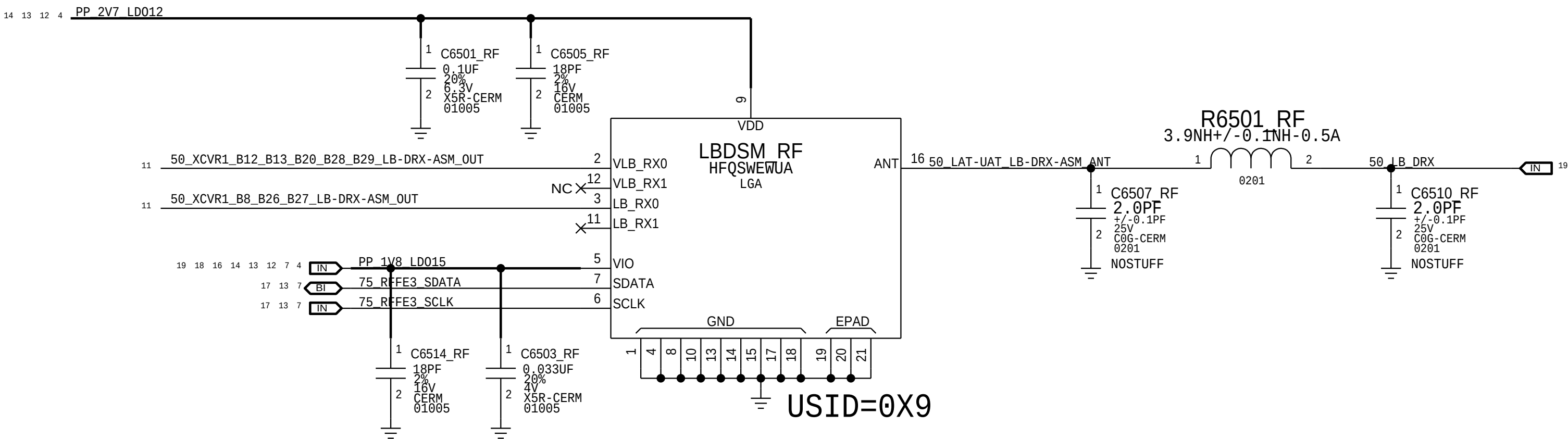


UPPER ANTENNA COUPLER

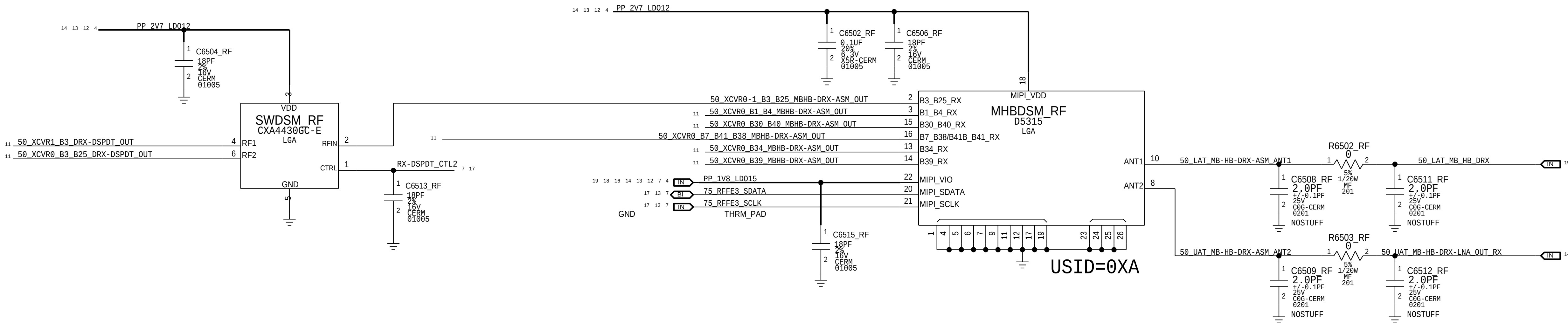


DIVERSITY RECEIVE

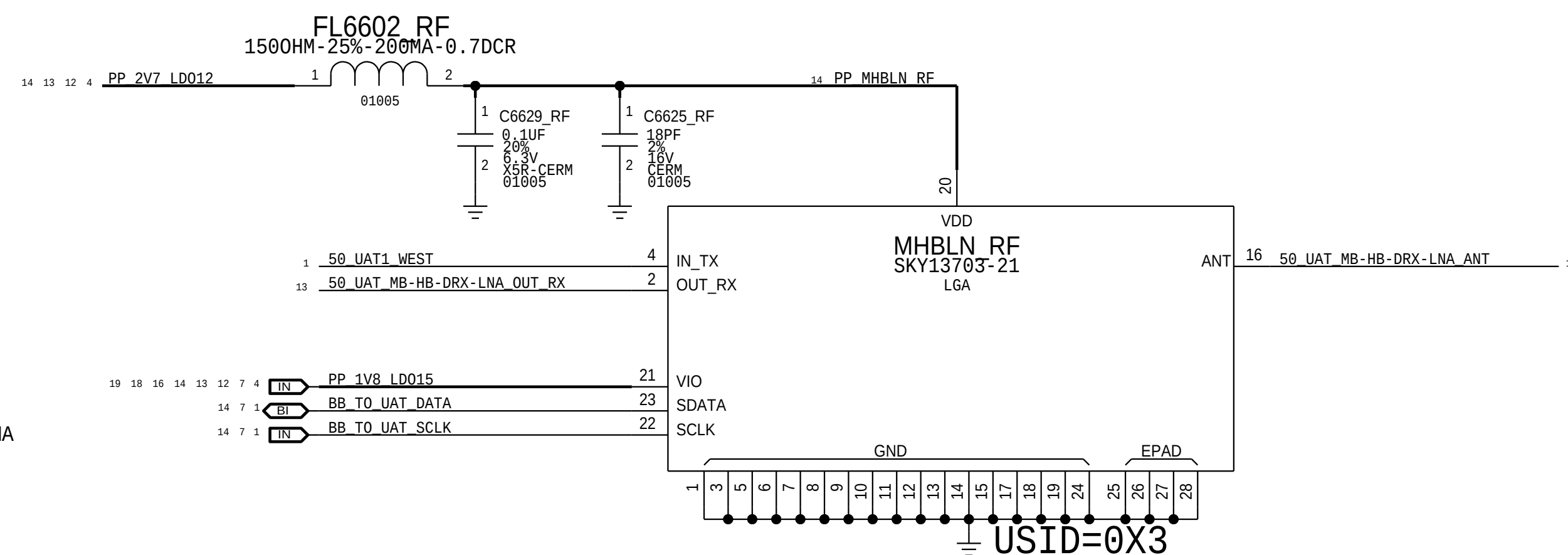
LB DRX ASM



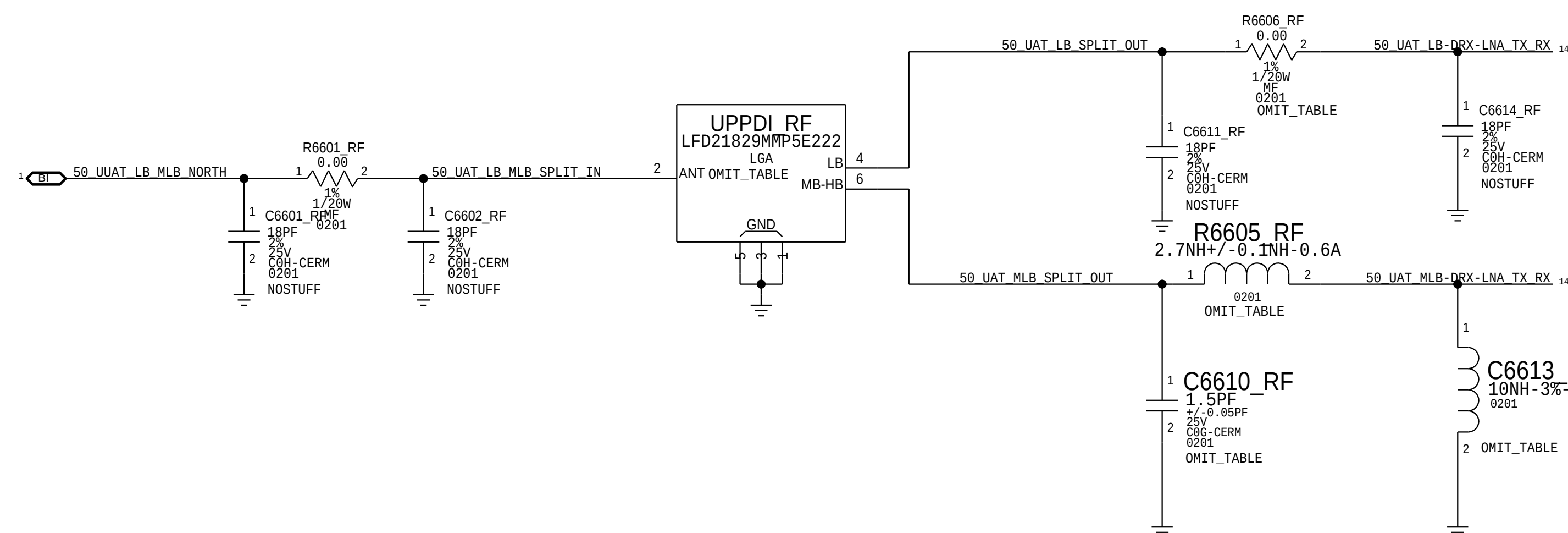
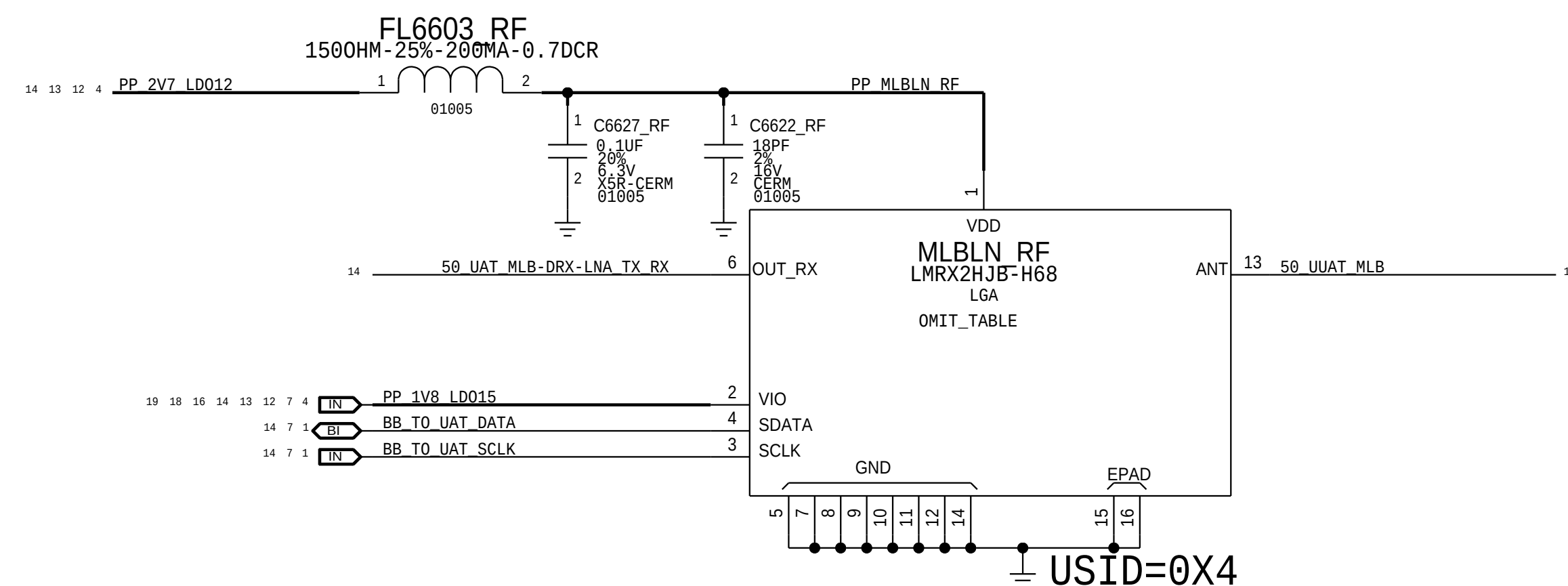
MB HB DRX ASM



LB DRX LNA

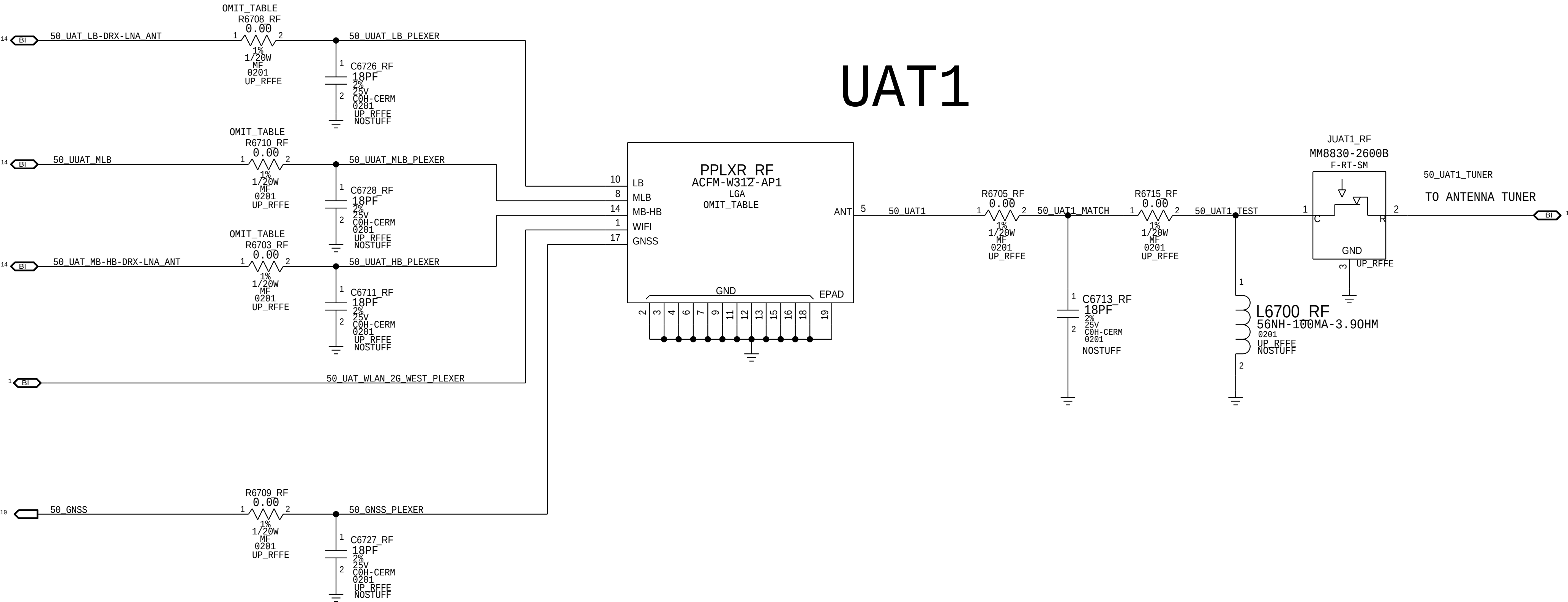


MLB DRX LNA

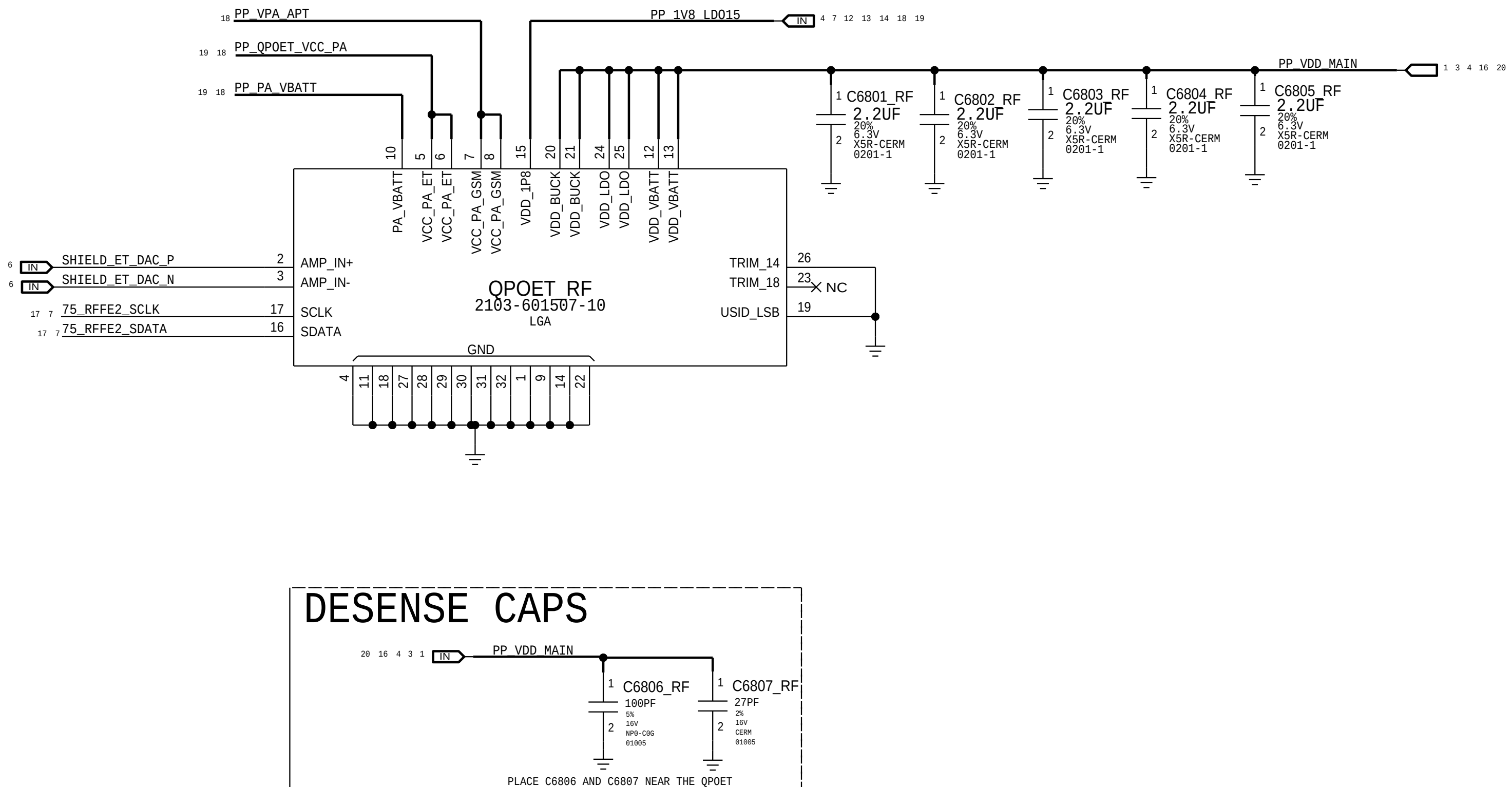


UPPER ANTENNA FEEDS

UAT1

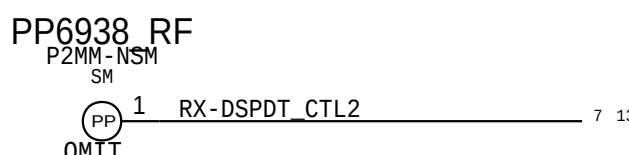
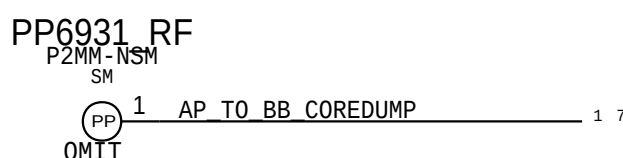
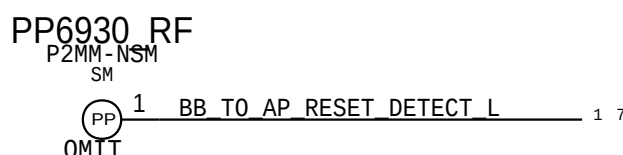
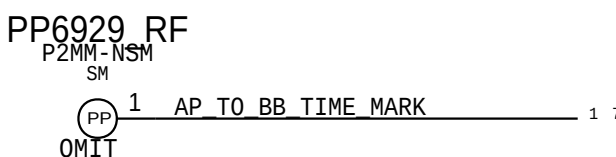
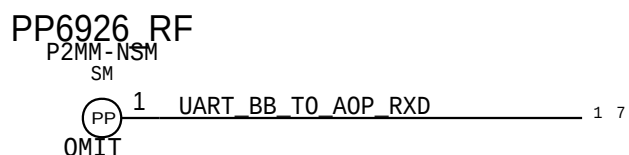
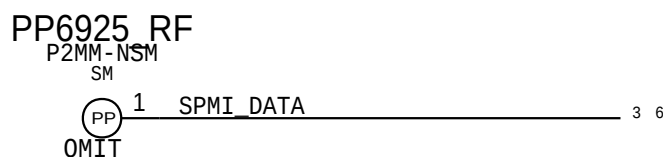
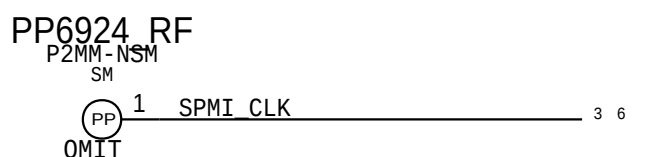
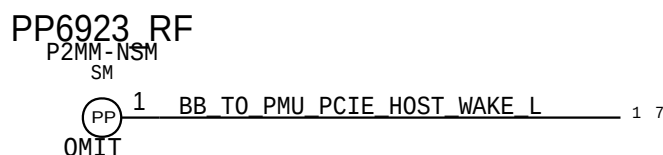
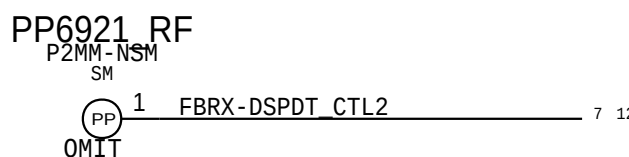
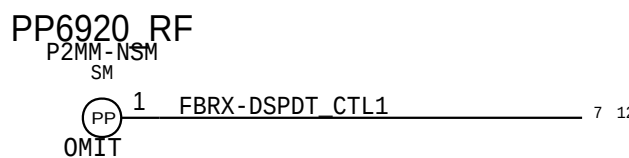
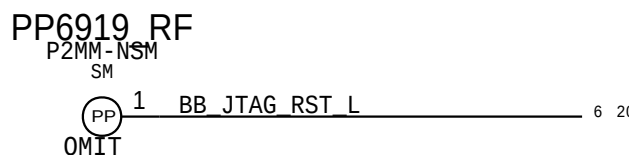
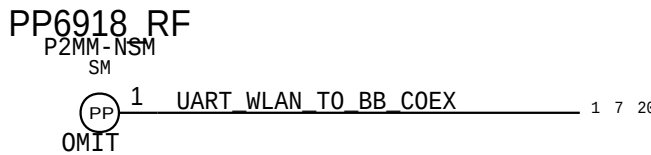
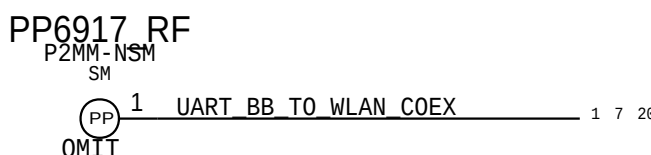
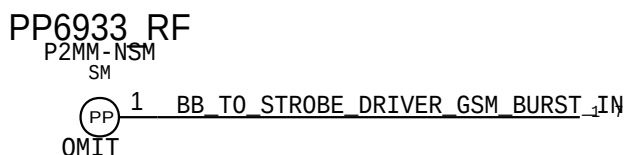
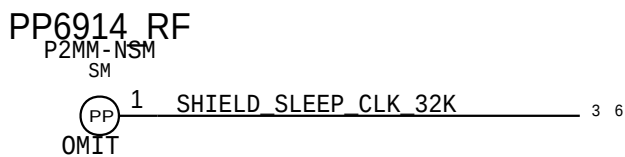
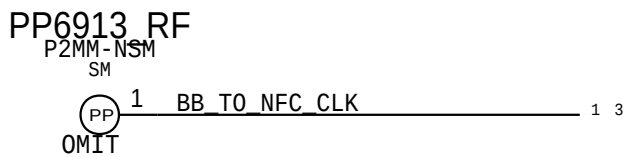
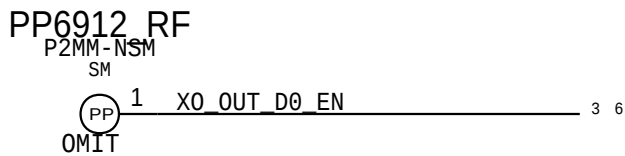
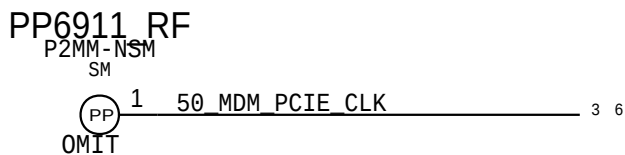
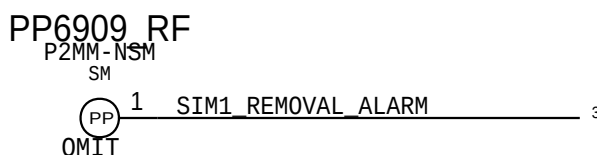
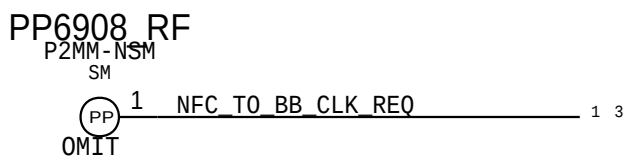
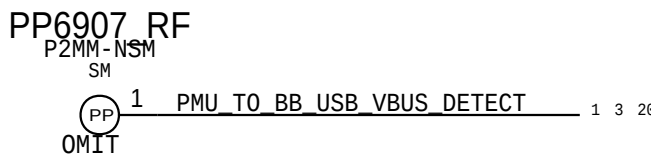
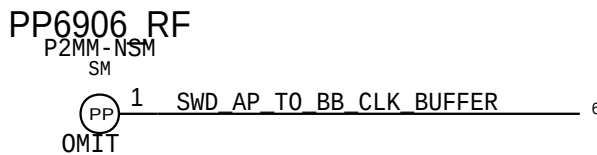
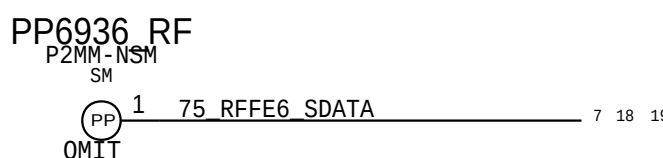
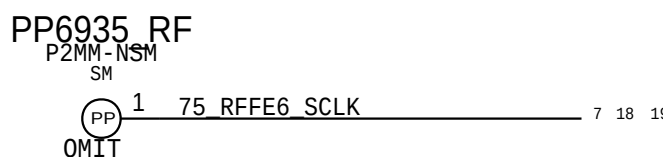
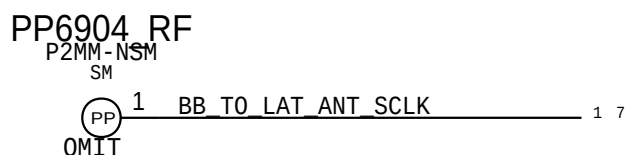
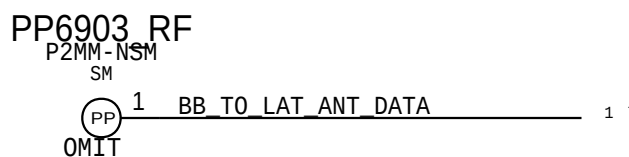
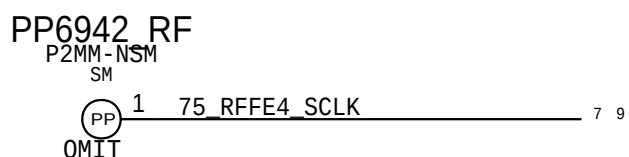
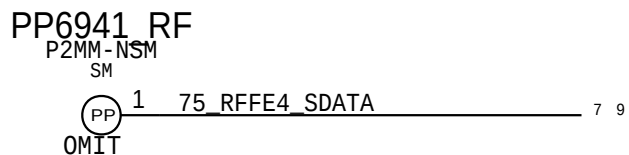
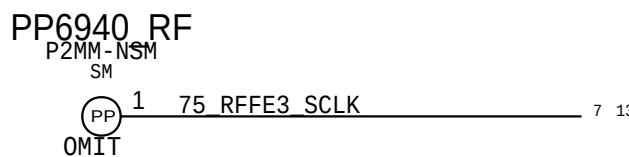
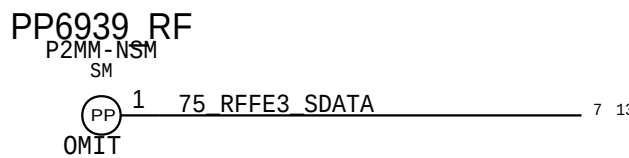
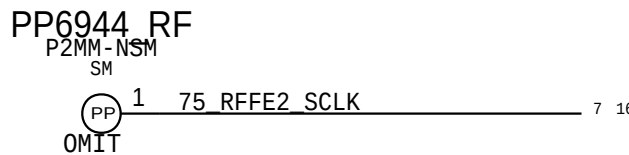
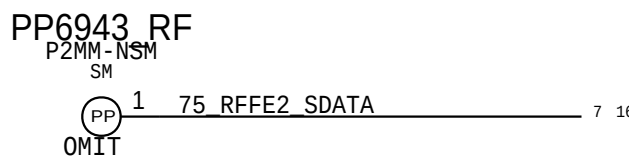
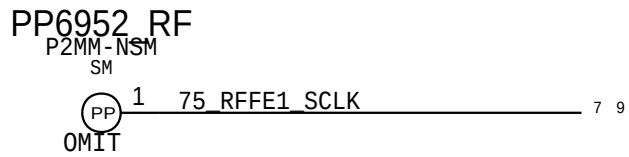
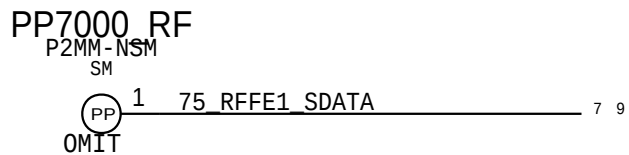


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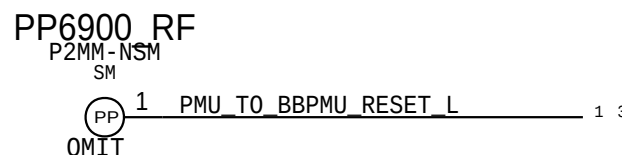
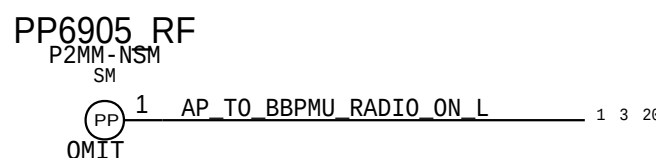
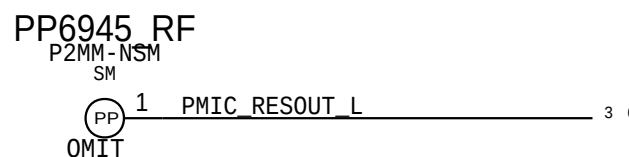


MLB TEST POINTS

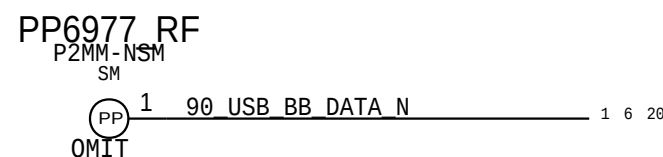
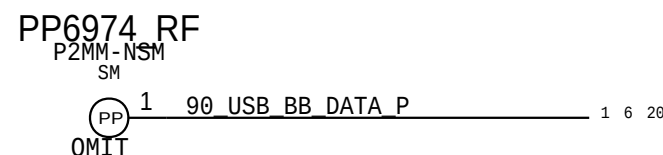
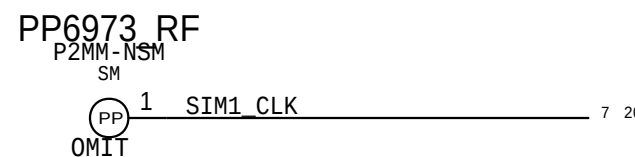
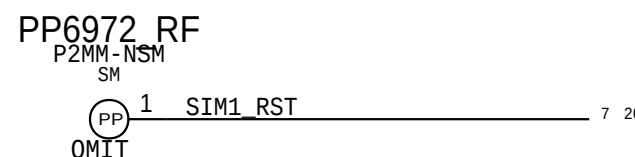
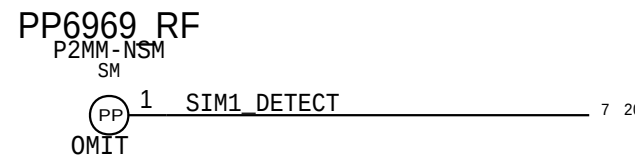
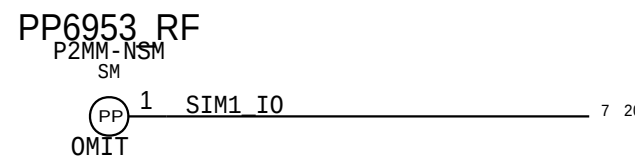
BASEBAND



BBPMU

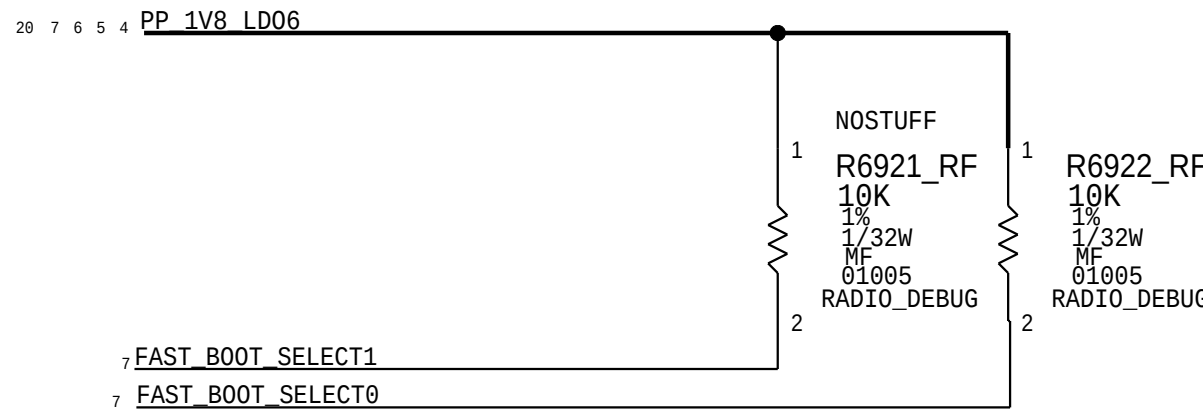


SIM

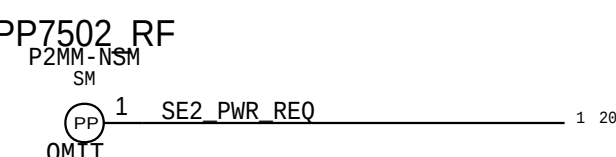
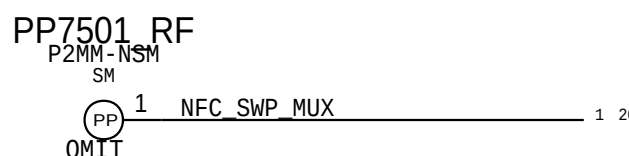
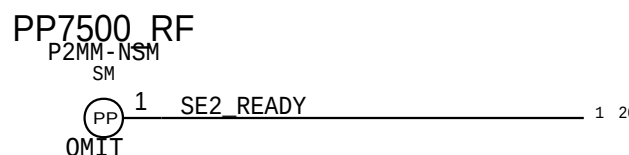
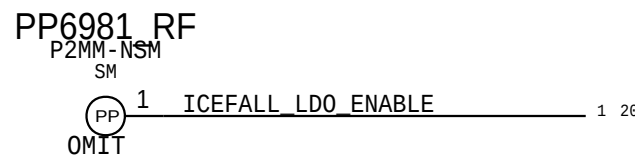
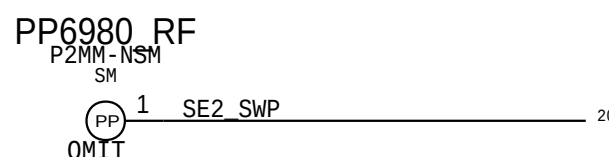
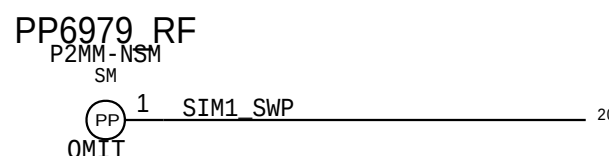
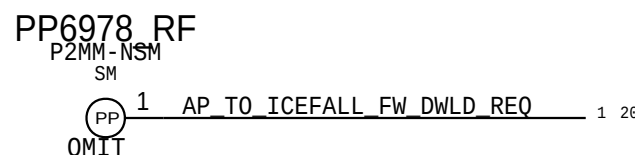


BOOT CONFIG

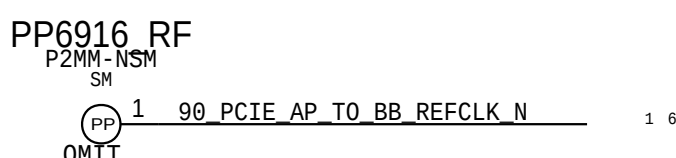
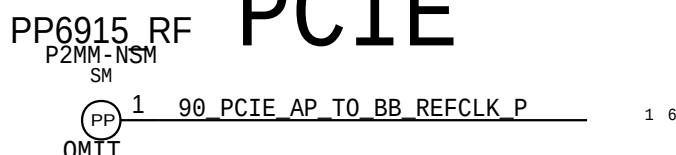
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USB = 0X2



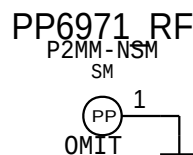
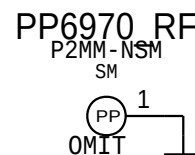
ICEFALL



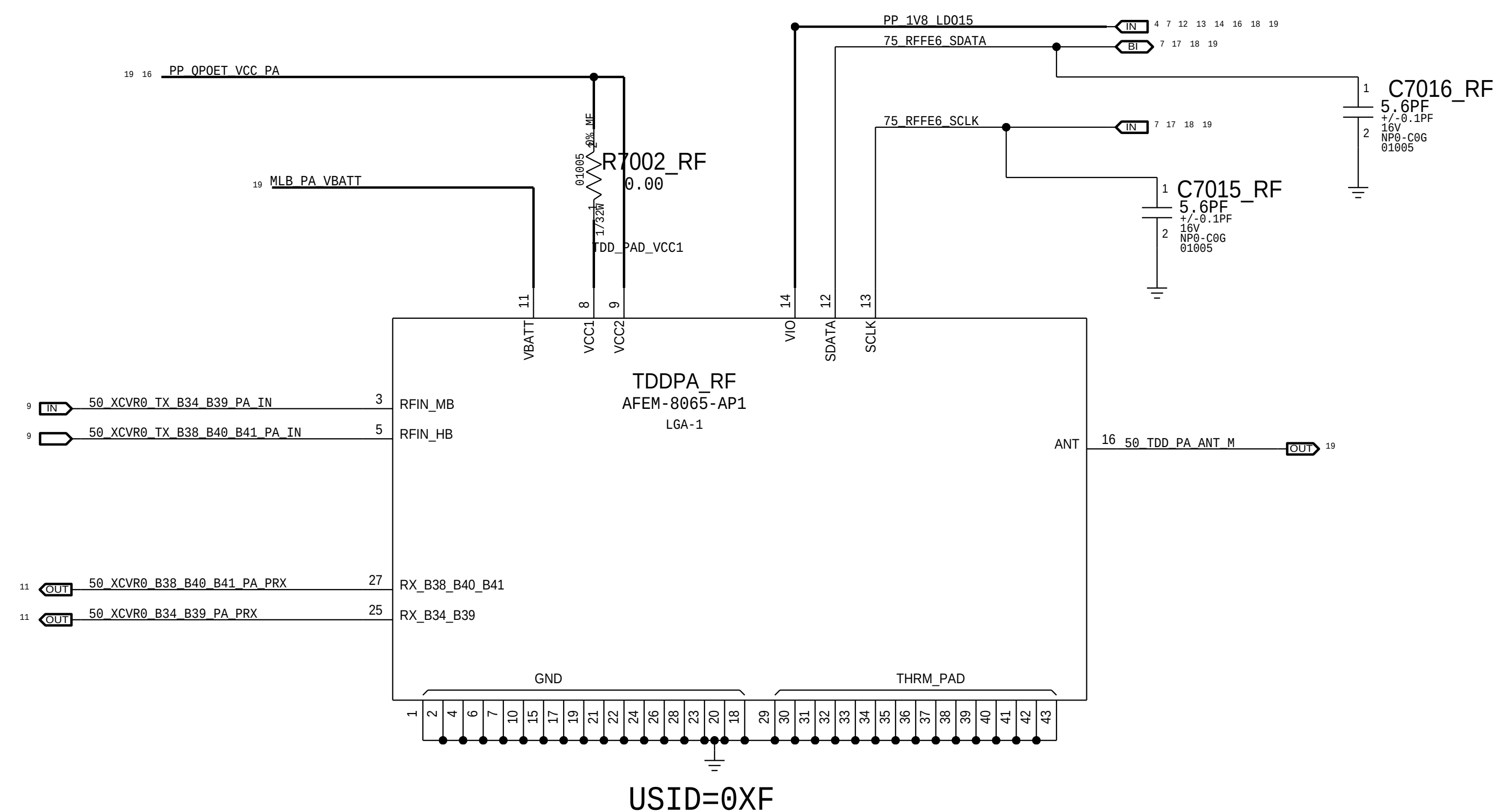
PCIE



PCIE GND



2G PA

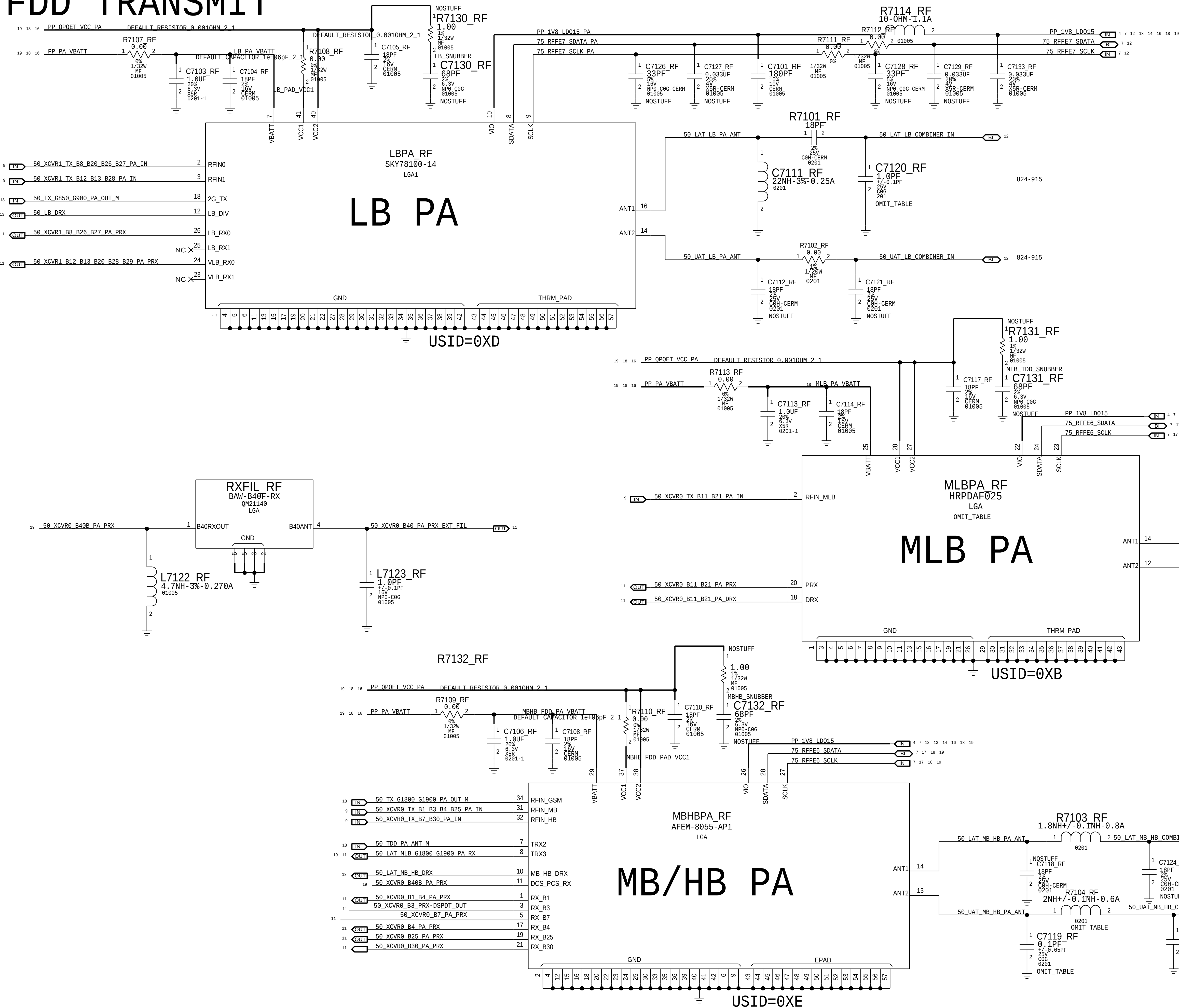




LB PA

MLB PA

MB/HB PA



ICEFALL

SWP MUX

ICEFALL LDO

DEBUG CONNECTOR

SIM CARD CONNECTOR

